Option Pricing PDEs with validation and calibration

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Abstract

Option pricing lies at the heart of quantitative finance, and Partial Differential Equations (PDEs) provide one of its most powerful and flexible computational frameworks. This report derives, solves, and validates PDE-based pricers for four distinct models – Black–Scholes, Heston stochastic volatility, multi-asset correlated Black–Scholes, and the Merton Jump-Diffusion process (PIDE) – together with an American option formulated as a Linear Complementarity Problem. For each model, a dedicated finite-difference scheme is implemented: an implicit scheme for the 1D Black–Scholes PDE, an Alternating Direction Implicit (ADI) method for the 2D Heston PDE, a full 2D finite-difference solver for multi-asset options, and an operator-splitting approach for the Merton PIDE. All solvers are systematically validated against analytical closed-form solutions where available, Monte Carlo simulations (including the Longstaff–Schwartz algorithm for American options), and a suite of five binomial/trinomial tree methods. Models are then calibrated on live market data: implied volatility calibration for Black–Scholes, MSE-minimisation on the option chain for Heston and Merton, and the CBOE equicorrelation methodology for multi-asset products. Calibration results highlight the impact of option-chain liquidity on parameter identifiability – a finding illustrated by contrasting a successful calibration on AAPL with a degenerate one on TotalEnergies (TTE). The report concludes with real-life option pricing on AAPL and NVDA, discussing the numerical limitations identified during validation and their amplification under market-calibrated parameters.

Keywords: PDE, PIDE, American option, Black–Scholes, Heston stochastic volatility, Merton Jump-Diffusion, Free Boundary Problem, Finite Difference Method, ADI scheme, Operator Splitting, Monte Carlo, Longstaff–Schwartz, Binomial Trees, Market Calibration, Implied Volatility, CBOE Equicorrelation.

1 Introduction

Motivation

The price of a derivative contract is, under mild regularity conditions, the discounted expectation of its payoff under a risk-neutral probability measure. Translating this statement into a computable number requires both a model for the dynamics of the underlying asset and a numerical method to evaluate the expectation. Partial Differential Equations provide an elegant and efficient route: by Feynman–Kac theory, the risk-neutral pricing problem is equivalent to solving a PDE backward in time from the payoff at maturity to the option value at inception. The PDE framework is particularly attractive because it naturally handles path-independent payoffs, provides direct access to option Greeks through differentiation of the numerical solution, and offers rigorous error control through the theory of finite-difference schemes.

The workhorse of derivative pricing remains the Black–Scholes model, derived under the assumption of constant volatility and continuous asset dynamics. Yet practitioners have long known that this model fails to reproduce the implied-volatility smile observed in equity option markets. Two of the most influential extensions are (i) the Heston model, which replaces constant volatility by a mean-reverting stochastic variance process, thereby capturing the skew and kurtosis of log-returns, and (ii) the Merton Jump-Diffusion model, which augments the continuous diffusion with a compound Poisson process to account for sudden price jumps – an empirical feature of equity markets well documented since the seminal work of [**merton1976option**]. For multi-asset derivatives, a correlated multi-dimensional

Black–Scholes PDE governs the joint dynamics of the underlyings, with the correlation structure playing a central role in the pricing of spread, basket, and exchange options. Finally, American-style options require a Free Boundary Problem formulation – a Linear Complementarity Problem (LCP) – in which the early-exercise constraint introduces a nonlinear projection step at each time layer of the numerical scheme.

Scope and Objectives

This report pursues four main objectives:

1. **Derivation.** Starting from the risk-neutral dynamics of each model, derive rigorously the PDE or PIDE that governs the option value, using Itô's lemma and the martingale condition under the risk-neutral measure $\mathbb{Q}$.

2. **Numerical resolution.** Design and implement a finite-difference solver tailored to each PDE: an implicit scheme for the 1D Black–Scholes equation, an ADI Craig–Sneyd scheme for the 2D Heston equation, a full 2D finite-difference solver with Crank–Nicolson time-stepping for multi-asset options, and an operator-splitting method for the Merton PIDE. American options are handled by projecting the European PDE solution onto the early-exercise constraint at each time step.

3. **Validation.** Benchmark every PDE solver against independent pricing methods: closed-form analytical formulas (Black–Scholes, Heston characteristic function inversion, Merton infinite series, Margrabe exchange formula, Kirk basket approximation), Monte Carlo simulation (standard and Longstaff–Schwartz for American options), and five binomial/trinomial tree schemes (CRR, Rendleman–Bartter, Binomial Tian, Trinomial Tian, Joshi).

4. **Market calibration and pricing.** Calibrate all models on live option chains (AAPL, NVDA, TTE, SPY) using, respectively, implied-volatility fitting, MSE minimisation, and the CBOE equicorrelation methodology. Price a representative set of European and American products – vanilla puts and calls, exchange options, spread options, and basket options – and discuss the numerical findings in light of the calibrated parameters.

Structure of the Report

The remainder of this report is organised as follows.

Section 2 derives the five PDEs/PIDEs under study: the Black–Scholes PDE (§2.1), the Heston PDE (§2.2), the multi-asset PDE (§2.3), the Merton Jump-Diffusion PIDE (§2.4), and the American option LCP (§2.5). All derivations are carried out under the risk-neutral measure $\mathbb{Q}$.

Section 3 presents the finite-difference methods used to solve each equation: the fully implicit scheme (§3.1), the ADI method on non-uniform grids (§3.2), the 2D multidimensional scheme (§3.3), operator splitting for the PIDE (§3.4), and the early-exercise projection for American options (§3.5).

Section 4 introduces the benchmark methods used to validate the PDE solvers: analytical solutions (§4.1), Monte Carlo and the Longstaff–Schwartz algorithm (§4.2), and binomial/trinomial trees (§4.3).

Section 5 presents the calibration procedures: risk-free rate bootstrapping and implied-volatility fitting for Black–Scholes (§5.1), MSE calibration for Heston (§5.2), CBOE equicorrelation for the multi-asset model (§5.3), and Peak-over-Threshold estimation for Merton (§5.4).

Section 6 reports all numerical results: PDE validation under manually fixed parameters (§6.1), followed by real-life pricing under market-calibrated parameters (§6.2).

Appendices provide supporting material on martingale theory and the Girsanov theorem (Appendix A.1), an introduction to Merton Jump-Diffusion processes (Appendix A.2), and the proof of the MJD PIDE (Appendix A.3).

Remark 1 (Scope)**.** All derivations and computations are performed under the risk-neutral measure $\mathbb{Q}$. We consider non-dividend-paying stocks throughout; the extension to continuous dividend yields is straightforward but introduces well-posed issues when the yield exceeds the risk-free rate, and is therefore left out of scope. This report is intended as an academic exercise; all results are cited to the best of the author's ability and have not been peer-reviewed.

2 PDE's and mathematical frameworks

This first chapter will study some well-known and well-established partial differential equations used in order to price vanilla options on stocks. We will be studying five different PDE's :

1. Original **Black-Scholes PDE** for Geometric Brownian Motion (GBM) dynamics of a single stock for a European-style option
2. PDE for a single stock following **Heston dynamics** (stochastic volatility) for a European-style option
3. **Multi-dimensional PDE** for a European-style option depending on more than one underlying stock
4. A Partial Integral Diffential Equation (PIDE) for an underlying stock following a **Merton Jump-Diffusion** (MJD) **process**
5. A PDE for an **american-style option** (path-dependant) for an underlying stock following GBM dynamics

Reader will notice that these P(I)DE are each derived in a similar way. It is recommended to pay attention of the steps followed in the first derivation for a better understanding of the next ones.
Some important remarks that are worth mentionning in order to define the precise framework we are working in :

- We are working with European options for four first cases. The only exception is the last equation that studies the price of an american option where we will have to deal with early exercise rule.
- We are working with non-dividend paying stock. If one desires to include dividend in the model, a possibility is to model it as a continuous divident yield and to substract it from the risk-free rate. This solution has two drawbacks : It is assumed constant and continuous and the studied problems become ill-posed if the continuous dividend yield is greater than risk-free rate. So we will stick with non-dividend paying stock.
- Pricing universe requires to work within risk-neutral measure. The dynamics of each underlying asset in this section is presented in its "risk-neutral dynamics". The proof of the american option in section 2.5 shows that even when we model an asset with its "real-world dynamics", the real-world return is not taken into account in the equation driving the option value.
- This chapter will often use Itô's and Martingale theory. A short summary of the theory of martingales is available in appendix A.1 while Itô's lemma will be mentionned when needed.

2.1 Model 1 : Black-Scholes PDE

Context

We are dealing with our first model to price options. This model will make strong assumptions but will provide a good intuition on how the dynamics of the underlying asset influence the price of the option at each time instant. Among the Black-Scholes assumptions, the one of constant volatility is probably one of the most irrealistic. It is obvious that volatility is not constant over time. Later on in this report, we will explain the concept of *"implied volatility"* which is an easy solution to deal with this wrong assumption in trading desks. For now, let us see how option are priced in the easiest and most well-known framework of pricing universe.

Underlying asset dynamics

We have an asset following a Geometric Brownian Motion (GBM) :

$$\frac{dS_t}{S_t} = r\,dt + \sigma dW_t^{\mathbb{Q}} \tag{2.1.1}$$

that can be rewritten as the following in order to identify an ito process :

$$dS_t = \underbrace{rS_t}_{K_t}\,dt + \underbrace{\sigma S_t}_{H_t}\,dW_t^{\mathbb{Q}} \tag{2.1.2}$$

Equation driving option value

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} + rS_t \frac{\partial V}{\partial S_t} - rV = 0 \tag{2.1.3}$$

Proof

Black-Scholes PDE can be derived by constructing a portfolio that will hedge away the risk. This is the principle of risk-neutral pricing : the value of a contract can be seen as the initial value needed to "fully" hedge the risk. However, in the next sections we will use a different approach to build PDE's driving option values. These derivations will analyze what one can call the *"drift"* of the process driving the option value. In fact, stochastic processes driving contracts value have two components : a *drift* and a *random*. The reason why we only pay attention to drift requires to understand martingale theory. An introduction and more complete explaination can be found in appendix A.1.
We start this proof by recalling the ito lemma in its differential form for a function of an ito process X_t :

Ito's Lemma :

$$df(X_t, t) = \left(\frac{\partial f}{\partial t} + \frac{1}{2}(H_t)^2 \frac{\partial^2 f}{\partial X_t^2} + K_t \frac{\partial f}{\partial X_t} \right) dt + H_t \frac{\partial f}{\partial X_t} dW_t \tag{2.1.4}$$

for an ito process of the form :

$$dX_t = K_t \, dt + H_t \, dW_t \tag{2.1.5}$$

Where

$$\int_0^T |K_s| ds < +\infty \tag{2.1.6}$$

$$\int_0^T |H_s|^2 ds < +\infty \tag{2.1.7}$$

Proof :

1. Dynamics of the asset under $\mathbb{Q}$:

$$\frac{dS_t}{S_t} = r \, dt + \sigma dW_t^{\mathbb{Q}} \tag{2.1.8}$$

2. European style option value :

$$V(t, S_t) \tag{2.1.9}$$

3. Ito's Lemma applied to the option value function :

$$dV = \left[\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} + r S_t \frac{\partial V}{\partial S_t} \right] dt + \sigma \, S_t \frac{\partial V}{\partial S_t} dW_t^{\mathbb{Q}} \tag{2.1.10}$$

4. Under risk-neutral measure, all assets discounted by a risk-free cash account B_t are martingale[1]. Thus :

$$\mathbb{E}^{\mathbb{Q}} \left[dV_t | \mathcal{F}_t \right] = rV dt \tag{2.1.11}$$

5. By equalling drifts of both equations (2.1.10) and (2.1.11), we get the desired PDE :

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} + r S_t \frac{\partial V}{\partial S_t} - rV = 0 \tag{2.1.12}$$

[1]Reader will find a reminder about martingale theory in Appendix A.1 and see why we state that "all assets discounted by risk-free cash account B_t are martingles".

2.2 Model 2 : Heston PDE

We can now start our (slightly more complex) derivation of the PDE under Heston dynamics.

Context

As introduced in the previous model, the assumption of constant volatility is rarely accurate. This is one of the reason of the introduction of this Heston model. In this model, the volatility is a mean-reverting stochastic process. This means that volatility is not constant over time, varies in a random way, but always tends to come back to its long-term mean value. In terms of statistical properties of the returns, Heston model helps building models that catch the fat tails of log-returns distributions (Kurtosis of returns) which is important in risk-management when we want our model to be able to reproduce extreme event.

Underlying asset dynamics

$$\begin{cases} \frac{dS_t}{S_t} = r\,dt + \sqrt{\nu_t} dW_t^{\mathbb{Q},1} \\ d\nu_t = \kappa(\theta - \nu_t)dt + \xi\sqrt{\nu_t} dW_t^{\mathbb{Q},2} \end{cases} \tag{2.2.1}$$

where

- ν_t is the spot variance process
- $\sigma_t = \sqrt{\nu_t}$ is the spot volatility
- κ is the "speed" of the mean-reverting process
- θ is the long-term average volatility
- ξ is the volatility of the stochastic process followed by the stock volatility
- We assume $W_t^{Q,1}$ and $W_t^{Q,2}$ are independent Brownian motions

Equation driving option value

$$rC_t = \frac{\partial C}{\partial t}dt + \frac{\partial C}{\partial S_t} rS_t + \frac{\partial C}{\partial \nu_t}\kappa(\theta - \nu_t) + \frac{\partial^2 C}{\partial (S_t)^2}\nu_t S_t^2 + \frac{1}{2}\frac{\partial^2 C}{\partial (\nu_t)^2}\xi^2\nu_t + \frac{\partial^2 C}{\partial S_t \partial \nu_t} S_t \xi \nu_t \tag{2.2.2}$$

Proof

We start this proof by recalling the Itô's lemma for multivariate process.
We assume to have n Itô's processes $X_t^1, \ldots, X_t^n$ defined over an interval $0 \leq t \leq T$. Each of these processes has a dimension p. Again, we give this lemma in its differential form for a function $f \in C^{2,1}$

Multivariate Itô's Lemma :

$$df(t, X_t^1, \dots, X_t^n) = \frac{\partial f}{\partial t}dt + \sum_{i=1}^{n} \frac{\partial f}{\partial X_t^i} dX_t^i + \frac{1}{2}\sum_{i,j=1}^{n} \frac{\partial^2 f}{\partial X_t^i \partial X_t^j} d\langle X_t^i, X_t^j\rangle\Big|_t \tag{2.2.3}$$

Where :

$$d\langle X_t^i, X_t^j\rangle\Big|_t = \left(H_t^i\right)^\top H_t^j \, dt \tag{2.2.4}$$

$$dX_t^i = K_t^i \, dt + \sum_{k=1}^{p} H_t^{i,k} dW_t^k \tag{2.2.5}$$

Where the following must hold for each Itô process :

$$\int_0^T |K_s| ds < +\infty \tag{2.2.6}$$

$$\int_0^T ||H_s||^2 ds < +\infty \tag{2.2.7}$$

Proof :

1. Dynamics of the asset under $\mathbb{Q}$

$$\begin{cases} \frac{dS_t}{S_t} = r\,dt + \sqrt{\nu_t} dW_t^{\mathbb{Q},1} \\ d\nu_t = \kappa(\theta - \nu_t)dt + \xi\sqrt{\nu_t} dW_t^{\mathbb{Q},2} \end{cases} \tag{2.2.8}$$

2. European style option value :

$$C(t, S_t, \nu_t) \tag{2.2.9}$$

3. Ito's Lemma applied to the option value function :

$$dC(t, S_t, \nu_t) = \frac{\partial C}{\partial t}dt + \left(\frac{\partial C}{\partial S_t}dS_t + \frac{\partial C}{\partial \nu_t}d\nu_t\right) + \frac{1}{2}\left(\frac{\partial^2 C}{\partial (S_t)^2} d\langle S_t, S_t\rangle\Big|_t + \frac{\partial^2 C}{\partial (\nu_t)^2} d\langle \nu_t, \nu_t\rangle\Big|_t\right) + \frac{\partial^2 C}{\partial S_t \partial \nu_t} d\langle S_t, \nu_t\rangle\Big|_t \tag{2.2.10}$$

Where

$$\begin{aligned} dS_t &= rS_t\,dt + \sqrt{\nu_t} S_t dW_t^{\mathbb{Q},1} \\ d\nu_t &= \kappa(\theta - \nu_t)dt + \xi\sqrt{\nu_t} dW_t^{\mathbb{Q},2} \\ d\langle S_t, S_t\rangle\Big|_t &= \nu_t S_t^2 dt \\ d\langle \nu_t, \nu_t\rangle\Big|_t &= \xi^2 \nu_t dt \\ d\langle S_t, \nu_t\rangle\Big|_t &= S_t \xi \nu_t dt \end{aligned}$$

4. We analyze the drift of $dC(t, S_t, \nu_t)$:

$$dC = \frac{\partial C}{\partial t}dt + \left(\frac{\partial C}{\partial S_t} r S_t\, dt + \frac{\partial C}{\partial S_t}\sqrt{\nu_t} S_t dW_t^{\mathbb{Q},1} + \frac{\partial C}{\partial \nu_t}\kappa(\theta - \nu_t)dt + \frac{\partial C}{\partial \nu_t}\xi\sqrt{\nu_t} dW_t^{\mathbb{Q},2}\right) + \frac{1}{2}\frac{\partial^2 C}{\partial (S_t)^2}\nu_t S_t^2 dt + \frac{1}{2}\frac{\partial^2 C}{\partial (\nu_t)^2}\xi^2 \nu_t dt + \frac{\partial^2 C}{\partial S_t \partial \nu_t} S_t \xi \nu_t dt \quad (2.2.11)$$

5. Under risk-neutral measure, all assets discounted by a risk-free cash account B_t are martingales. Thus :

$$\mathbb{E}^{\mathbb{Q}}\left[dC_t | \mathcal{F}_t\right] = rC_t\, dt \quad (2.2.12)$$

6. By equalling drifts of both equations (2.2.11) and (2.2.12), we get the desired PDE :

$$rC_t = \frac{\partial C}{\partial t}dt + \frac{\partial C}{\partial S_t} r S_t + \frac{\partial C}{\partial \nu_t}\kappa(\theta - \nu_t) + \frac{\partial^2 C}{\partial (S_t)^2}\nu_t S_t^2 + \frac{1}{2}\frac{\partial^2 C}{\partial (\nu_t)^2}\xi^2 \nu_t + \frac{\partial^2 C}{\partial S_t \partial \nu_t} S_t \xi \nu_t \quad (2.2.13)$$

Reader should pay attention this case of study, we assume the brownian motions to be independant. One can do the exact same computations with dependant brownian motion using following trick :

$$dW_t^1 = \rho\, dW_{t,corr}^1 + \sqrt{1-\rho^2} dW_{t,corr}^2$$
$$dW_t^2 = \rho\, dW_{t,corr}^2 + \sqrt{1-\rho^2} dW_{t,corr}^1$$

This would result in

$$d\langle S_t, \nu_t\rangle|_t = S_t \xi \nu_t \rho\, dt$$

Giving

$$rC_t = \frac{\partial C}{\partial t}dt + \frac{\partial C}{\partial S_t} r S_t + \frac{\partial C}{\partial \nu_t}\kappa(\theta - \nu_t) + \frac{\partial^2 C}{\partial (S_t)^2}\nu_t S_t^2 + \frac{1}{2}\frac{\partial^2 C}{\partial (\nu_t)^2}\xi^2 \nu_t + \frac{\partial^2 C}{\partial S_t \partial \nu_t} S_t \xi \nu_t \rho$$

From $\mathbb{P}$ to $\mathbb{Q}$ in Heston framework :
Suppose the underlying moves according to following real-world dynamcics :

$$\begin{cases} \frac{dS_t}{S_t} = \mu\, dt + \sqrt{\nu_t} dW_t^{1,\mathbb{P}} \\ d\nu_t = \kappa^{\mathbb{P}} \left(\theta^{\mathbb{P}} - \nu_t\right) dt + \xi^{\mathbb{P}} \sqrt{\nu_t} dW_t^{2,\mathbb{P}} \\ \rho^{\mathbb{P}} = dW_t^{1,\mathbb{P}} dW_t^{2,\mathbb{P}} \end{cases} \tag{2.2.14}$$

According to [47], the Girsanov theorem to go from $\mathbb{P}$ to $\mathbb{Q}$ gives :

$$\begin{cases} dW_t^{1,\mathbb{Q}} = dW_t^{1,\mathbb{P}} + \frac{\mu - r}{\sqrt{\nu_t}} dt \\ dW_t^{2,\mathbb{Q}} = dW_t^{2,\mathbb{P}} + \frac{\lambda}{\xi^{\mathbb{P}}} \sqrt{\nu_t} dt \end{cases}$$

Which leads to the risk-free dynamics introduced in equation (2.2.1), written in a different style as :

$$\begin{cases} \frac{dS_t}{S_t} = r\, dt + \sqrt{\nu_t} dW_t^{1,\mathbb{Q}} \\ d\nu_t = \kappa^{\mathbb{Q}} \left(\theta^{\mathbb{Q}} - \nu_t\right) dt + \xi^{\mathbb{Q}} \sqrt{\nu_t} dW_t^{2,\mathbb{Q}} \\ \rho^{\mathbb{Q}} = dW_t^{1,\mathbb{Q}} dW_t^{2,\mathbb{Q}} = \rho^{\mathbb{P}} \\ \kappa^{\mathbb{Q}} = \kappa^{\mathbb{P}} \\ \theta^{\mathbb{Q}} = \frac{\kappa^{\mathbb{P}} \theta^{\mathbb{P}}}{\kappa^{\mathbb{P}} + \lambda} \end{cases} \tag{2.2.15}$$

2.3 Model 3 : Multi-Asset PDE

We suppose our option now depends on n different assets. Each asset follows a GBM.

Context

Both previous models allowed to price european options where the payoff depended on only one underlying asset S_t. We introduced multivariate Itô's lemma when the asset dynamics was driven by two stochastic processes. Now, we will allow our option to depend on multiple underlying assets $S_t^1, S_t^2, \ldots$. We will restrict each asset dynamic to follow a GBM and not a more complex Heston model. For curious readers, multi-asset options with *"Merton-Jump-Diffusions"* dynamics are introduced in section 11.5 of [15]

Underlying asset dynamics

$$\underbrace{\begin{pmatrix} \frac{dS_t^1}{S_t^1} \\ \vdots \\ \frac{dS_t^n}{S_t^n} \end{pmatrix}}_{\underline{S_t}} = \underbrace{\begin{pmatrix} r_1 \\ \vdots \\ r_n \end{pmatrix}}_{\underline{r}} dt + \underbrace{\begin{pmatrix} \sigma_{11} & \sigma_{12} & \ldots & \sigma_{1n} \\ \sigma_{21} & \ddots & & \\ \vdots & & & \\ \sigma_{n1} & & & \sigma_{nn} \end{pmatrix}}_{\underline{\underline{\Sigma}}} \underbrace{\begin{pmatrix} dW_t^{\mathbb{Q},1} \\ \vdots \\ dW_t^{\mathbb{Q},n} \end{pmatrix}}_{\underline{dW_t}} \tag{2.3.1}$$

Equation driving option value

$$rC_t = \frac{\partial C}{\partial t} + \sum_{i=1}^{n} \frac{\partial C}{\partial S_t^i} r_i S_t^i dt + \frac{1}{2} \sum_{i,j=1}^{n} \frac{\partial^2 C}{\partial S_t^i \partial S_t^j} S_t^i S_t^j e_i^\top \underline{\underline{\Sigma}}\, \underline{\underline{\Sigma}}^\top e_j \tag{2.3.2}$$

Where

- $e_i \in \mathbb{R}^n$ is the i-th standard basis vector
- $\underline{\underline{\Sigma}}$ is Cholesky decomposition of covariance matrix of the assets return.

Proof

1. Dynamics of assets under $\mathbb{Q}$ are given by equation (2.3.1)
2. European-style option value :
$$C(t, S_t^1, \ldots, S_t^n) \tag{2.3.3}$$
3. Ito's Lemma applied to the option value function :
$$dC = \frac{\partial C}{\partial t} dt + \sum_{i=1}^{n} \frac{\partial C}{\partial S_t^i} dS_t^i + \frac{1}{2} \sum_{i,j=1}^{n} \frac{\partial^2 C}{\partial S_t^i \partial S_t^j} d\langle S_t^i, S_t^j \rangle \Big|_t \tag{2.3.4}$$
where
$$\begin{aligned} dS_t^i &= r_i S_t^i dt + e_i^\top \underline{\underline{\Sigma}}\, \underline{dW_t} \\ &= r_i S_t^i dt + \sigma_{i1}\, S_t^1\, dW_t^{\mathbb{Q},1} + \cdots + \sigma_{in}\, S_t^n\, dW_t^{\mathbb{Q},n} \\ &= r_i S_t^i dt + \sum_{k=1}^{n} \sigma_{ik}\, S_t^i\, dW_t^{\mathbb{Q},k} \end{aligned}$$

$$
\begin{aligned}
d\langle S_t^i, S_t^j\rangle\Big|_t &= (H_t^i)^\top H_t^j dt \\
&= S_t^i S_t^j \, e_i^\top \underline{\underline{\Sigma}}\,\underline{\underline{\Sigma}}^\top e_j \\
&= S_t^i S_t^j \, (\underline{\underline{\Sigma\Sigma}}^\top)_{\{ij\}}
\end{aligned}
$$

where

$$
H_t^i = S_t^i \underline{\underline{\Sigma}}^\top e_i
$$

4. We rewrite previous equation as :

$$
dC = \frac{\partial C}{\partial t}dt + \sum_{i=1}^n \frac{\partial C}{\partial S_t^i}\left(r_i S_t^i dt + \sum_{k=1}^n \sigma_{ik} dW_t^{\mathbb{Q},k}\right) + \frac{1}{2}\sum_{i,j=1}^n S_t^i S_t^j \frac{\partial^2 C}{\partial S_t^i \partial S_t^j} e_i^\top \underline{\underline{\Sigma}}\,\underline{\underline{\Sigma}}^\top e_j \tag{2.3.5}
$$

And we analyze the drift of $dC(t, S_t^1, \dots, S_t^n)$:

$$
dC = \frac{\partial C}{\partial t}dt + \sum_{i=1}^n \frac{\partial C}{\partial S_t^i}\left(r_i S_t^i dt + \sum_{k=1}^n \sigma_{ik} dW_t^{\mathbb{Q},k}\right) + \frac{1}{2}\sum_{i,j=1}^n S_t^i S_t^j \frac{\partial^2 C}{\partial S_t^i \partial S_t^j} e_i^\top \underline{\underline{\Sigma}}\,\underline{\underline{\Sigma}}^\top e_j dt \tag{2.3.6}
$$

5. Under risk-neutral measure, all assets discounted by a risk-free cash account B_t are martingale. Thus :

$$
\mathbb{E}^{\mathbb{Q}}\left[dC_t | \mathcal{F}_t\right] = rC_t \, dt \tag{2.3.7}
$$

6. By equalling drifts of both equations (2.3.6) and (2.3.7), we get the desired PDE :

$$
rC_t = \frac{\partial C}{\partial t} + \sum_{i=1}^n \frac{\partial C}{\partial S_t^i} r_i S_t^i dt + \frac{1}{2}\sum_{i,j=1}^n S_t^i S_t^j \frac{\partial^2 C}{\partial S_t^i \partial S_t^j} e_i^\top \underline{\underline{\Sigma}}\,\underline{\underline{\Sigma}}^\top e_j \tag{2.3.8}
$$

Result for option with two underlyings :
Further in this report, we will show how to solve this PDE for a 2-D case (ie, option depending on two underlyings). So we adapt the previous general result for a 2-D PDE.
Underlying assets dynamics :

$$
\begin{bmatrix} \frac{dS_t^1}{S_t^1} \\ \frac{dS_t^2}{S_t^2} \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} dt + \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix} \begin{bmatrix} dW_t^{1,\mathbb{Q}} \\ dW_t^{2,\mathbb{Q}} \end{bmatrix} \tag{2.3.9}
$$

Equation driving option value :

$$
\begin{aligned}
rC_t = \frac{\partial C}{\partial t} &+ r_1 S_t^1 \frac{\partial C}{\partial S^1} + r_2 S_t^2 \frac{\partial C}{\partial S^2} + \frac{(\sigma_{11}^2 + \sigma_{12}^2)\,(S_t^1)^2}{2}\frac{\partial^2 C}{\partial (S^1)^2} \\
&+ \frac{(\sigma_{21}^2 + \sigma_{22}^2)\,(S_t^2)^2}{2}\frac{\partial^2 C}{\partial (S^2)^2} + (\sigma_{11}\sigma_{12} + \sigma_{21}\sigma_{22})\, S_t^1 S_t^2 \frac{\partial^2 C}{\partial S^1 \partial S^2}
\end{aligned} \tag{2.3.10}
$$

2.4 Model 4 : Merton Jump-Diffusion PIDE

Context

We will now move on to slightly more complex dynamics for the underlying asset. Yet, this simple and intuitive modification will lead to much more complex numerical resolutions (see next chapter).
The idea behind this model is that the evolution of our asset price through time is not always continous as the Geometric Brownian Motion implicitly assumes. Here, jumps in the asset price are introduced. This can be observed for example after unexpected bad results are announced by a listed company and where we observe a stock price that lost 5% before market opening. In this example, asset value jumped. A classical way to do that is to assume that jumps follow a renewal process (hereafter, a Poisson process) and that log-size of these jumps are normally distributed (Merton Model).

Underlying asset dynamics

$$S_t = S_0 \exp(L_t) \tag{2.4.1}$$

where L_t is a Lévy-Process. An example could be :

$$L_t = \gamma t + \sigma W_t + Y_t \tag{2.4.2}$$

where

- $Y_t = \sum_{i=1}^{N_t} \log(y_t)$ Is a compound Poisson process
- $N_t \sim Po(\lambda t)$ Is the number of jumps that happened and follows a poisson process
- $\log(y_t) \sim \mathcal{N}(\mu_y, \delta_y)$ Is the absolute log-size of the jump from S_t to $S_{t+dt} = y_t S_t$
- $\gamma = r - \frac{\sigma^2}{2} - \lambda\kappa$
- λ Is the intensity of the jumps
- $\kappa = \exp\left(\mu_y + \frac{\delta_y^2}{2}\right) - 1$

For a more detailed comprehension of the intuition behind this model, the reader is invited to go appendix A.2 where an introduction is given on Merton-Jump-Diffusion processes.
Note that some authors would rather write the dynamics as $S_t = S_0 \exp(rt + X_t)$ and $\gamma = -\frac{\sigma^2}{2} - \lambda\kappa$.

Equation driving option value

$$rC_t = \underbrace{\frac{\partial C}{\partial t} + rS_t\frac{\partial C}{\partial S_t} + \frac{\sigma^2 S_t^2}{2}\frac{\partial^2 C}{\partial S_t^2}}_{*_1} + \underbrace{\int_{\mathbb{R}} \left[C(t, S_t e^y) - C(t, S_t) - S_t(e^y - 1)\frac{\partial C_t}{\partial S_t}\right]\nu(dy)}_{*_2} \tag{2.4.3}$$

Where we can intuitively understand terms $*_1$ as the diffusion part (attentive reader will recognize part of Black-Scholes PDE there) and $*_2$ as the part related to the jumps.

Proof

The development of the proof to find this PIDE is taken from [15] and a summary of this proof can be found in A.3. We recommend any curious reader to look at the an excellent introduction done there about jump processes in finance !

2.5 American Option PDE : a Linear Complementary Problem

Context

Here, the assumption of no yield dividend is particularly important. In fact, the PDE for American options is the exact same as the one for European options. But the numerical resolution must be adapted to the constraint of possibility of early exercise. We will also only consider put options for this model. As it can be shown, in the spectific case of no divident yield, both american and european call options have same value.
Eventhough we do not have a different PDE to solve, we will end up with a so-called *Linear Complementary Problem* (LCP).
As we will deal with american options, we invite the reader to get familiar with the difference between european and american put and call options on the following intuitive pricer developped by Wolfram Language and Mathematica :
Andrzej Kozlowski (2008), "Standard American and European Options" Wolfram Demonstrations Project. demonstrations.wolfram.com/StandardAmericanAndEuropeanOptions/

Underlying asset dynamics

In order to keep things easy, we go back to an asset following a Geometric Brownian Motion.

$$\frac{dS_t}{S_t} = r\,dt + \sigma\,dW_t \tag{2.5.1}$$

Equation driving option value

We define following quantities and operators :

- $V(t, S_t)$ is the value of the american option
- $\mathcal{L} \equiv \frac{\sigma^2 S_t^2}{2}\frac{\partial^2}{\partial S_t^2} + rS\frac{\partial}{\partial S_t} - r$ an operator
- $\phi(t, S)$ the payoff function

The LCP for american option pricing is :

$$\left(\frac{\partial V}{\partial t} + \mathcal{L}V\right)(V - \phi) = 0 \qquad \text{such that} \begin{cases} \frac{\partial V}{\partial t} + \mathcal{L}V \geq 0 \\ V - \phi \geq 0 \end{cases} \tag{2.5.2}$$

Where, and it is important to notice it, we have at least one stric equality out of the two inequalities described just above.

Proof

Hereafter, we show the developpement made in [5] to obtain the LCP in the specific case of a put option. In this proof, we assume our stock to follow a GBM within real-world measure. It will be shown, as mentionned in the introduction of this chapter, that the real-world return does not influence the equation driving option value over time. So we start with the dynamics of our underlying asset :

$$\frac{dS_t}{S_t} = \mu\,dt + \sigma\,dW_t \tag{2.5.3}$$

As we restrict ourselve to american put option, we will first get an intuition at what would happen in we introduced the "early exercise rule" for european put options. This will help us define two specific regions for our underlying value.

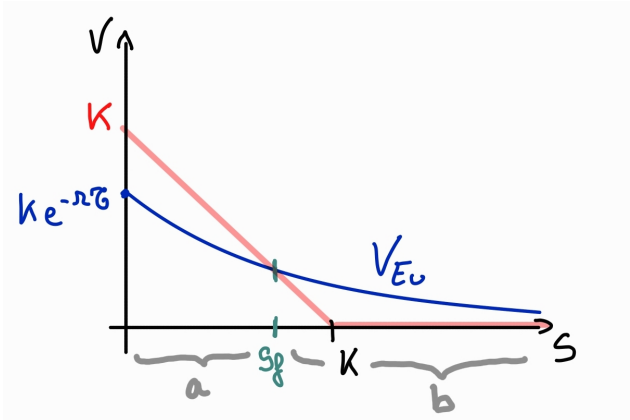


Figure 1: Schematic european payoff - put option

We analyze what would happen if early exercise rule was applied to figure 1 :

- **For small S :** let us say we take S close to 0, we have a value V_{eu} for the option. If we exercise the put instantaneously, we receive $K - S$ as the payoff $(K - S)_+$.
- We can make an arbitrage and have $K - \underbrace{S}_{\simeq 0} - V_{eu} > 0$.
- This is to explain that the values of European put options can not be the one of American put options.

The reasoning above suggests that, for a American option (wether it is a put or not),we must have the following inequality respected at any time:

$$V(t, S_t) - \phi(S) \geq 0 \tag{2.5.4}$$

So we guess that we should obtain a value function of the following form :

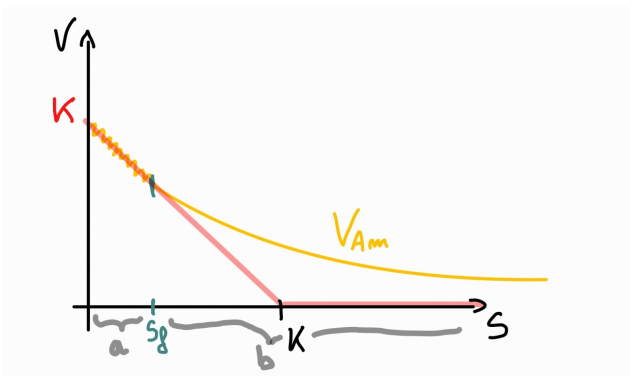


Figure 2: Schematic american payoff - put option

Where :

- For $S < S_f$, we have $V_{am} = \phi = (K - S)_+$
- For $S > S_f$, we have V driven by the classical Black-Scholes PDE : $\frac{\partial V}{\partial t} + \mathcal{L}V = 0$

We can now write a first part of our problem formulation :

1. Region (a) - if $S \leq S_f(t)$: option should be exercised and $V_{am} = (K - S)_+$
2. Region (b) - if $S > S_f(t)$: option should be held and $V_{am} > (K - S)_+$
3. We see that $0 \leq S_f(t) \leq K = S_F(T)$

We just demonstrated one of both inequalities that need to be satisfied at any time :

$$V - \phi \geq 0 \tag{2.5.5}$$

Problem is, this "boundary" $S_f(t)$ is unknown but essential to solve the problem. This is why this kind of problem is called ***"Free Boundary Problem"***.
We will now prove second inequality.
We have our asset following its real-world dynamics :

$$dS_t = \mu\, S_t\, dt + \sigma\, S_t\, dW_t \tag{2.5.6}$$

We create a risk-free portfolio Π(ie, we would like this portfolio to earn risk-free rate r). It is composed of one option and $-\Delta_t$ units of underlying asset. This term Δ_t is in fact $\frac{\partial V}{\partial S_t}$.

$$\Pi_t = V_t - \frac{\partial V}{\partial S_t} S_t \tag{2.5.7}$$

What is the variation of this portfolio $d\Pi_t$? We are trying to develop the following quantity :

$$d\Pi_t = dV_t - \Delta_t dS_t \tag{2.5.8}$$

First, we apply one-dimensional Itô's lemma to the option value function V :

$$dV_t = \left(\frac{\partial V_t}{\partial t} + \mu S_t \frac{\partial V_t}{\partial S_t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial V_t}{\partial S_t^2} \right) dt + \sigma S_t \frac{\partial V_t}{\partial S_t} dW_t \tag{2.5.9}$$

Putting (2.5.6) and (2.5.9) into (2.5.8), we obtain :

$$d\Pi_t = \left(\frac{\partial V_t}{\partial t} + \mu S_t \frac{\partial V_t}{\partial S_t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial V_t}{\partial S_t^2} \right) dt + \sigma S_t \frac{\partial V_t}{\partial S_t} dW_t - \mu\, S_t \frac{\partial V_t}{\partial S_t}\, dt - \sigma\, S_t \frac{\partial^2 V_t}{\partial S_t^2}\, dW_t$$

And we are left with

$$d\Pi_t = \left(\frac{\partial V_t}{\partial t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V_t}{\partial S_t^2} \right) dt \tag{2.5.10}$$

From here, we will show that :

- **(Part I)** The portfolio can not earn more than risk-free rate : ie, we can not have $d\Pi_t > r\, \Pi_t\, dt$, no matter the region we are in.
- **(Part II)** If we are in region (a), the following must hold : $d\Pi_t \leq r\, \Pi_t\, dt$
- **(Part III)** If we are in region (b), the following must hold : $d\Pi_t = r\, \Pi_t\, dt$

Why can the portfolio not earn more than risk-free rate ? (Part I)
An arbitrage profit would then be possible. We show it with following strategy.
At t_0, you :

1. borrow $\Pi_{t_0}\$$ from "Bank 1" at risk-free rate.
2. Pay V_{t_0} to the market to get $1V_{t_0}$ option
3. Short $a_{t_0}S_{t_0}$ part of the stock to the market and get paid $a_{t_0}S_{t_0}\$$

These three steps did not cost you any money at the moment. You received $\Pi_{t_0}\$$ from "Bank 1" and building the portfolio cost you exactly the same by definition. The cash flows are represented in next figure.

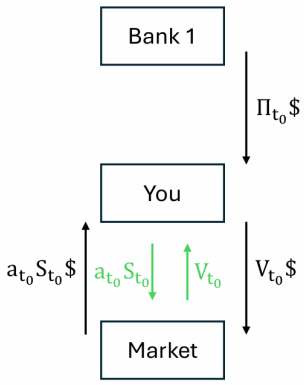


Figure 3: **(Part I)** Arbitrage - cash flows at t_0

After $t_1 = t_0 + dt$, you :

1. Have your portfolio value that grew from Π_{t_0} to $\Pi_{t_1} = \Pi_{t_0} + d\Pi_{t_0}$. You close your positions.
2. Owe the bank $\Pi_{t_0} + r\Pi_{t_0}dt$

The cash flows at t_1 are then :

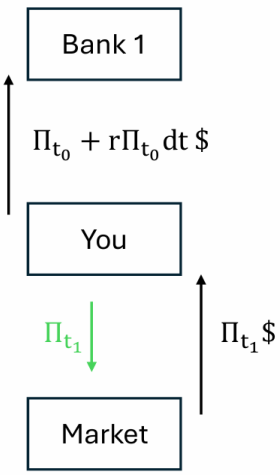


Figure 4: **(Part I)** Arbitrage - cash flows at t_1

Summing up your own cash flows at t_0 and t_1, you make a profit of $\Pi_{t_1} - (\Pi_{t_0} + r\Pi_{t_0}dt)$.

$$\text{Profit} = \Pi_{t_1} - (\Pi_{t_0} + r\Pi_{t_0}dt) = \Pi_{t_0} + d\Pi_{t_0} - (\Pi_{t_0} + r\Pi_{t_0}dt) = d\Pi_{t_0} - r\Pi_{t_0}dt > 0$$

Note that this reasoning is not influenced by the region. If we are in region (a), we exercise the option directly and make the profit at the moment. If we are in region (b), we wait until the stock falls back in region (a) which is guaranteed at maturity.

So, at the moment, we know that $d\Pi_t \leq r\,\Pi_t\,dt$.

If we are in region (a), the following must hold : $\mathbf{d\Pi_t \leq r\,\Pi_t\,dt}$

In region (a), we know that it is optimal to exercise immediately. So the value of the option is equal to its payoff :

$$V(t, S_t) = (K - S)_+ = K - S$$

We differentiate with respect to S

$$\frac{\partial V_t}{\partial S_t} = -1$$

Putting this inside portfolio definition :

$$\begin{aligned}\Pi_t &= V_t - (-1)S \\ &= K - S + S \\ &= K\end{aligned}$$

As the Portfolio value is constant, we have $d\Pi_t = 0$. And we have well the inequality :

$$d\Pi_t = 0 \leq r\,\Pi_t\,dt \tag{2.5.11}$$

If we are in region (b), the following must hold : $\mathbf{d\Pi_t = r\,\Pi_t\,dt}$

In region (b), exercising the option is not optimal. So, if one sells the option, the buyer will not exercise it right away. Therefore the option "survives" during the infinitesimal time interval dt. As it is not optimal to exercise the option and we have seen that $d\Pi_t \leq r\,\Pi_t\,dt$, we necesseraly have a strict equality :

$$d\Pi_t = r\,\Pi_t\,dt$$

By replacing terms in $d\Pi_t$ and Π_t, we see that in region (b), the following equality must hold :

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} + rS_t \frac{\partial V}{\partial S_t} - rV = 0 \tag{2.5.12}$$

Summary of what we showed so far.

No matter the region, we must have

$$V - \phi \geq 0$$

In region (a), we must have

$$V(t, S_t) = \phi$$

In region (b), we must have

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} + rS_t \frac{\partial V}{\partial S_t} - rV = 0$$

The last remaining problem is that these condition depend on the boundary $S_f(t)$. To eliminate dependencies to that free boundary, we formulate the problem as the following LCP :

$$\left(\frac{\partial V}{\partial t} + \mathcal{L}V\right)(V - \phi) = 0 \qquad \text{such that} \begin{cases} \frac{\partial V}{\partial t} + \mathcal{L}V \geq 0 \\ V - \phi \geq 0 \end{cases} \tag{2.5.13}$$

We can check that this LCP checks every needed condition :

- If $V - \phi = 0$, then $\frac{\partial V}{\partial t} + \mathcal{L}V \geq 0$ and we are well in region (a)
- if $\frac{\partial V}{\partial t} + \mathcal{L}V = 0$, then $V - \phi > 0$ and we are in region (b)
- At all times, $V - \phi \geq 0$

3 Numerical methods for studied P(I)DE's

Now that we obtained the equations (PDE's, PIDE and LCP) that we have to solve, we will present numerical methods to solve these. Numerical resolutions often comes with stability conditions that are not discussed in this work but that can be usually found in cited papers. This section is intended to give one numerical resolution for each problem. The first solution for the Black-Scholes model will be an implicit scheme for finite differences. Heston PDE will then be solved via an *"Alternative Direction Method" (ADI)* method. ADI method can be seen as a particular case of *"Operator Splitting" methods*. The reader will find a great introduction to intuition behind operator splitting techniques on this Github page [32]. Operator splitting method can be thought of a way of taking advantage of numerical resolution algorithms well suited for each part of the problem we are solving. For example, for the Merton Jump-Diffusion PIDE, we will use operator splitting so that we can solve the diffusion part with a given algorithm and leave the jump part to be solved with a better suited algorithm. We start this chapter with a short summary of what we have at the moment. For now, we are left with equations describing the evolution on an option value over time with respect to some pre-defined parameters such as risk-free rate, volatility, etc. These problems are well-posed but depend on what is called a "terminal condition". For an option, its fair value is known at maturity because, at that time, its value function is strictly equal to the payoff function. So, we are invastigating how final values (terminal values) are driving fair prices today. For this reason, we will start with a change of variable to turn our terminal-condition problem into an initial-condition problem.

$$\tau = T - t \qquad \text{(Time to maturity)} \tag{3.0.1}$$

Each next subsection will follow the same structure :

1. A **reminder** about dynamics and equation of the studied problem
2. The **Domain** we are working on
3. The **Boundary conditions** of the problem and explaination of these conditions.
4. **Methods** used to solve the problem
5. An easy **Case of study** where we apply the numerical methods to studied problem

It is important to note that some of these "boundary conditions" are actually properties of the specific derivative product we are working with and not actual conditions. Only "initial condition" are effectively condition that we need to impose to obtain a well-defined problem and unique solution. These initial conditions will be the payoff function applied to each point of the considered boundary. Otherwise, it could mean that for identical payoff function, we could set different boundary conditions and obtain different price (which does not make sens, one single payoff must give one unique price).

3.1 Model 1 : Finite differences - Implicit scheme

Reminder

Dynamics :

$$\frac{dS_t}{S_t} = r\,dt + \sigma\,dW_t \tag{3.1.1}$$

PDE :

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 V}{\partial S_t^2} + rS_t \frac{\partial V}{\partial S_t} - rV = 0 \tag{3.1.2}$$

As mentionned in the introduction of this chapter, we do a change of variable $\tau = T - t$ and obtain :

$$\frac{\partial V}{\partial \tau} - \frac{1}{2}\sigma^2 S_\tau^2 \frac{\partial^2 V}{\partial S^2} - rS_\tau \frac{\partial V}{\partial S} - rV = 0 \tag{3.1.3}$$

Domain

As in the following section, we start by defining the computational domain on which the Black–Scholes PDE will be solved:

- The contract has a maturity of T years, hence the time variable satisfies $\tau \in [0, T]$.
- The underlying asset price will be assumed to lie in the following spatial domain. For numerical purposes, and following [17], we truncate the domain by assuming a maximum asset value of four times the strike price:

$$S \in [0, S_{\max}], \qquad S_{\max} = 4K.$$

The time axis is divided into M equidistant intervals, and the stock price axis into N equidistant intervals. If we set $(S_{\max}, T) = (4K,\, T)$, we obtain the following domain:

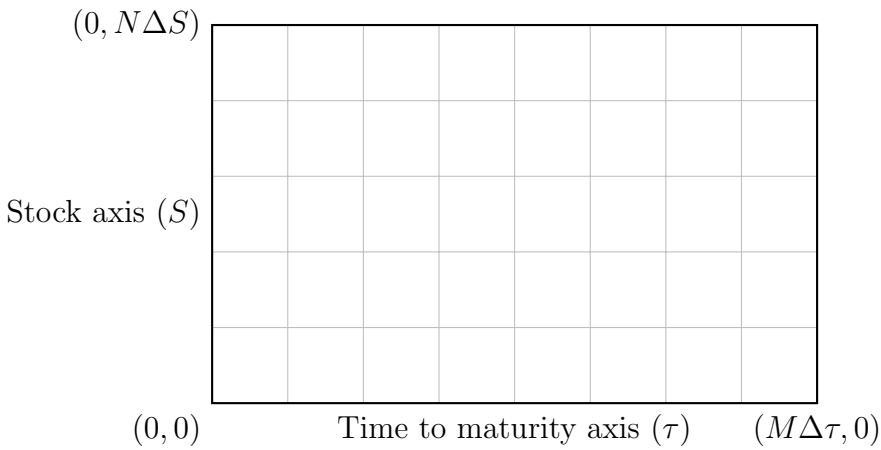


Figure 5: Black-Scholes PDE domain – illustration.

Boundary conditions

We will apply the following initial and boundary conditions according to the development made in [17].

European call option

- **Payoff at maturity (initial condition):**

$$V(S,\, \tau = 0) = \phi(S).$$

- **Lower boundary**[2]**:**

$$V(0, \tau) = 0, \qquad \forall\, \tau \in [0, T].$$

- **Upper boundary:**

$$\frac{V(S, \tau)}{S} \;\to\; 1 \quad \text{as} \quad S \to \infty.$$

Having defined the domain, grid, and boundary conditions, we are now in a position to apply finite-difference methods. In this report, we focus on a classical approach: the **implicit method**. As this method can be heavy in terms of computational timing, author refers to [24] for readers curious to improve their numerical performances.

Method

Following the development in [17], we discretize the time derivative using a backward difference, and the spatial (underlying) derivatives using first- and second-order central differences.

Spatial and Temporal Discretization

Consider a function $V : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ and a uniform mesh with spacing ΔS in the asset direction and $\Delta\tau$ in the time direction. We define the following finite-difference approximations:

$$\frac{\partial V}{\partial \tau}(n\Delta S,\, (m+1)\Delta\tau) \approx \frac{v_n^{m+1} - v_n^m}{\Delta\tau}, \tag{3.1.4}$$

$$\frac{\partial V}{\partial S}(n\Delta S,\, (m+1)\Delta\tau) \approx \frac{v_{n+1}^{m+1} - v_{n-1}^{m+1}}{2\,\Delta S}, \tag{3.1.5}$$

$$\frac{\partial^2 V}{\partial S^2}(n\Delta S,\, (m+1)\Delta\tau) \approx \frac{v_{n+1}^{m+1} - 2v_n^{m+1} + v_{n-1}^{m+1}}{(\Delta S)^2}. \tag{3.1.6}$$

The stencils associated with these approximations are illustrated in Figures 6– 8.

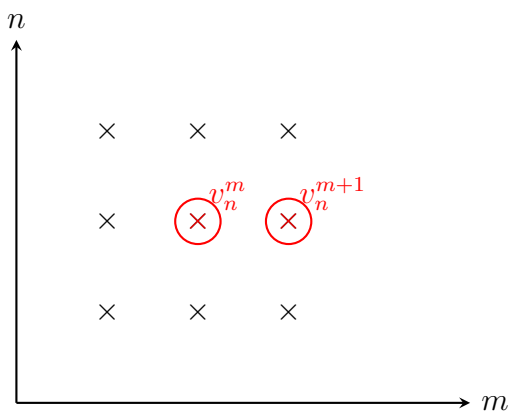


Figure 6: Stencil for the time derivative approximation (3.1.4).

[2]As mentionned in ***Remark 12.2*** in [15], this "condition" is a property of the solution for the particular case of some payoff functions (here a call). Each time we present this as a boundary condition, keep in mind it is only a property of the solution in a particular case! This is NOT a condition on the solution!

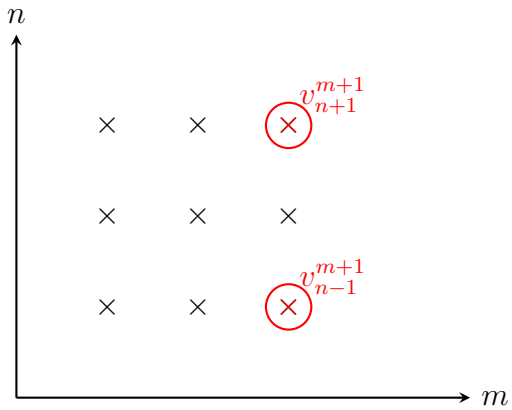


Figure 7: Stencil for the first spatial derivative approximation (3.1.5).

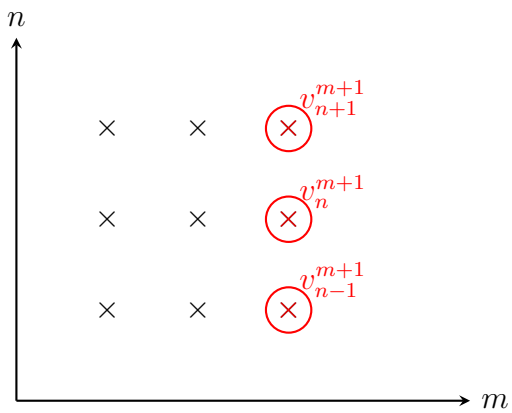


Figure 8: Stencil for the second spatial derivative approximation (3.1.6).

Recurrence Relation

Let $\alpha = \sigma\Delta\tau^2$ and $\beta = r\Delta\tau$. Substituting approximations (3.1.4)–(3.1.6) into (3.1.3), and rearranging terms, we obtain the following recurrence relation:

$$v_n^m = \frac{1}{2}(\beta n - \alpha n^2)\, v_{n-1}^{m+1} + (1 + \beta + \alpha n^2)\, v_n^{m+1} - \frac{1}{2}(\beta n + \alpha n^2)\, v_{n+1}^{m+1}. \tag{3.1.7}$$

Matrix Formulation

With boundary conditions specified and the problem reformulated as an initial value problem (with initial conditions defined by the payoff function), we seek to compute v^{m+1} from the known values v^m. We introduce a tridiagonal matrix A and a right-hand side vector

b^m as follows:

$$b^m = v^m - z^{m+1} = \begin{pmatrix} v_1^m \\ \vdots \\ v_{N-1}^m \end{pmatrix} - \begin{pmatrix} l_0\, v_0^{m+1} \\ 0 \\ \vdots \\ 0 \\ u_N\, v_N^{m+1} \end{pmatrix},$$

$$\begin{aligned} u_n &= -\frac{1}{2}(\beta(n-1) + \alpha(n-1)^2), \quad n = 1, \dots, N-1, \\ d_n &= 1 + \beta + \alpha n^2, \quad n = 2, \dots, N, \\ l_n &= \frac{1}{2}(\beta(n+1) - \alpha(n+1)^2), \quad n = 0, \dots, N-2, \end{aligned}$$

$$A = \begin{pmatrix} d_1 & u_2 & 0 & \cdots & 0 \\ l_1 & d_2 & u_3 & & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & & \ddots & \ddots & u_{N-1} \\ 0 & \cdots & 0 & l_{N-2} & d_{N-1} \end{pmatrix}.$$

The vector z^{m+1} encodes the boundary conditions at each time step. Thus, at each time step, the option values are computed by solving the tridiagonal system:

$$A\, v^{m+1} = b^m, \tag{3.1.8}$$

where A is the corresponding tridiagonal coefficient matrix.

Solution Algorithm

Thomas Algorithm for (3.1.8)

Forward sweep:
for $n = 2, \dots, N-1$ **do**
  $w \leftarrow l_{n-1}/d_{n-1}$
  $d_n \leftarrow d_n - w \cdot u_n$
  $b_n^m \leftarrow b_n^m - w \cdot b_{n-1}^m$
end for
Back substitution:
$v_{N-1}^{m+1} \leftarrow b_{N-1}^m / d_{N-1}$
for $n = N-2, \dots, 1$ **do**
  $v_n^{m+1} \leftarrow (b_n^m - u_{n+1} \cdot v_{n+1}^{m+1})/d_n$
end for

Case of study

3.2 Model 2 : ADI Finite differences and non-uniform grids

Alternating Direction Implicit method (ADI) is a sort of "operator splitting method" where the idea is to solve a 3D differential problem by using three consecutive 1D problems (one in S_t-direction, one in ν_t-direction followed by a time-step iteration). The numerical resolution presented here follow the ones developped in [25], [31] and [19]. This resolution will use a special grid as foundation to solve the problem. If one desires to have an implementation with equidistant points and for the same Heston PDE, we refer to the work done by Antoni Smolski in [40].

Reminder

Dynamics :

$$\begin{cases} \frac{dS_t}{S_t} = r\,dt + \sqrt{\nu_t} dW_t^{\mathbb{Q},1} \\ d\nu_t = \kappa(\theta - \nu_t)dt + \xi\sqrt{\nu_t} dW_t^{\mathbb{Q},2} \end{cases} \tag{3.2.1}$$

Equation :

$$rC_t = \frac{\partial C}{\partial t}dt + \frac{\partial C}{\partial S_t} rS_t + \frac{\partial C}{\partial \nu_t}\kappa(\theta - \nu_t) + \frac{\partial^2 C}{\partial (S_t)^2}\nu_t S_t^2 + \frac{1}{2}\frac{\partial^2 C}{\partial (\nu_t)^2}\xi^2\nu_t + \frac{\partial^2 C}{\partial S_t \partial \nu_t} S_t \xi \nu_t \tag{3.2.2}$$

As mentionned in the introduction of this chapter, we do a change of variable $\tau = T - t$ and obtain :

$$\frac{\partial C}{\partial \tau} = \frac{\partial C}{\partial S_\tau} rS_\tau + \frac{\partial C}{\partial \nu_\tau}\kappa(\theta - \nu_\tau) + \frac{\partial^2 C}{\partial (S_\tau)^2}\nu_\tau S_\tau^2 + \frac{1}{2}\frac{\partial^2 C}{\partial (\nu_\tau)^2}\xi^2\nu_\tau + \frac{\partial^2 C}{\partial S_\tau \partial \nu_t} S_\tau \xi \nu_\tau - rC_\tau \tag{3.2.3}$$

Domain and boundary conditions

As in the previous section, we start by defining the computational domain on which we will solve Heston PDE :

- The contract has a maturity of T years, hence the time variable $\tau \in [0, T]$
- The underlying asset price and spot variance will be assumed to be in the following spatial domain :

$$(S_\tau, \nu_\tau) \in [0, S_{max}] \times [0, V]$$

If we set $(S_{max}, V, T) = (8, 5, 4)$, we get the following domain :

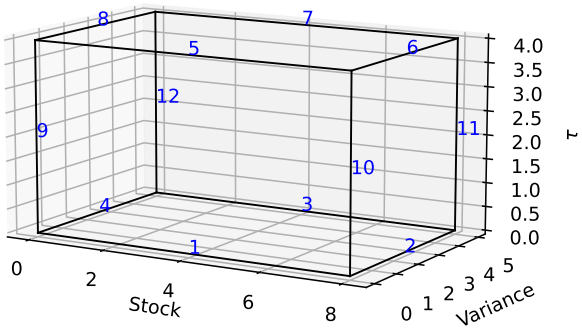


Figure 9: Heston PDE domain - illustration

We will apply the following initial and boundary conditions according to the developpment made in [35] :

European call option

- On edges 1,2,3,4 and surface defined between them : $C(S, \nu, 0) = \phi(S, K)$
- On edges 4,8,9,12 and surface defined between them : $C(0, \nu, \tau) = 0$
- On edges 2,6,10,11 and surface defined between them : $\frac{\partial C}{\partial S}(S_{max}, \nu, \tau) = 1$
- On deges 3,7,11,12 and surface defined between them : $C(S, V, \tau) = S$

European Put option

- On edges 1,2,3,4 and surface defined between them : $C(S, \nu, 0) = \phi(S, K)$
- On edges 4,8,9,12 and surface defined between them : $C(0, \nu, \tau) = 0$
- On edges 2,6,10,11 and surface defined between them : $\frac{\partial C}{\partial S}(S_{max}, \nu, \tau) = 1$
- On deges 3,7,11,12 and surface defined between them : $C(S, V, \tau) = Ke^{-r\tau}$

For more complex products boundary conditions, see [35].
In this section, we will use *"non-uniform mesh"* in S-direction and ν-direction in order to reduce the computational cost of the method. The discretization in time will follow a uniform mesh. The goal is to obtain many mesh points in the neighborhood of $S = K$ and $\nu = 0$

S-direction

Let $m_1 \in \mathbb{N}_{0,1}$ and $c \in \mathbb{R}_0^{+}$[3]. We define a set of equidistant points as $\zeta_0 < \zeta_1 < \cdots < \zeta_{m_1}$ that are defined as :

$$\zeta_i = \sinh^{-1}\left(\frac{-K}{c}\right) + i\dot{\Delta}\zeta \quad 0 \leq i \leq m_1 \tag{3.2.4}$$

[3]in fact, we take $c = K/5$

where

$$\Delta\zeta = \frac{1}{m_1}\left[\sinh^{-1}\left(\frac{S_{max}-K}{c}\right) - \sinh^{-1}\left(\frac{-K}{c}\right)\right] \tag{3.2.5}$$

The non-uniform mesh $0 = S_0 < S_1 < \cdots < S_{m_1} = S_{max}$ is given by :

$$S_i = K + c\sinh(\zeta_i), \quad 0 \leq i \leq m_1 \tag{3.2.6}$$

We illustrate how this gives a non-equidistant set of points in the next figure.

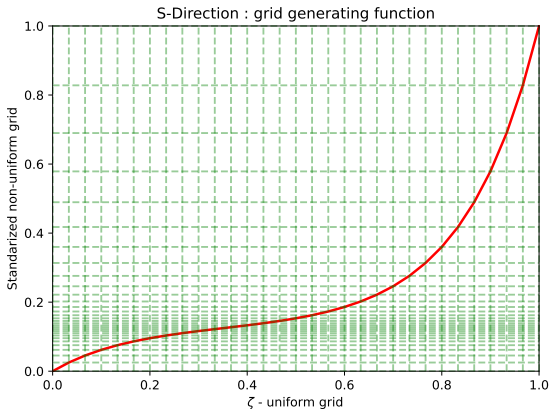


Figure 10: Standardized non-uniform grid in S direction

nu-direction

Let $m_2 \in \mathbb{N}_{0,1}$ and $d \in \mathbb{R}_0^+$ [4]. We define a set of equidistant points as $\eta_0 < \eta_1 < \cdots < \eta_{m_2}$ that are defined as $\eta_j = j\Delta\eta$ where

$$\Delta\eta = \frac{1}{m_2}\sinh^{-1}\left(\frac{V}{d}\right)$$

The non-uniform mesh $0 = \nu_0 < \nu_1 < \cdots < \nu_{m2} = V$ is given by :

$$\nu_i = d\sinh(\eta_j) \tag{3.2.7}$$

Combining two non-uniform meshes, we obtain our two-dimensional non-uniform grid at each time step :

[4] In fact, we will take $d = V/500$

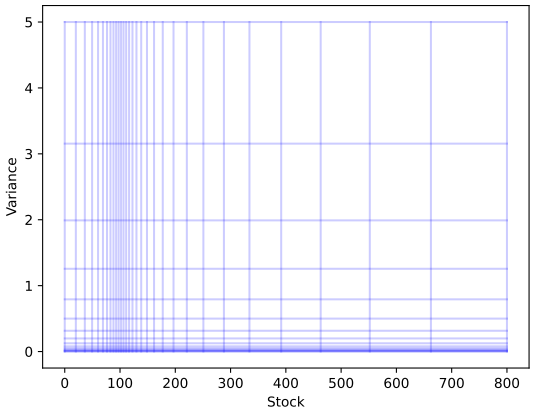


Figure 11: Heston PDE : Non-Uniform grid at each time step

We denote this grid by the set :

$$\mathcal{G} = \{(S_i, \nu_j) : 1 \leq i \leq m_1, 0 \leq j \leq m_2 - 1\} \tag{3.2.8}$$

As this grid is defined for each time step, we can imagine a problem defined with only three time steps t_1, t_2, t_3 for which the whole grid could be represented as :

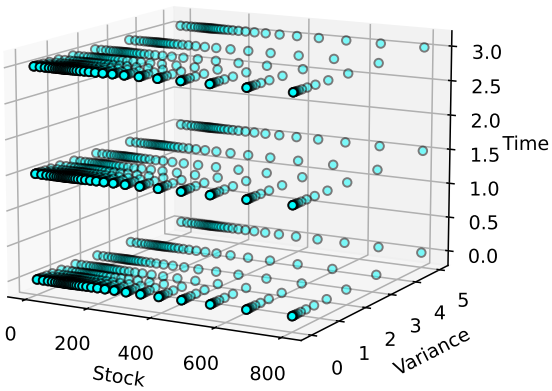


Figure 12: Heston PDE : Non-Uniform grid

Method

We begin by presenting the spatial discretization scheme used at each time step, followed by the ADI (Alternating Direction Implicit) scheme for time discretization.

Spatial Discretization

Consider a function $f : \mathbb{R} \to \mathbb{R}$ and mesh points $x_0 < x_1 < \cdots < x_m$ with spacing $\Delta x_i = x_i - x_{i-1}$. We define three finite difference approximations for the first derivative:

$$f'(x_i) \approx \alpha_{i,-2} f(x_{i-2}) + \alpha_{i,-1} f(x_{i-1}) + \alpha_{i,0} f(x_i), \tag{3.2.9}$$
$$f'(x_i) \approx \beta_{i,-1} f(x_{i-1}) + \beta_{i,0} f(x_i) + \beta_{i,1} f(x_{i+1}), \tag{3.2.10}$$
$$f'(x_i) \approx \gamma_{i,0} f(x_i) + \gamma_{i,1} f(x_{i+1}) + \gamma_{i,2} f(x_{i+2}), \tag{3.2.11}$$

with coefficients:

$$\alpha_{i,-2} = \frac{\Delta x_i}{\Delta x_{i-1}(\Delta x_{i-1} + \Delta x_i)}, \qquad \alpha_{i,-1} = -\frac{\Delta x_{i-1} + \Delta x_i}{\Delta x_{i-1}\Delta x_i}, \quad \alpha_{i,0} = \frac{\Delta x_{i-1} + 2\Delta x_i}{\Delta x_i(\Delta x_{i-1} + \Delta x_i)},$$
$$\beta_{i,-1} = -\frac{\Delta x_{i+1}}{\Delta x_i(\Delta x_i + \Delta x_{i+1})}, \qquad \beta_{i,0} = \frac{\Delta x_{i+1} - \Delta x_i}{\Delta x_i \Delta x_{i+1}}, \qquad \beta_{i,1} = \frac{\Delta x_i}{\Delta x_{i+1}(\Delta x_i + \Delta x_{i+1})},$$
$$\gamma_{i,0} = -\frac{2\Delta x_{i+1} + \Delta x_{i+2}}{\Delta x_{i+1}(\Delta x_{i+1} + \Delta x_{i+2})}, \qquad \gamma_{i,1} = \frac{\Delta x_{i+1} + \Delta x_{i+2}}{\Delta x_{i+1}\Delta x_{i+2}}, \quad \gamma_{i,2} = -\frac{\Delta x_{i+1}}{\Delta x_{i+2}(\Delta x_{i+1} + \Delta x_{i+2})}.$$

For the second derivative, we use the centered difference formula:

$$f''(x_i) \approx \delta_{i,-1} f(x_{i-1}) + \delta_{i,0} f(x_i) + \delta_{i,1} f(x_{i+1}), \tag{2.10}$$

where

$$\delta_{i,-1} = \frac{2}{\Delta x_i(\Delta x_i + \Delta x_{i+1})}, \qquad \delta_{i,0} = -\frac{2}{\Delta x_i \Delta x_{i+1}}, \qquad \delta_{i,1} = \frac{2}{\Delta x_{i+1}(\Delta x_i + \Delta x_{i+1})}.$$

The mixed partial derivative $\frac{\partial^2 f}{\partial x \partial y}$ at (x_i, y_j) is approximated by:

$$\frac{\partial^2 f}{\partial x \partial y}(x_i, y_j) \approx \sum_{k,l=-1}^{1} \beta_{i,k} \hat{\beta}_{j,l} f\left(x_{i+k}, y_{j+l}\right), \tag{3.2.12}$$

where $\hat{\beta}_{j,l}$ denotes the beta coefficients in the y-direction.

Applying these approximations to the Heston PDE, we obtain the discretized partial derivatives:

$$\begin{aligned}
\frac{\partial C}{\partial S}(S_i, \nu_j) &= \beta_{i,-1} C(S_{i-1}, \nu_j) + \beta_{i,0} C(S_i, \nu_j) + \beta_{i,1} C(S_{i+1}, \nu_j), \\
\frac{\partial C}{\partial \nu}(S_i, \nu_j) &= \hat{\beta}_{j,-1} C(S_i, \nu_{j-1}) + \hat{\beta}_{j,0} C(S_i, \nu_j) + \hat{\beta}_{j,1} C(S_i, \nu_{j+1}), \\
\frac{\partial^2 C}{\partial S^2}(S_i, \nu_j) &= \delta_{i,-1} C(S_{i-1}, \nu_j) + \delta_{i,0} C(S_i, \nu_j) + \delta_{i,1} C(S_{i+1}, \nu_j), \\
\frac{\partial^2 C}{\partial \nu^2}(S_i, \nu_j) &= \hat{\delta}_{j,-1} C(S_i, \nu_{j-1}) + \hat{\delta}_{j,0} C(S_i, \nu_j) + \hat{\delta}_{j,1} C(S_i, \nu_{j+1}), \\
\frac{\partial^2 C}{\partial S \partial \nu}(S_i, \nu_j) &= \sum_{k,l=-1}^{1} \beta_{i,k} \hat{\beta}_{j,l} C\left(S_{i+k}, \nu_{j+l}\right).
\end{aligned}$$

Time Discretization

The time discretization employs uniform time steps $\Delta\tau$. The temporal derivative is approximated by a first-order finite difference:

$$\frac{\partial C}{\partial \tau} \approx \frac{C(S_i, \nu_j, \tau_{n+1}) - C(S_i, \nu_j, \tau_n)}{\Delta \tau}. \tag{3.2.13}$$

Iterative Solution Procedure

With boundary conditions specified and the problem reformulated as an initial value problem (with initial conditions defined by the payoff function), we seek to compute $C(S_i, \nu_j, \tau_{n+1})$ from the known values $C(S_i, \nu_j, \tau_n)$.

At each iteration, our 2D grid is $\underline{C}(S, \nu, \tau_n) = \{C(S_i, \nu_j, \tau_n)\}_{i=0,j=0}^{m_1,m_2}$. Define the solution vector U^n containing all grid values $C(S_i, \nu_j, \tau_n)$ for $i \in \{0, \ldots, m_1\}$ and $j \in \{0, \ldots, m_2\}$. The vector U^n has size $m = (m_1 + 1) \times (m_2 + 1)$, with indexing:

$$U^n[k] = C(S_i, \nu_j, \tau_n) \quad \text{where} \quad k = i \cdot m_2 + j.$$

Our goal is to express the update from U^n to U^{n+1} using matrix notation. We introduce a matrix A (decomposed into A_0, A_1, A_2) and a vector b (decomposed into b_0, b_1, b_2), then present the Douglas and Craig-Sneyd iteration algorithms.

Substituting the spatial and temporal discretizations into the Heston PDE yields:

$$\begin{aligned}
\frac{C(S_i, \nu_j, \tau_{n+1}) - C(S_i, \nu_j, \tau_n)}{\Delta\tau} = {} & (\beta_{i,-1} C(S_{i-1}, \nu_j, \tau_n) + \beta_{i,0} C(S_i, \nu_j, \tau_n) + \beta_{i,1} C(S_{i+1}, \nu_j, \tau_n))\, r S_i \\
& + \Big(\hat{\beta}_{j,-1} C(S_i, \nu_{j-1}, \tau_n) + \hat{\beta}_{j,0} C(S_i, \nu_j, \tau_n) \\
& \qquad + \hat{\beta}_{j,1} C(S_i, \nu_{j+1}, \tau_n)\Big)\, \kappa(\theta - \nu_j) + (\delta_{i,-1} C(S_{i-1}, \nu_j, \tau_n) \\
& \qquad + \delta_{i,0} C(S_i, \nu_j, \tau_n) + \delta_{i,1} C(S_{i+1}, \nu_j, \tau_n))\, \frac{1}{2} \nu_j S_i^2 \\
& + \Big(\hat{\delta}_{j,-1} C(S_i, \nu_{j-1}, \tau_n) + \hat{\delta}_{j,0} C(S_i, \nu_j, \tau_n) \\
& \qquad + \hat{\delta}_{j,1} C(S_i, \nu_{j+1}, \tau_n)\Big)\, \frac{1}{2} \xi^2 \nu_j \\
& + \left(\sum_{k,l=-1}^{1} \beta_{i,k} \hat{\beta}_{j,l} C\left(S_{i+k}, \nu_{j+l}, \tau_n\right) \right) \rho S_i \xi \nu_j - r C(S_i, \nu_j, \tau_n).
\end{aligned} \tag{3.2.14}$$

We express this iteration compactly as:

$$U^{n+1} = (I + \Delta\tau A) U^n + \Delta\tau b, \tag{3.2.15}$$

where $A = A_0 + A_1 + A_2$ and $b = b_0 + b_1 + b_2$.

Matrix Construction: A_0, A_1, A_2

The matrix A is decomposed into three sparse matrices representing different derivative terms:

- A_0: mixed derivative terms $\frac{\partial^2 C}{\partial S \partial \nu}$
- A_1: S-direction terms (derivatives with respect to S)
- A_2: ν-direction terms (derivatives with respect to ν)

The discount term $-rC(S_i, \nu_j, \tau)$ is split equally between A_1 and A_2.

A_0 Construction Algorithm

```
for i = 1, ..., m_1 - 1 do
    for j = 1, ..., m_2 - 1 do
        k ← i · m_2 + j
        for d_i ∈ {-1, 0, 1} do
            for d_j ∈ {-1, 0, 1} do
                k' ← (i + d_i) · m_2 + (j + d_j)
                A_0[k, k'] ← ρξν_j S_i β_{i,d_i} β̂_{j,d_j}
            end for
        end for
    end for
end for
```

A_1 Construction Algorithm

```
for i = 1, ..., m_1 - 1 do
    for j = 1, ..., m_2 - 1 do
        k ← i · m_2 + j
        A_1[k, (i - 1) · m_2 + j] += (1/2) ν_j S_i^2 δ_{i,-1} + r S_i β_{i,-1}
        A_1[k, i · m_2 + j] += (1/2) ν_j S_i^2 δ_{i,0} + r S_i β_{i,0} - r/2
        A_1[k, (i + 1) · m_2 + j] += (1/2) ν_j S_i^2 δ_{i,1} + r S_i β_{i,1}
    end for
end for
```

A_2 Construction Algorithm

```
for i = 1, ..., m_1 - 1 do
    for j = 1, ..., m_2 - 1 do
        k ← i · m_2 + j
        A_2[k, i · m_2 + (j - 1)] += (1/2) ξ^2 ν_j δ̂_{j,-1} + κ(θ - ν_j) β̂_{j,-1}
        A_2[k, i · m_2 + j] += (1/2) ξ^2 ν_j δ̂_{j,0} + κ(θ - ν_j) β̂_{j,0} - r/2
        A_2[k, i · m_2 + (j + 1)] += (1/2) ξ^2 ν_j δ̂_{j,1} + κ(θ - ν_j) β̂_{j,1}
    end for
end for
```

These algorithms produce sparse matrices with the sparsity patterns shown in Figure 13.

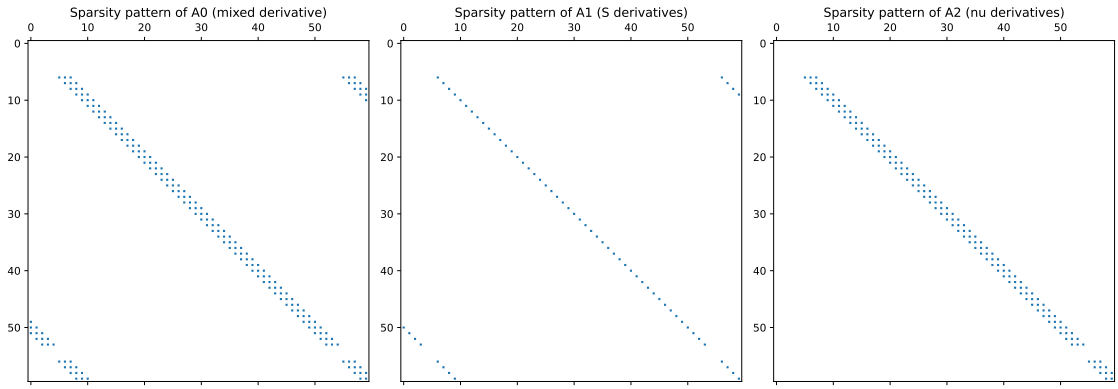


Figure 13: Sparsity patterns of intermediate matrices A_0, A_1, and A_2 (zoomed view)

The vectors b_0, b_1, b_2 are constructed analogously to incorporate boundary conditions and are omitted here for brevity.

ADI Time-Stepping Schemes

The ADI method alternates between implicit treatment of different spatial directions, enabling efficient solution of the resulting tridiagonal systems. We present two variants: Douglas and Craig-Sneyd. The formulations below are based on [19].

Douglas Scheme	**Craig-Sneyd Scheme**
$Y_0 = U^n + \Delta\tau(AU^n + b)$ $Y_1 = (I - \Theta\Delta\tau A_1)^{-1}(Y_0 - \Theta\Delta\tau A_1 U^n)$ $Y_2 = (I - \Theta\Delta\tau A_2)^{-1}(Y_1 - \Theta\Delta\tau A_2 U^n)$ $U^{n+1} = Y_2$	$Y_0 = U^n + \Delta\tau(AU^n + b)$ $Y_1 = (I - \Theta\Delta\tau A_1)^{-1}(Y_0 - \Theta\Delta\tau A_1 U^n)$ $Y_2 = (I - \Theta\Delta\tau A_2)^{-1}(Y_1 - \Theta\Delta\tau A_2 U^n)$ $Y_3 = Y_0 + \frac{1}{2}\Delta\tau A_0(Y_2 - U^n)$ $Y_4 = (I - \Theta\Delta\tau A_1)^{-1}(Y_3 - \Theta\Delta\tau A_1 U^n)$ $Y_5 = (I - \Theta\Delta\tau A_2)^{-1}(Y_4 - \Theta\Delta\tau A_2 U^n)$ $U^{n+1} = Y_5$

Table 1: Douglas and Craig-Sneyd ADI iteration schemes

The parameter Θ controls the implicitness of the scheme and is typically set to $\Theta = \frac{1}{2}$ (Crank-Nicolson), though other values may be chosen to balance stability and accuracy. The Craig-Sneyd scheme includes an additional correction step to better handle the mixed derivative term A_0, improving accuracy for problems with significant cross-derivative effects. More ADI schemes and more complete analysis of these techniques can, again, be found in [25], [31].

3.3 Model 3 : Finite differences - multidimensional

We will only treat the special case of a two-dimensional option (ie, an option that depends only on $S_{1,T}$ and $S_{2,T}$). For higher order cases, we refer the reader to [33].
What we will present here is an adaptation of the ADI method present in previous section. As we are working an option depending only on two underlying assets, we can see that the spot variance process in Heston model will be replaced by a new stochastic process describing the second underlying evolution over time.

Reminder

Dynamics :

$$\begin{bmatrix} \frac{dS_t^1}{S_t^1} \\ \frac{dS_t^2}{S_t^2} \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} dt + \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix} \begin{bmatrix} dW_t^{1,\mathbb{Q}} \\ dW_t^{2,\mathbb{Q}} \end{bmatrix} \tag{3.3.1}$$

PDE :

$$\begin{aligned} rC_t = \frac{\partial C}{\partial t} + r_1 S_t^1 \frac{\partial C}{\partial S^1} + r_2 S_t^2 \frac{\partial C}{\partial S^2} + \frac{\left(\sigma_{11}^2 + \sigma_{12}^2\right)\left(S_t^1\right)^2}{2} \frac{\partial^2 C}{\partial (S^1)^2} \\ + \frac{\left(\sigma_{21}^2 + \sigma_{22}^2\right)\left(S_t^2\right)^2}{2} \frac{\partial^2 C}{\partial (S^2)^2} + \left(\sigma_{11}\sigma_{12} + \sigma_{21}\sigma_{22}\right) S_t^1 S_t^2 \frac{\partial^2 C}{\partial S^1 \partial S^2} \end{aligned} \tag{3.3.2}$$

As mentionned in the introduction of this chapter, we do a change of variable $\tau = T - t$ and obtain :

$$\begin{aligned} \frac{\partial C}{\partial \tau} = r_1 S_\tau^1 \frac{\partial C}{\partial S^1} + r_2 S_\tau^2 \frac{\partial C}{\partial S^2} + \frac{\left(\sigma_{11}^2 + \sigma_{12}^2\right)\left(S_\tau^1\right)^2}{2} \frac{\partial^2 C}{\partial (S^1)^2} \\ + \frac{\left(\sigma_{21}^2 + \sigma_{22}^2\right)\left(S_\tau^2\right)^2}{2} \frac{\partial^2 C}{\partial (S^2)^2} + \left(\sigma_{11}\sigma_{12} + \sigma_{21}\sigma_{22}\right) S_\tau^1 S_\tau^2 \frac{\partial^2 C}{\partial S^1 \partial S^2} - rC_\tau \end{aligned} \tag{3.3.3}$$

Domain and boundary conditions

As in the previous section, we start by defining the computational domain on which we will solve Heston PDE :

- The contract has a maturity of T years, hence the time variable $\tau \in [0, T]$
- The two underlying assets price will be assumed to be in the following spatial domain :

$$(S_\tau^1, S_\tau^2) \in [0, S_{max}^1] \times [0, S_{max}^2]$$

If we set $(S_{max}^1, S_{max}^2, T) = (8, 5, 4)$, we get the following domain :

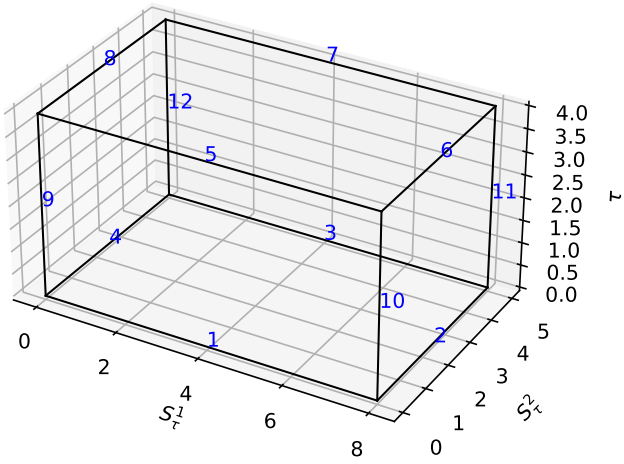


Figure 14: Multi asset PDE domain - illustration

Hereafter, we present the boundary conditions for several types of option according to the developpment made in [12].

Exchange option : The payoff of an exchange option is defined as :

$$\phi(S^1, S^2) = \left(S^1 - S^2\right)_+$$

- On edges 1,2,3,4 and surface defined between them : $C(S^1, S^2, 0) = \phi(S^1, S^2)$
- On edges 4,8,9,12 and surface defined between them : $C(0, S^2, \tau) = 0$
- On edges 1,5,9,10 and surface defined between them : $C(S^1, 0, \tau) = S^1$
- On edges 2,6,10,11 and surface defined between them : $C(S^1_{max}, S^2, \tau) = \phi(S^1_{\max}, S^2)$
- On edges 3,7,11,12 and surface defined between them : $C(S^1, S^2_{\max}, \tau) = \phi(S^1, S^2_{\max})$

Call Spread option : The payoff of a *call* spread option is defined as :

$$\phi(S^1, S^2, K) = \left(S^1 - S^2 - K\right)_+$$

- On edges 1,2,3,4 and surface defined between them : $C(S^1, S^2, 0) = \phi(S^1, S^2, K)$
- On edges 4,8,9,12 and surface defined between them : $C(0, S^2, \tau) = (-Ke^{-r\tau} - S^2)_+$
- On edges 1,5,9,10 and surface defined between them : $C(S^1, 0, \tau) = (S^1 - Ke^{-r\tau})_+$
- On edges 2,6,10,11 and surface defined between them : $C(S^1_{\max}, S^2, \tau) = (S^1_{\max} - S^2 - Ke^{-r\tau})_+$
- On edges 3,7,11,12 and surface defined between them : $C(S^1, S^2_{\max}, \tau) = (S^1 - S^2_{\max} - Ke^{-r\tau})_+$

Put Spread option : The payoff of a *put* spread option is defined as :

$$\phi(S^1, S^2, K) = \left(K - (S^1 - S^2)\right)_+$$

- On edges 1,2,3,4 and surface defined between them : $C(S^1, S^2, 0) = \phi(S^1, S^2, K)$
- On edges 4,8,9,12 and surface defined between them : $C(0, S^2, \tau) = (Ke^{-r\tau} + S^2)_+$
- On edges 1,5,9,10 and surface defined between them : $C(S^1, 0, \tau) = (Ke^{-r\tau} - S^1)_+$
- On edges 2,6,10,11 and surface defined between them : $C(S^1_{\max}, S^2, \tau) = (Ke^{-r\tau} - (S^1_{\max} - S^2))_+$
- On edges 3,7,11,12 and surface defined between them : $C(S^1, S^2_{\max}, \tau) = (Ke^{-r\tau} - (S^1 - S^2_{\max}))_+$

Call Basket option : The payoff of a *call* basket option is defined as :

$$\phi(w_1, w_2, S^1, S^2, K) = \left(\sum_{i=1}^{2} w_i S^i - K\right)_+$$

- On edges 1,2,3,4 and surface defined between them : $C(S^1, S^2, 0) = \phi(w_1, w_2, S^1, S^2, K)$
- On edges 4,8,9,12 and surface defined between them : $C(0, S^2, \tau) = (w_2 S^2 - Ke^{-r\tau})_+$
- On edges 1,5,9,10 and surface defined between them : $C(S^1, 0, \tau) = (w_1 S^1 - Ke^{-r\tau})_+$
- On edges 2,6,10,11 and surface defined between them :

$$C(S^1_{\max}, S^2, \tau) = \left(w_1 S^1_{\max} + w_2 S^2 - Ke^{-r\tau}\right)_+$$

- On edges 3,7,11,12 and surface defined between them :

$$C(S^1, S^2_{\max}, \tau) = \left(w_1 S^1 + w_2 S^2_{\max} - Ke^{-r\tau}\right)_+$$

Put Basket option : The payoff of a *put* basket option is defined as :

$$\phi(w_1, w_2, S^1, S^2, K) = \left(K - \sum_{i=1}^{2} w_i S^i\right)_+$$

- On edges 1,2,3,4 and surface defined between them : $C(S^1, S^2, 0) = \phi(w_1, w_2, S^1, S^2, K)$
- On edges 4,8,9,12 and surface defined between them : $C(0, S^2, \tau) = (Ke^{-r\tau} - w_2 S^2)_+$
- On edges 1,5,9,10 and surface defined between them : $C(S^1, 0, \tau) = (Ke^{-r\tau} - w_1 S^1)_+$
- On edges 2,6,10,11 and surface defined between them :

$$C(S^1_{\max}, S^2, \tau) = \left(Ke^{-r\tau} - (w_1 S^1_{\max} + w_2 S^2)\right)_+$$

- On edges 3,7,11,12 and surface defined between them :

$$C(S^1, S^2_{\max}, \tau) = \left(Ke^{-r\tau} - (w_1 S^1 + w_2 S^2_{\max})\right)_+$$

We now present the mesh we will solve our problem in. Here, an advantage of what we developped for Heston ADI is that we just need to modify the second axis and slightly adapt the formulae for the discretizations of S-directions. We replace the discretization in the ν-direction by an exact same one as for the S-direction.
In previous section, equations (3.2.4), (3.2.5), (3.2.6) depended on the strike K. Generally speaking, for options depending on multiple underlyings, the payoff is defined as a function of the difference between a common strike and a function of both underlyings : $\phi(S_T^1, S_T^2, K) = f(S_T^1, S_T^2) - K$. For example, a European basket call option payoff is expressed as :

$$\phi(S_T^1, \dots, S_T^N, K) = \left(\sum_{i=1}^{N} w_i S_T^i - K \right)_+$$

So it does not make sens any more to "center" our refined grid on S-direction around the strike only. Instead, we choose to center our refined grid aroung the initial value of the corresponding stock and to play with parameter c for a more or less refined grid around initial value $S_0^{1\,or\,2}$. The formulae for spatial discretizations are thus :
S^1-direction :

Let $m_1 \in \mathbb{N}_{0,1}$ and $c_1 \in \mathbb{R}_0^+$. We define a set of equidistant points as $\zeta_0 < \zeta_1 < \cdots < \zeta_{m_1}$ that are defined as :

$$\zeta_i = \sinh^{-1}\left(\frac{-S_0^1}{c_1}\right) + i\dot{\Delta}\zeta \quad 0 \le i \le m_1$$

where

$$\Delta\zeta = \frac{1}{m_1}\left[\sinh^{-1}\left(\frac{S_{max}^1 - S_0^1}{c_1}\right) - \sinh^{-1}\left(\frac{-S_0^1}{c_1}\right)\right]$$

The non-uniform mesh $0 = S_0^1 < S_1^1 < \cdots < S_{m_1}^1 = S_{max}^1$ is given by :

$$S_i^1 = S_0^1 + c_1 \sinh(\zeta_i), \quad 0 \le i \le m_1 \tag{3.3.4}$$

S^2-direction : Let $m_2 \in \mathbb{N}_{0,1}$ and $c_2 \in \mathbb{R}_0^+$. We define a set of equidistant points as $\zeta_0 < \zeta_1 < \cdots < \zeta_{m_2}$ that are defined as :

$$\zeta_i = \sinh^{-1}\left(\frac{-S_0^2}{c_2}\right) + i\dot{\Delta}\zeta \quad 0 \le i \le m_2$$

where

$$\Delta\zeta = \frac{1}{m_2}\left[\sinh^{-1}\left(\frac{S_{max}^2 - S_0^2}{c_2}\right) - \sinh^{-1}\left(\frac{-S_0^2}{c_2}\right)\right]$$

The non-uniform mesh $0 = S_0^2 < S_1^2 < \cdots < S_{m_2}^2 = S_{max}^2$ is given by :

$$S_i^2 = S_0^2 + c_2 \sinh(\zeta_i), \quad 0 \le i \le m_2 \tag{3.3.5}$$

If we set $S_0^1 = 50$ and $S_0^2 = 150$, we thus have a spatial grid of the following form :

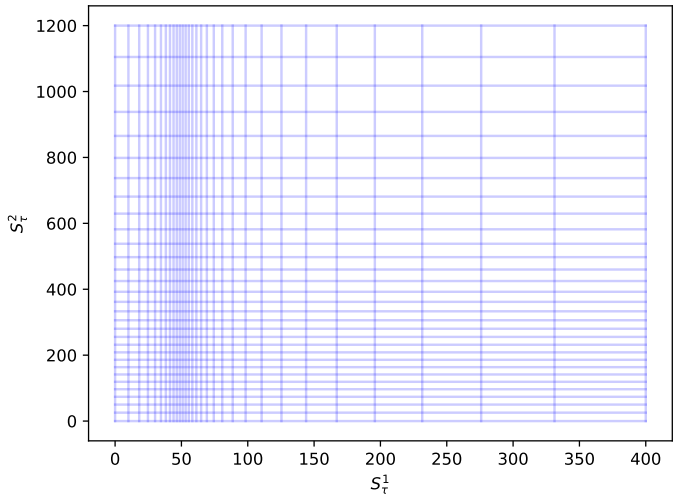


Figure 15: Multi Asset PDE : Non-Uniform grid at each ime step

And, as previously, we can imagine a problem defined with only three time steps t_1, t_2, t_3 for which the whole grid could be represented as :

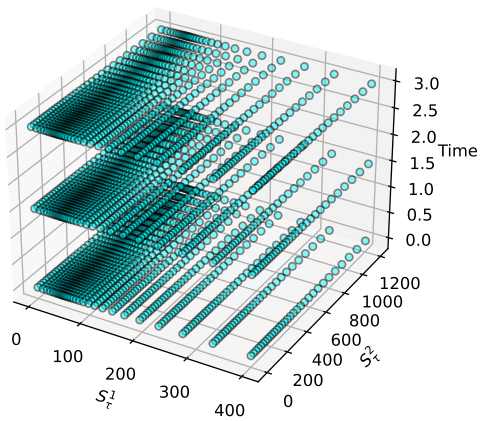


Figure 16: Multi asset PDE : Non-Uniform grid

Method

We will adapt the developpement made in Heston model section to this new non-uniform grid and slightly different PDE. The step will remain identical :

- spatial discretization
- time discretization
- Iterative solution procedure
- Matrices A_0, A_1, A_2 construction

Spatial Discretization

We consider the same definitions for the coefficient $\alpha_{i,\dots}, \beta_{i,\dots}, \delta_{i,\dots}, \gamma_{i,\dots}$ and the same difference schemes to approximate $f'(x_i), f''(x_i), \frac{\partial^2 f}{\partial x \partial y}$'. Applying these approximation to the 2-underlyings PDE, we obtain the discretized partial derivatives :

$$\begin{aligned}
\frac{\partial C}{\partial S^1}(S_i^1, S_j^2) &= \beta_{i,-1} C(S_{i-1}^1, S_j^2) + \beta_{i,0} C(S_i^1, S_j^2) + \beta_{i,1} C(S_{i+1}^1, S_j^2) \\
\frac{\partial C}{\partial S^2}(S_i^1, S_j^2) &= \hat{\beta}_{j,-1} C(S_i^1, S_{j-1}^2) + \hat{\beta}_{j,0} C(S_i^1, S_j^2) + \hat{\beta}_{j,1} C(S_i^1, S_{j+1}^2) \\
\frac{\partial^2 C}{\partial (S^1)^2}(S_i^1, S_j^2) &= \delta_{i,-1} C(S_{i-1}^1, S_j^2) + \delta_{i,0} C(S_i^1, S_j^2) + \delta_{i,1} C(S_{i+1}^1, S_j^2) \\
\frac{\partial^2 C}{\partial (S^2)^2}(S_i^1, S_j^2) &= \hat{\delta}_{j,-1} C(S_i^1, S_{j-1}^2) + \hat{\delta}_{j,0} C(S_i^1, S_j^2) + \hat{\delta}_{j,1} C(S_i^1, S_{j+1}^2) \\
\frac{\partial^2 C}{\partial S^1 \partial S^2}(S_i^1, S_j^2) &= \sum_{k,l=-1}^{1} \beta_{i,k} \hat{\beta}_{j,l} C(S_{i+k}^1, S_{j+l}^2)
\end{aligned}$$

Time Discretiaztion

Again, the time discretization employs uniform time steps $\Delta\tau$ and the temporal derivative is approximated with :

$$\frac{\partial C}{\partial \tau} \approx \frac{C(S_i^1, S_j^2, \tau_{n+1} - C(S_i^1, S_j^2, \tau_n}{\Delta\tau}$$

Iterative Solution Procedure

The iteration solution procedure follows the one described for Heston PDE resolution. We rewrite the same developpement as previous section with adapted notation for this 2-underlying PDE.

With boundary conditions specified and the problem reformulated as an initial value problem (with initial conditions defined by the payoff function), we seek to compute $C(S_i^1, S_j^2, \tau_{n+1})$ from the known values $C(S_i^1, S_j^2, \tau_n)$.

At each iteration, our 2D grid is $\underline{C}(S^1, S^2, \tau_n) = \{C(S_i^1, \S_j^2, \tau_n)\}_{i=0,j=0}^{m_1,m_2}$. Define the solution vector U^n containing all grid values $C(S_i^1, S_j^2, \tau_n)$ for $i \in \{0, \dots, m_1\}$ and $j \in \{0, \dots, m_2\}$. The vector U^n has size $m = (m_1 + 1) \times (m_2 + 1)$, with indexing:

$$U^n[k] = C(S_i^1, S_j^2, \tau_n) \quad \text{where} \quad k = i \cdot m_2 + j.$$

Our goal is to express the update from U^n to U^{n+1} using matrix notation. We introduce a matrix A (decomposed into A_0, A_1, A_2) and a vector b (decomposed into b_0, b_1, b_2), then present the Douglas and Craig-Sneyd iteration algorithms.

Substituting the spatial and temporal discretizations into the 2-underlyings PDE yields:

$$\begin{aligned}
&\frac{C(S_i^1, S_j^2, \tau_{n+1} - C(S_i^1, S_j^2, \tau_n}{\Delta\tau} = r_1 S_i^1 \left(\beta_{i,-1} C(S_{i-1}^1, S_j^2) + \beta_{i,0} C(S_i^1, S_j^2) + \beta_{i,1} C(S_{i+1}^1, S_j^2)\right) \\
&+ r_2 S_j^2 \left(\hat{\beta}_{j,-1} C(S_i^1, S_{j-1}^2) + \hat{\beta}_{j,0} C(S_i^1, S_j^2) + \hat{\beta}_{j,1} C(S_i^1, S_{j+1}^2)\right) \\
&+ \frac{(\sigma_{11}^2 + \sigma_{12}^2)\,(S_i^1)^2}{2} \left(\delta_{i,-1} C(S_{i-1}^1, S_j^2) + \delta_{i,0} C(S_i^1, S_j^2) + \delta_{i,1} C(S_{i+1}^1, S_j^2)\right) \\
&+ \frac{(\sigma_{21}^2 + \sigma_{22}^2)\,(S_j^2)^2}{2} \left(\hat{\delta}_{j,-1} C(S_i^1, S_{j-1}^2) + \hat{\delta}_{j,0} C(S_i^1, S_j^2) + \hat{\delta}_{j,1} C(S_i^1, S_{j+1}^2)\right) \\
&+ (\sigma_{11}\sigma_{12} + \sigma_{21}\sigma_{22})\, S_i^1 S_j^2 \left(\sum_{k,l=-1}^{1} \beta_{i,k} \hat{\beta}_{j,l} C(S_{i+k}^1, S_{j+l}^2)\right) - rC_\tau
\end{aligned} \tag{3.3.6}$$

We express this iteration compactly as :

$$U^{n+1} = (I + \Delta\tau A)U^n + \Delta\tau b,$$

where $A = A_0 + A_1 + A_2$ and $b = b_0 + b_1 + b_2$.
Matrix Construction: A_0, A_1, A_2
The matrix A is decomposed into three sparse matrices representing different derivative terms :

- A_0: mixed derivative terms $\frac{\partial^2 C}{\partial S^1 \partial S^2}$
- A_1: S^1-direction terms (derivatives with respect to S^1)
- A_2: §^{2}-direction terms (derivatives with respect to S^2)

The discount term $-rC(S_i, \nu_j, \tau)$ is split equally between A_1 and A_2.

A_0 Construction Algorithm

```
for i = 1, ..., m_1 - 1 do
    for j = 1, ..., m_2 - 1 do
        k ← i · m_2 + j
        for d_i ∈ {-1, 0, 1} do
            for d_j ∈ {-1, 0, 1} do
                k' ← (i + d_i) · m_2 + (j + d_j)
                A_0[k, k'] += (σ_11 * σ_12 + σ_21 * σ_22) * S_i^1 * S_j^2 β_{i,d_i} β̂_{j,d_j}
            end for
        end for
    end for
end for
```

A_1 Construction Algorithm

```
for i = 1, ..., m_1 - 1 do
    for j = 1, ..., m_2 - 1 do
        k ← i · m_2 + j
        A_1[k, (i - 1) · m_2 + j] += r * S_i^1 * β_{i,-1} + ((σ_11^2 + σ_12^2) * (S_i^1)^2 / 2) * δ_{i,-1}
        A_1[k, i · m_2 + j] += r * S_i^1 * β_{i,0} + ((σ_11^2 + σ_12^2) * (S_i^1)^2 / 2) * δ_{i,0}
        A_1[k, (i + 1) · m_2 + j] += r * S_i^1 * β_{i,1} + ((σ_11^2 + σ_12^2) * (S_i^1)^2 / 2) * δ_{i,1}
    end for
end for
```

A_2 Construction Algorithm

for $i = 1, \ldots, m_1 - 1$ **do**
  for $j = 1, \ldots, m_2 - 1$ **do**
    $k \leftarrow i \cdot m_2 + j$
    $A_2[k, i \cdot m_2 + (j-1)] \mathrel{+}= r * S_j^2 * \hat{\beta}_{j,-1} + \frac{\left(\sigma_{21}^2 + \sigma_{22}^2\right) * (S_j^2)^2}{2} * \hat{\delta}_{j,-1}$
    $A_2[k, i \cdot m_2 + j] \mathrel{+}= r * S_j^2 * \hat{\beta}_{j,0} + \frac{\left(\sigma_{21}^2 + \sigma_{22}^2\right) * (S_j^2)^2}{2} * \hat{\delta}_{j,0}$
    $A_2[k, i \cdot m_2 + (j+1)] \mathrel{+}= r * S_j^2 * \hat{\beta}_{j,1} + \frac{\left(\sigma_{21}^2 + \sigma_{22}^2\right) * (S_j^2)^2}{2} * \hat{\delta}_{j,1}$
  end for
end for

These algorithms produce sparse matrices as for Heston model with exact same sparsity patterns shown in figure 13. Only the values of the entries do differ.
Finally, the **ADI Time-Stepping schemes** are strictly identical to the ones used for Heston solution in previous section.

3.4 Model 4 : Operator splitting for PIDE

Context

We now consider a PIDE that incorporates an integral term arising from jumps in the underlying asset price. The numerical scheme implemented here follows closely the approach described in [7], and a broader study of finite-difference methods for PIDEs can be found in [16].

We first rewrite the PIDE in terms of the log-price (or *log-moneyness*) variable and then derive the final PIDE that is solved numerically using an operator-splitting finite-difference/FFT-convolution scheme.

Under the Merton jump-diffusion model, the (undiscounted) option price $C(t, S)$ satisfies the PIDE

$$rC = \frac{\partial C}{\partial t} + rS\frac{\partial C}{\partial S} + \frac{\sigma^2 S^2}{2}\frac{\partial^2 C}{\partial S^2} + \int_{\mathbb{R}} \left[C(t, Se^y) - C(t, S) - S(e^y - 1)\frac{\partial C}{\partial S}\right] \nu(dy),$$

where $\nu(dy) = \lambda f_Y(y)\, dy$ is the Lévy jump measure with intensity λ and jump-size density f_Y.

We introduce the log-price variable

$$x = \log(S),$$

and define $u(t, x) := C(t, e^x)$. Standard chain-rule manipulations yield

$$\begin{cases} rS\dfrac{\partial C}{\partial S} = r\dfrac{\partial u}{\partial x}, \\ \dfrac{\sigma^2 S^2}{2}\dfrac{\partial^2 C}{\partial S^2} = \dfrac{\sigma^2}{2}\left(\dfrac{\partial^2 u}{\partial x^2} - \dfrac{\partial u}{\partial x}\right), \\ C(t, Se^y) = u(t, x + y). \end{cases}$$

Substituting into the original PIDE gives

$$ru = \frac{\partial u}{\partial t} + \left(r - \frac{\sigma^2}{2}\right)\frac{\partial u}{\partial x} + \frac{\sigma^2}{2}\frac{\partial^2 u}{\partial x^2} + \int_{\mathbb{R}} \left[u(t, x + y) - u(t, x) - (e^y - 1)\frac{\partial u}{\partial x}\right] \nu(dy).$$

Using $\nu(dy) = \lambda f_Y(y)\, dy$, we separate the integral:

$$\int_{\mathbb{R}} \left[u(t, x + y) - u(t, x) - (e^y - 1)\frac{\partial u}{\partial x}\right] \nu(dy) = \int_{\mathbb{R}} u(t, x + y)\, \nu(dy) - \lambda u(t, x) \underbrace{\int_{\mathbb{R}} f_Y(y)\, dy}_{=1}$$
$$- \lambda \underbrace{\int_{\mathbb{R}} (e^y - 1) f_Y(y)\, dy}_{\kappa} \frac{\partial u}{\partial x}.$$

Here

$$\kappa := \int_{\mathbb{R}} (e^y - 1) f_Y(y)\, dy = \mathbb{E}[e^Y - 1].$$

Collecting terms, we obtain

$$ru = \frac{\partial u}{\partial t} + \left(r - \frac{\sigma^2}{2} - \lambda\kappa\right)\frac{\partial u}{\partial x} + \frac{\sigma^2}{2}\frac{\partial^2 u}{\partial x^2} + \int_{\mathbb{R}} u(t, x + y)\, \nu(dy) - \lambda u.$$

Rewriting this as an evolution equation for u, we get

$$\frac{\partial u}{\partial t} = \left(r - \frac{\sigma^2}{2} - \lambda\kappa\right)\frac{\partial u}{\partial x} + \frac{\sigma^2}{2}\frac{\partial^2 u}{\partial x^2} + \int_{\mathbb{R}} u(t, x+y)\,\nu(dy) - (r+\lambda)u.$$

Introducing the time-to-maturity variable $\tau = T - t$, so that $u(\tau, x) := u(T - \tau, x)$, and noting that $u_\tau = -u_t$, we obtain

$$\frac{\partial u}{\partial \tau} = \left(r - \frac{\sigma^2}{2} - \lambda\kappa\right)\frac{\partial u}{\partial x} + \frac{\sigma^2}{2}\frac{\partial^2 u}{\partial x^2} + \int_{\mathbb{R}} u(\tau, x+y)\,\nu(dy) - (r+\lambda)u. \tag{3.4.1}$$

Since the integral in (3.4.1) is taken over the whole real line, we must truncate it for numerical purposes. We thus consider the truncated PIDE

$$\boxed{\frac{\partial u}{\partial \tau} = \left(r - \frac{\sigma^2}{2} - \lambda\kappa\right)\frac{\partial u}{\partial x} + \frac{\sigma^2}{2}\frac{\partial^2 u}{\partial x^2} + \int_{-B_1}^{B_2} u(\tau, x+y)\,\nu(dy) - (r+\lambda)u, \qquad (3.4.2)}$$

where $B_1, B_2 > 0$ are truncation bounds chosen large enough to capture most of the jump mass.

Domain and boundary conditions

We define the computational domain as follows:

- The contract has maturity T, hence the time variable satisfies $\tau \in [0, T]$.
- The *effective* spatial domain on which we compute the option price is
$$x_\tau \in [A_1, A_2].$$
- To evaluate the truncated integral in (3.4.2), we must extend the spatial domain, so that
$$x_\tau \in [A_1 - B_1, A_2 + B_2].$$

Even though we only need the solution on $[A_1, A_2]$, the enlargement by B_1, B_2 is necessary to evaluate the shifted values $u(\tau, x+y)$ for $y \in [-B_1, B_2]$.

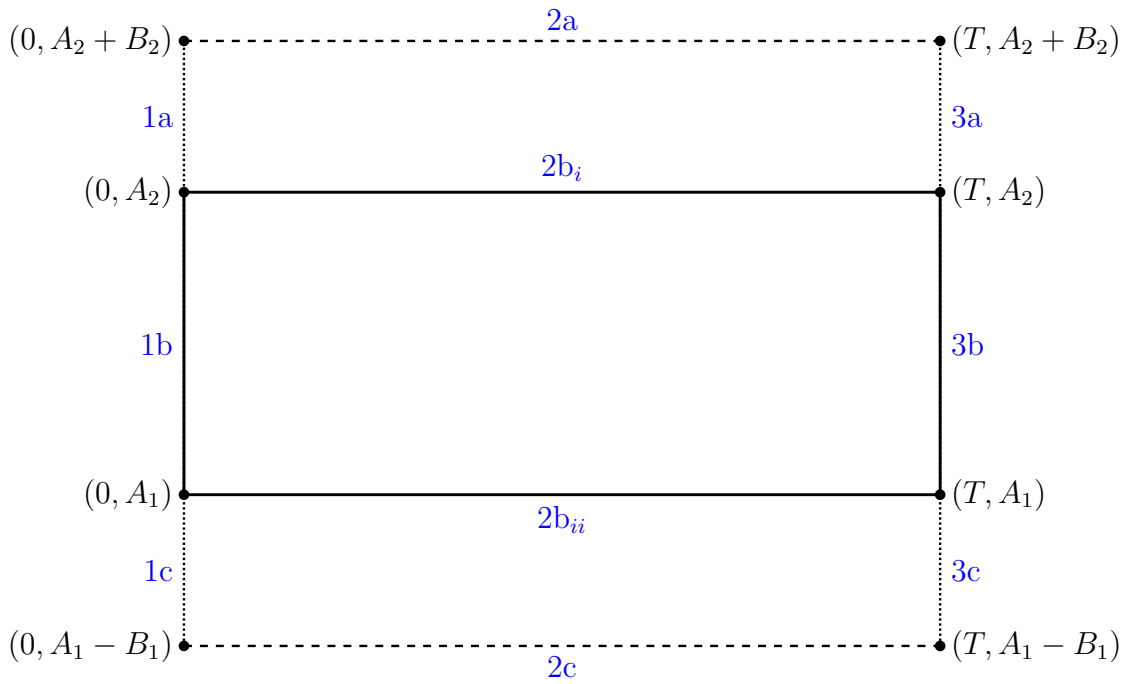


Figure 17: MJD PIDE computational domain in (τ, x).

We use a uniform grid in x (not drawn on Figure 17), exactly as in the Black–Scholes PDE case. Let Δx be the spatial step and $\Delta\tau$ the time step.

We impose the following boundary and terminal conditions.

For a **European call**:

- At maturity (edges $1a, 1b, 1c$):
$$u(0, x) = C(T, e^x) = (e^x - K)_+.$$
- For large x (edges $2a, 2b_i$ and the surface between them):
$$u(\tau, x) = e^x - Ke^{-r\tau},$$
which is the usual linear asymptotic of a call for $S \to \infty$.
- For small x (edges $2b_{ii}, 2c$ and the surface between them):
$$u(\tau, x) = 0,$$
corresponding to $C \to 0$ as $S \to 0$.

For a **European put**:

- At maturity (edges $1a, 1b, 1c$):
$$u(0, x) = C(T, e^x) = (K - e^x)_+.$$
- For small x (edges $2b_{ii}, 2c$ and the surface between them):
$$u(\tau, x) = Ke^{-r\tau},$$
which is the asymptotic behavior of a put as $S \to 0$.
- For large x (edges $2a, 2b_i$ and the surface between them):
$$u(\tau, x) = 0,$$
corresponding to $C \to 0$ as $S \to \infty$.

Choice of A_1, A_2, B_1, B_2

Following [7], the "inner" spatial domain $[A_1, A_2]$ is chosen in log-space exactly as in the Black–Scholes PDE section, with the change of variable $x = \log S$. For example, for strike K we may set

$$A_1 = \log\left(\frac{K}{3}\right), \qquad A_2 = \log(3K).$$

The additional extension used to evaluate the truncated integral is chosen to cover three standard deviations of the jump component. For a compound Poisson process with log-jump size $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$ and intensity λ, the variance of the jump component over a horizon τ is

$$\sigma_\Upsilon^2(\tau) = \mathrm{Var}\left[\sum_{i=1}^{N_\tau} Y_i\right] = \lambda(\mu_Y^2 + \sigma_Y^2)\,\tau.$$

We evaluate this at $\tau = 1$ and define the number of additional grid points by

$$e_p = \left\lfloor \frac{3\sigma_\Upsilon(\tau = 1)}{\Delta x} \right\rfloor,$$

so that

$$B_1 = e_p\,\Delta x = B_2.$$

In the implementation, this is exactly the role of the parameter `extraP` on the log-price grid.

Method

The PIDE (3.4.2) consists of a local diffusion-advection-reaction part and a non-local jump integral. We use an operator-splitting (IMEX) approach: the local operator is treated implicitly in τ, while the jump term is treated explicitly via convolution.

Integral resolution. The truncated integral

$$\int_{-B_1}^{B_2} u(\tau, x + y)\,\nu(dy) \tag{3.4.3}$$

is a discrete convolution in the log-price variable when evaluated on a uniform grid. More precisely, if we represent the values of u on the extended grid by a vector and discretize ν into weights $\{\nu_k\}$, then the discrete integral corresponds to the convolution $(u * \nu)_n$. As B_1, B_2 grow large, (3.4.3) approaches a full-line convolution. This suggests the use of FFT-based convolution, which is significantly more efficient than direct quadrature for large grids; see, e.g., [23] for background on convolution theory.

Spatial and time discretization. We consider a uniform grid $x_n = x_{\min} + n\Delta x$, $n = 0, \ldots, N-1$, extended by `extraP` $= e_p$ points on each side. Time-to-maturity is discretized as $\tau_m = m\Delta\tau$, $m = 0, \ldots, M-1$, with $\Delta\tau = T/(M-1)$. We denote by $v_n^m \approx u(\tau_m, x_n)$.

Using standard centered finite differences in space and a one-step implicit scheme in time for the local operator, we approximate

$$\frac{\partial u}{\partial \tau}(n\Delta x, (m+1)\Delta\tau) \approx \frac{v_n^{m+1} - v_n^m}{\Delta\tau}, \tag{3.4.4}$$

$$\frac{\partial u}{\partial x}(n\Delta x, (m+1)\Delta\tau) \approx \frac{v_{n+1}^{m+1} - v_{n-1}^{m+1}}{2\Delta x}, \tag{3.4.5}$$

$$\frac{\partial^2 u}{\partial x^2}(n\Delta x, (m+1)\Delta\tau) \approx \frac{v_{n+1}^{m+1} - 2v_n^{m+1} + v_{n-1}^{m+1}}{(\Delta x)^2}. \tag{3.4.6}$$

Iterative solution procedure

We write the PIDE in the form

$$u_\tau = \mathcal{L}_{\text{loc}} u + \mathcal{J} u,$$

with

$$\mathcal{L}_{\text{loc}} u = \left(r - \frac{\sigma^2}{2} - \lambda\kappa\right) u_x + \frac{\sigma^2}{2} u_{xx} - (r+\lambda)u, \qquad \mathcal{J} u = \int_{-B_1}^{B_2} u(\tau, x+y)\, \nu(dy).$$

We treat $\mathcal{L}_{\text{loc}}$ implicitly in τ and $\mathcal{J}$ explicitly. Using the discretizations (3.4.4)–(3.4.6), the scheme at grid point (x_n, τ_{m+1}) reads

$$\frac{v_n^{m+1} - v_n^m}{\Delta\tau} = \mathcal{L}_{\text{loc}} v_n^{m+1} + \lambda \int_{-B_1}^{B_2} v^m(x_n + y)\, \tilde{\nu}(dy), \tag{3.4.7}$$

where $\tilde{\nu}$ denotes the discretized jump measure and we have absorbed the reaction term $-(r+\lambda)v$ into $\mathcal{L}_{\text{loc}}$. In the implementation, we define

$$m := \lambda\kappa = \lambda\left(\mathbb{E}[e^Y] - 1\right) = \lambda\left(e^{\mu_J + \frac{1}{2}\sigma_J^2} - 1\right)$$

and write the drift as $r - m - \dfrac{1}{2}\sigma^2$, which matches the Python code.

The discrete local operator is

$$\mathcal{L}_{\text{loc}} v_n^{m+1} = \left(r - m - \frac{\sigma^2}{2}\right) \frac{v_{n+1}^{m+1} - v_{n-1}^{m+1}}{2\Delta x} + \frac{\sigma^2}{2} \frac{v_{n+1}^{m+1} - 2v_n^{m+1} + v_{n-1}^{m+1}}{(\Delta x)^2} - (r+\lambda) v_n^{m+1}.$$

Rearranging (3.4.7), we obtain the linear system

$$\left(I - \Delta\tau(\mathcal{L}_{\text{loc}})\right) \mathbf{v}^{m+1} = \mathbf{v}^m + \Delta\tau\, \boldsymbol{J}\mathbf{v}^m - \mathbf{o}^m, \tag{3.4.8}$$

where $\mathbf{v}^m$ collects interior values v_n^m, $\boldsymbol{J}$ is the discrete jump-convolution operator, and $\mathbf{o}^m$ collects boundary contributions. In the code, this system is implemented as

$$D\mathbf{v}^{m+1} = \mathbf{v}^m + \Delta\tau\, \mathbf{j}^m - \mathbf{o}^m,$$

with D tridiagonal and $\mathbf{j}^m$ the FFT-based jump term.

The tridiagonal matrix D has entries

$$\ell = \frac{\Delta\tau}{2}\left(\frac{r - m - \frac{\sigma^2}{2}}{\Delta x} - \frac{\sigma^2}{(\Delta x)^2}\right),$$
$$d = 1 + \Delta\tau\left(\frac{\sigma^2}{(\Delta x)^2} + r + \lambda\right),$$
$$u = -\frac{\Delta\tau}{2}\left(\frac{r - m - \frac{\sigma^2}{2}}{\Delta x} + \frac{\sigma^2}{(\Delta x)^2}\right),$$

The jump weights $\boldsymbol{\nu} \in \mathbb{R}^{2\texttt{extraP}+1}$ are obtained by differencing the Gaussian CDF on the extended grid. For log-normal jumps with $Y \sim \mathcal{N}(\mu_J, \sigma_J^2)$, which corresponds to

$$\nu_k = \lambda\left[\Phi\left(\frac{(k+0.5)\Delta x - \mu_J}{\sigma_J}\right) - \Phi\left(\frac{(k-0.5)\Delta x - \mu_J}{\sigma_J}\right)\right],$$

where Φ is the standard normal CDF and k indexes the discrete jump offsets.

Solution algorithm. The numerical algorithm is as follows:

1. Construct the extended log-price grid

$$x \in [x_{\min} - \texttt{extraP}\,\Delta x, x_{\max} + \texttt{extraP}\,\Delta x]$$

   with $\texttt{extraP} = e_p$.

2. Impose the payoff at $\tau = 0$ on the entire extended grid:

$$V(x,0) = \begin{cases} (e^x - K)_+ & \text{for a call,} \\ (K - e^x)_+ & \text{for a put.} \end{cases}$$

3. Impose Dirichlet asymptotics on the ghost layers for all τ:
   - Call: $V(\tau, x) \equiv 0$ for $x \le A_1 - B_1$, and $V(\tau, x) \approx e^x - Ke^{-r\tau}$ for $x \ge A_2 + B_2$.
   - Put: $V(\tau, x) \approx Ke^{-r\tau}$ for $x \le A_1 - B_1$, and $V(\tau, x) \equiv 0$ for $x \ge A_2 + B_2$.
4. Precompute and LU-factorize the tridiagonal matrix D once.
5. For each time step $m = 0, \ldots, M-2$:
   (a) Compute boundary offsets at the current time level m:

$$o_1^m = \ell\, V(x_{\text{left ghost}}, \tau_m), \quad o_{N-1}^m = u\, V(x_{\text{right ghost}}, \tau_m).$$

   In code:

```
offset[0]   = a * V[extraP, i]
offset[-1]  = c * V[-1 - extraP, i]
```

   (b) Evaluate the explicit jump term via FFT convolution:

```
V_jump = V[extraP+1:-extraP-1, i]
         + dt * signal.fftconvolve(V[:, i], nu, mode="valid")
```

which corresponds to

$$\mathbf{j}^m \approx \boldsymbol{J}\mathbf{v}^m.$$

(c) Form the right-hand side:

$$\mathbf{rhs}^m = \mathbf{v}^m + \Delta\tau\,\mathbf{j}^m - \mathbf{o}^m.$$

(d) Solve for the interior values at the next time level:

$$D\,\mathbf{v}^{m+1} = \mathbf{rhs}^m.$$

6. At $\tau = T$ (last column), interpolate in x to obtain $C(0, S_0)$ at $x = \log S_0$.

This IMEX scheme combines an implicit Euler step for the local diffusion-advection-reaction operator with an explicit FFT-based convolution for the jump part, and matches exactly the implementation given in the Python code.

3.5 American option : Early exercise rule as a projection on the European solver

In Section 2.5 we derived the linear complementarity problem (LCP) for the value of an American option,

$$\left(\frac{\partial V}{\partial t} + \mathcal{L}V\right)(V - \phi) = 0, \qquad \frac{\partial V}{\partial t} + \mathcal{L}V \geq 0, \qquad V - \phi \geq 0,$$

with at least one of the two inequalities holding with equality at every point in time and space.[27, 20] After spatial discretisation and time reversal $\tau = T - t$, this variational inequality becomes a semi-discrete partial differential complementarity problem (PDCP) of the form

$$U'(\tau) \;\geq\; A\,U(\tau) + g, \qquad U(\tau) \;\geq\; \Phi, \qquad (U(\tau) - \Phi)^\top (U'(\tau) - A\,U(\tau) - g) = 0,$$

where $U(\tau)$ collects the option values on the spatial grid, A is the discretisation matrix of $\mathcal{L}$ and Φ the discretised payoff ϕ.[27, 1]

A backward Euler or θ–scheme in time applied to the *European* problem yields at each time step a linear system

$$B\,U^n = c^{n-1},$$

with $B = I - \theta\Delta\tau A$ and c^{n-1} depending on U^{n-1} and the boundary data.[27, 20] Interpreted in the American setting, $U^n_{\text{cont}} := B^{-1}c^{n-1}$ is the discrete *continuation value*, i.e. the value obtained if early exercise were not allowed at time level τ_n. The LCP then reads, in fully discrete form,

$$B\,U^n \;\geq\; c^{n-1}, \qquad U^n \;\geq\; \Phi, \qquad (B\,U^n - c^{n-1})_i (U^n_i - \Phi_i) = 0 \quad \forall i,$$

which is a standard linear complementarity problem.[26, 27]

A large class of efficient methods for American options observes that, under standard monotonicity assumptions on the finite–difference/ADI discretisation (essentially that $-A$ is an M–matrix), the discrete complementarity conditions can be enforced by a simple *projection* at each time step:[26, 27, 20]

$$U^n_i \;=\; \max(U^n_{\text{cont},i},\, \Phi_i) \quad \text{for every grid node } i.$$

This is the idea behind projected SOR, Brennan–Schwartz type algorithms, and the Ikonen–Toivanen (IT) splitting and ADI–IT schemes that are now standard for American options.[26, 27, 20]

Application to the different models. *Model 1 – Black–Scholes (1D).* The implicit finite–difference scheme of Section 3.1 already computes v^{m+1} from v^m by solving a tridiagonal linear system for the European option.[1] For an American put, we reuse *exactly* the same time–stepping and, at each time level and interior node,

$$v^m_n \;\leftarrow\; \max(v^m_{\text{cont},n},\, \phi(S_n)).$$

For a non–dividend-paying stock, American and European calls have the same value, so this projection does not change the solution for calls, as explained in Section 2.5.[1]

Model 2 – Heston (2D, ADI). The Heston American put satisfies a 2D PDCP with the same differential operator as in the European case.[27, 20] Haentjens and in 't Hout adapt

ADI schemes by first performing an ADI time step for the semidiscrete PDE to obtain a provisional continuation value $\bar{U}_{ij}^{n+1}$ and then enforcing the early–exercise constraint through an IT–type fractional step:[20]

$$U_{ij}^{n+1} \leftarrow \max(\bar{U}_{ij}^{n+1},\, \phi(S_i)) \quad \text{for all } (S_i, \nu_j) \text{ on the grid.}$$

In our notation this is again the rule $U^{n+1} = \max(U_{\text{cont}}^{n+1}, \Phi)$, now applied pointwise on the 2D non–uniform Cartesian grid.

Model 3 – Multi–asset GBM (2D). The multi–asset European PDE in Section 2.3 leads, after spatial discretisation, to a high–dimensional linear ODE system $U' = AU + g$. American payoffs under this dynamics are governed by the same operator together with the obstacle $U \geq \Phi$, hence the same LCP structure as in the 1D case.[1] Componentwise splitting methods and ADI–IT schemes in the literature on multi–asset American options proceed exactly as above: one European time step (e.g. ADI) to compute U_{cont}^{n+1}, followed by the pointwise projection $U_{ij}^{n+1} = \max(U_{\text{cont},ij}^{n+1}, \Phi_{ij})$ at each (S_i^1, S_j^2).[2]

Model 4 – Merton Jump–Diffusion PIDE. For Merton jump–diffusion, the European problem is a PIDE that we already solve via operator splitting into a diffusion part and a (FFT–based) integral part.[2, 1] The American analogue is a partial integro–differential complementarity problem (PIDCP) with the same generator and the constraint $V \geq \phi$. Recent work by Boen and in 't Hout shows that operator–splitting schemes (IMEX and ADI) can be extended to such PIDCPs by combining one full PIDE time step (diffusion + jumps) with an IT–type projection step:[2]

$$u_k^{n+1} \leftarrow \max(u_{\text{cont},k}^{n+1},\, \phi(S_k)) \quad \text{for each spatial node } k.$$

Thus, also in the jump–diffusion setting, we can reuse the European operator–splitting solver and impose early exercise by a pointwise projection onto the payoff.

Summary. For all models considered in this report where the European price is obtained by solving a linear PDE/PIDE with Markov state variables (S), (S, ν) or (S^1, S^2), the American problem is obtained by the same spatial and temporal discretisation, interpreted as a continuation operator, followed by the projection

$$V^n \leftarrow \max(V_{\text{cont}}^n,\, \phi)$$

at each backward time level and grid node. Under standard monotonicity conditions on the discretisation, this projection step is equivalent to enforcing the discrete LCP and is the basis of modern splitting methods such as IT–splitting and ADI–IT for American options.[26, 27, 20, 2]

4 Validation methods

In this section, we will give the fundations of some other pricing techniques that will be used to validate the price given by methods used to solve P(I)DE's described previously. This section goal is to give a first introduction to these pricing techniques. Reader will find in section 6 how they are applied in practice for specific situations.

4.1 Analytical solutions - Vanilla options

We will first focus on standard European calls and puts options for different models (Black-Scholes, Heston and Merton Jump Diffusion). We will then give results for American style put options. In this section, we present some results that allow to obtain (semi-)analytical solutions to compute the price of european call options. Obviously, the call-put parity allows to deduce put price, no matter the model we are working with.

$$\text{Call-Put parity :} \qquad C_t + Ke^{-r\tau} = P_t + S_t \tag{4.1.1}$$

Black-Scholes analytical solution

According to [22], the well known solution for a European call in Black-Scholes framework[5] can be written as :

$$C(S,t) = e^{-q\tau} S_t \Phi(d_1) - e^{-r\tau} K \Phi(d_2) \tag{4.1.2}$$

Where :

- $d_1 = \frac{\log\left(\frac{S_t}{K}\right) + (r - q + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}$
- $d_2 = d_1 - \sigma\sqrt{\tau}$

Heston "semi-analytical" solution

Heston models offers what one can call "semi-analytical" solutions for european-style options where the underlying asset is assumed to follow Heston risk-neutral dynamics as described in section 2.2. Here, we present and combine the work done in [39] and [47] that allows to obtain semi-analytical solutions for a european call.

Result :

A call with following data :

- Initial underlying asset value : S_0
- Strike : K
- Initial spot variance : ν_0
- Time to maturity : τ

Have a fair value equals to :

$$C(S_0, K, \nu_0, \tau) = \frac{S_0 - Ke^{-r\tau}}{2} + \frac{1}{\pi}\int_0^\infty \mathcal{R}\left[e^{r\tau}\frac{\varphi(\phi - i)}{i\phi K^{i\phi}} - K\frac{\varphi(\phi)}{i\phi K^{i\phi}} d\phi\right] \tag{4.1.3}$$

where :

- $\varphi(X_0, K, \nu_0, \tau|\phi) = \exp(r\phi i\tau) S^{i\phi} \left[\frac{1 - ge^{d\tau}}{1-g}\right]^{-2a/\xi^2}$
  $\exp\left(\frac{a\tau}{\xi^2}(b^2 - \rho\xi\phi i + d) + \frac{\nu_0}{\xi^2}(b^2 - \rho\xi\phi i + d)\left[\frac{1-e^{d\tau}}{1-ge^{d\tau}}\right]\right)$
- $d = \sqrt{(\rho\xi\phi i - b)^2 + \xi^2(\phi i + \phi^2)}$

[5] This means the underlying asset follows a GBM as described in section 2.1

- $g = \frac{b-\rho\xi\phi i+d}{b-\rho i\phi i-d}$
- $a = \kappa\theta$
- $b = \kappa + \lambda$

Eventhough this might look like a complicated solution, it can easily implemented in any programming language and computed as easily. An implementation in python can be found in [47]
Note : This result is called "semi-analytical" because the integral (ie, Fourier transform) has no closed-form and needs to be computed numerically. Even with this numerical requirement, this elegant solution allows time-efficient computations compared to more complex numerical solutions such as Monte-Carlo simulations and many others.

Merton Jump-Diffusion (MJD) analytical solution

In the Merton (1976) jump-diffusion model, the (ex-dividend) stock price under the risk–neutral measure satisfies

$$\frac{\mathrm{d}S_t}{S_{t^-}} = (r - q - \lambda\kappa)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t + (J-1)\,\mathrm{d}N_t, \tag{4.1.4}$$

where

- r is the constant risk–free rate and q the continuous dividend yield,
- W_t is a standard Brownian motion,
- N_t is a Poisson process with intensity $\lambda > 0$,
- J is the (multiplicative) jump size, independent of W_t and N_t,
- the compensated drift term $\kappa := \mathbb{E}[J-1]$ ensures risk–neutrality.

Merton assumes that the jump sizes are lognormally distributed,

$$\ln J \sim \mathcal{N}(\mu_J, \delta^2), \qquad \Rightarrow \qquad \kappa = \mathbb{E}[J-1] = \exp\left(\mu_J + \frac{1}{2}\delta^2\right) - 1. \tag{4.1.5}$$

Let $\tau = T - t$ denote time to maturity and consider a European call with strike K and maturity T. Conditioning on the number of jumps on $[t, T]$, the terminal log–price is (conditionally) Gaussian. Hence the call price can be expressed as an (absolutely convergent) infinite sum of Black–Scholes–type terms [37]:

$$C^{\mathrm{MJD}}(S_t, K, \tau) = \sum_{n=0}^{\infty} \frac{e^{-\lambda\tau}(\lambda\tau)^n}{n!}\, C^{(n)}(S_t, K, \tau), \tag{4.1.6}$$

where $C^{(n)}$ is the conditional call price given that exactly n jumps occur before maturity.

Conditional on $N_T - N_t = n$, the terminal log–price is normal with

$$m_n = \ln S_t + \left(r - q - \lambda\kappa - \frac{1}{2}\sigma^2\right)\tau + n\,\mu_J, \qquad v_n = \sigma^2\tau + n\,\delta^2. \tag{4.1.7}$$

Standard lognormal calculus then yields

$$C^{(n)}(S_t, K, \tau) = e^{-r\tau}\left[\exp(m_n + \frac{1}{2}v_n)\,\Phi(d_1^{(n)}) - K\,\Phi(d_2^{(n)})\right], \tag{4.1.8}$$

with

$$d_1^{(n)} = \frac{m_n - \ln K + v_n}{\sqrt{v_n}}, \qquad d_2^{(n)} = d_1^{(n)} - \sqrt{v_n}. \tag{4.1.9}$$

Equations (4.1.6)–(4.1.8) provide an exact pricing formula for European calls under the MJD model. In practice, the infinite sum is truncated at some $n_{\max}$ such that the Poisson tail probability $\sum_{n>n_{\max}} e^{-\lambda\tau}(\lambda\tau)^n/n!$ is numerically negligible; typically $n_{\max}$ between 20 and 50 suffices for realistic parameter values [50, 43].

By put–call parity, the corresponding European put price satisfies

$$P^{\mathrm{MJD}}(S_t, K, \tau) = C^{\mathrm{MJD}}(S_t, K, \tau) - S_t e^{-q\tau} + K e^{-r\tau}. \tag{4.1.10}$$

Multi-asset options in the multi-dimensional Black–Scholes model

We consider d risky assets $S^{(1)}, \dots, S^{(d)}$ following correlated geometric Brownian motions under the risk–neutral measure, with constant volatilities σ_i, constant dividend yields q_i and instantaneous correlation matrix ρ_{ij}. In this setting, several important multi–asset payoffs admit closed–form or near closed–form pricing formulas.

Exchange options (Margrabe formula) An exchange option with maturity T and payoff

$$\Pi_T = (S_T^{(1)} - S_T^{(2)})^+ \tag{4.1.11}$$

gives the right to exchange one unit of asset 2 for one unit of asset 1 at time T. Margrabe [34] derived a closed–form expression, which is essentially a Black–Scholes formula with an effective volatility that incorporates correlation:

$$C^{\mathrm{ex}}(S_t^{(1)}, S_t^{(2)}, \tau) = S_t^{(1)} e^{-q_1\tau}\Phi(d_1) - S_t^{(2)} e^{-q_2\tau}\Phi(d_2), \tag{4.1.12}$$

where

$$\sigma_{12} = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}, \tag{4.1.13}$$

and

$$d_1 = \frac{\ln\left(\dfrac{S_t^{(1)}}{S_t^{(2)}}\right) + (q_2 - q_1 + \dfrac{1}{2}\sigma_{12}^2)\tau}{\sigma_{12}\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma_{12}\sqrt{\tau}. \tag{4.1.14}$$

This formula is a very convenient benchmark for validating two–asset PDE solvers with correlated underlyings.

Spread options (Kirk approximation) A (two–asset) spread call option has payoff

$$\Pi_T = (S_T^{(1)} - S_T^{(2)} - K)^+, \tag{4.1.15}$$

with constant strike $K \geq 0$. A widely used closed–form approximation, due to Kirk [30], expresses the spread option price in a Black–Scholes–like form. Define the forward prices

$$F_1 = S_t^{(1)} e^{(r-q_1)\tau}, \qquad F_2 = S_t^{(2)} e^{(r-q_2)\tau}, \tag{4.1.16}$$

the effective second forward

$$\tilde{F}_2 = F_2 + K e^{r\tau}, \tag{4.1.17}$$

and the weight

$$b = \frac{F_2}{\tilde{F}_2}. \tag{4.1.18}$$

Kirk's approximation for the spread call price is then

$$C^{\mathrm{spr}}(S_t^{(1)}, S_t^{(2)}, K, \tau) \approx e^{-r\tau}\left[F_1\Phi(\hat{d}_1) - \tilde{F}_2\Phi(\hat{d}_2)\right], \tag{4.1.19}$$

where the effective volatility is

$$\sigma_K = \sqrt{\sigma_1^2 - 2b\,\rho_{12}\,\sigma_1\sigma_2 + b^2\sigma_2^2}, \tag{4.1.20}$$

and

$$\hat{d}_1 = \frac{\ln\left(\frac{F_1}{\tilde{F}_2}\right) + \frac{1}{2}\sigma_K^2\tau}{\sigma_K\sqrt{\tau}}, \qquad \hat{d}_2 = \hat{d}_1 - \sigma_K\sqrt{\tau}. \tag{4.1.21}$$

Despite being an approximation, Kirk's formula is known to be highly accurate for many parameter regimes and is particularly useful for checking multi–asset PDE implementations [8, 21].

Basket options (lognormal approximation) A basket call option on d assets with weights w_i has payoff

$$\Pi_T = \left(\sum_{i=1}^{d} w_i S_T^{(i)} - K\right)^+. \tag{4.1.22}$$

In the general multi–asset Black–Scholes setting, no closed–form solution is available in terms of elementary functions. However, a classical and very effective approach is to approximate the basket $\sum_i w_i S_T^{(i)}$ by a single lognormal variable via moment–matching [46, 6]. Let

$$m_B = \mathbb{E}\left[\sum_{i=1}^{d} w_i S_T^{(i)}\right], \qquad v_B = \mathrm{Var}\left[\sum_{i=1}^{d} w_i S_T^{(i)}\right]. \tag{4.1.23}$$

Assuming $\sum_i w_i S_T^{(i)} \approx \tilde{S}_T$ with $\ln \tilde{S}_T \sim \mathcal{N}(\mu_B, \sigma_B^2)$ such that

$$m_B = \exp(\mu_B + \frac{1}{2}\sigma_B^2), \qquad v_B = (\exp(\sigma_B^2) - 1)\exp(2\mu_B + \sigma_B^2), \tag{4.1.24}$$

one solves for μ_B and σ_B^2 and then applies the Black–Scholes formula with underlying $\tilde{S}_t$ and volatility $\sigma_B/\sqrt{\tau}$. This yields a simple analytical approximation which is often sufficiently accurate for benchmarking purposes [14].

4.2 Monte-Carlo simulations and its variants

4.2.1 Monte-Carlo

Context

Monte-Carlo simulation is a powerful technique for option pricing that complements both PDE-based methods and analytical solutions. Unlike numerical PDE solvers which discretize the underlying space, Monte-Carlo methods work in the probability space by simulating many sample paths of the underlying asset and averaging the resulting option payoffs. This approach is particularly valuable for:

- **Validation**: Providing independent price estimates to verify outputs from PDE solvers and ensure numerical stability.
- **Flexibility**: Easily accommodating complex models (Heston, Merton Jump-Diffusion), path-dependent payoffs (Asian, Barrier), and multiple underlying assets with arbitrary correlations.
- **Robustness**: Functioning without the stability constraints that plague finite-difference schemes.

Despite these advantages, Monte-Carlo methods suffer from inherent computational cost and slow convergence relative to analytical solutions, making them a complementary tool rather than a primary pricing engine for European vanilla options.

Fundamental principle: Law of large numbers

How Monte-Carlo works

The pricing of a European option using Monte-Carlo is grounded in the martingale representation of option value under the risk-neutral measure Q. For a European call option with payoff $H(S_T)$ at maturity T, the no-arbitrage fair value at time $t = 0$ is:

$$V_0 = e^{-rT}\mathbb{E}^Q\left[H(S_T)\right],$$

where $\mathbb{E}^Q$ denotes expectation under the risk-neutral measure.

Since we cannot compute this expectation analytically for complex models, Monte-Carlo approximates it as an empirical mean over many simulated paths. Given M independent simulations of the underlying price trajectory, each generating a terminal value $S_T^{(m)}$, the Monte-Carlo estimator is:

$$\hat{V}_0^{\mathrm{MC}} = e^{-rT}\frac{1}{M}\sum_{m=1}^{M} H\left(S_T^{(m)}\right).$$

By the Law of Large Numbers, as $M \to \infty$, $\hat{V}_0^{\mathrm{MC}} \to V_0$ almost surely. In practice, M is finite (typically 10^4 to 10^6), so the estimator carries estimation error.

Why it works: The law of large numbers and martingale framework

The validity of Monte-Carlo rests on two pillars:

1. **Risk-Neutral Pricing**: All assets discounted by the risk-free account form martingales under Q. Thus the discounted payoff $e^{-rT}H(S_T)$ is a martingale, and by the optional stopping theorem, its value today is the expectation of this quantity.

2. **Law of Large Numbers**: If $X_1, X_2, \ldots, X_M$ are i.i.d. random variables with finite expectation μ, then
$$\frac{1}{M}\sum_{m=1}^{M} X_m \xrightarrow{a.s.} \mu \quad \text{as } M \to \infty.$$
Each path's discounted payoff $e^{-rT}H(S_T^{(m)})$ is an independent realization from the distribution of the discounted payoff, so averaging them converges to the true expectation.

This separation of the probability space and state space (stock price) allows Monte-Carlo to sidestep the mesh generation and boundary condition complexity inherent to PDE solvers.

What Monte-Carlo requires

Risk-neutral dynamics and numerical integration

To implement Monte-Carlo, we must specify the risk-neutral dynamics of the underlying. For a simple GBM[6]:

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t^Q,$$

which in discrete time (Euler scheme) becomes:

$$S_{t+\Delta t} = S_t \exp\left(r\Delta t + \sigma\sqrt{\Delta t}\,Z\right), \tag{4.2.1}$$

where $Z \sim \mathcal{N}(0,1)$ is a standard normal random variable.

For M paths and N time steps, one generates $M \times N$ independent standard normal draws, constructs each path from $t = 0$ to $t = T$, evaluates the payoff at maturity, discounts, and averages.

Input data and assumptions

Monte-Carlo requires:

- **Model parameters**: risk-free rate r, volatility σ, and (for complex models) stochastic volatility, jump intensity, correlation structure.
- **Time discretization**: number of steps N and step size $\Delta t = T/N$.
- **Path count**: number of simulations M.
- **Random number generation**: a pseudo-random or quasi-random number generator (Sobol, Halton for variance reduction).
- **Payoff function**: arbitrarily complex, including path-dependence (lookback, Asian, barrier).

Critically, unlike PDE methods, there is no domain truncation or boundary condition specification—the natural boundary of the probability space handles these implicitly.

[6]Note : Monte-Carlo algorithm has no restriction on the assumed underlying asset dynamics apart from being discrete.

Strengths of Monte-Carlo

1. **Model flexibility**: Easily handles Heston dynamics (stochastic volatility), Merton Jump-Diffusion (PIDE jumps), multi-asset options with arbitrary correlations, and path-dependent payoffs.
2. **Stability**: No CFL (Courant–Friedrichs–Lewy) stability constraints; choice of Δt and M is governed by accuracy, not stability.
3. **Parallelization**: Each of the M paths is independent, so Monte-Carlo parallelizes trivially across CPUs/GPUs (not presented in this report).
4. **Dimension scaling**: For European options (not early exercise), convergence is independent of the number of state variables, making it suitable for high-dimensional baskets and multi-asset derivatives.
5. **Validation and benchmarking**: Provides an independent price estimate to cross-check PDE solvers, analytical solutions, and calibration procedures.

Weaknesses and limitations

1. **Convergence rate**: Standard Monte-Carlo converges at rate $\mathcal{O}(1/\sqrt{M})$. To halve the error, one must quadruple the number of paths, making high-precision pricing computationally expensive.
2. **Variance**: The estimator $\hat{V}_0^{\mathrm{MC}}$ is noisy. Its standard error is

$$\mathrm{SE} = \frac{\sigma_{\mathrm{payoff}}}{\sqrt{M}},$$

   where σ_{payoff} is the standard deviation of the discounted payoff across paths. For short-dated or deep OTM options, this variance can be large.
3. **Bias from discretization**: Using finite time steps (Euler) introduces a discretization bias; one must refine Δt to control this (Milstein or higher-order schemes improve accuracy but add complexity).
4. **Slow for vanilla European options**: For European vanilla options, analytical solutions (Black-Scholes, Heston semi-analytical) are orders of magnitude faster.
5. **American options require special algorithms**: Early-exercise features demand backward induction (Longstaff-Schwartz algorithm), which adds computational layers.
6. **Greeks computation**: Computing sensitivities (Delta, Gamma, Vega) via Monte-Carlo requires either finite differences (expensive) or pathwise differentiation (requires smooth payoffs); it is less elegant than PDE-based Greeks.

Variance reduction techniques

To mitigate the slow $\mathcal{O}(1/\sqrt{M})$ convergence, practitioners use variance reduction techniques:

Antithetic variates

For each standard normal draw Z, also compute a path with $-Z$. The two payoffs are negatively correlated, reducing the overall variance. With M pairs instead of M singles, variance typically falls by 30–50%, effectively increasing sample efficiency.

Control variates

If a related instrument has a known price (e.g., the analytical Black-Scholes price of a nearby vanilla option), one can use it to reduce the variance of the target option. The estimator becomes:

$$\hat{V}^{\mathrm{CV}} = \hat{V}^{\mathrm{MC}} + \beta \left(V^{\mathrm{analytical}} - \hat{V}^{\mathrm{MC}}_{\mathrm{control}} \right),$$

where β is optimized to minimize the variance of the correction term. This can reduce variance by 50–90% for correlated instruments.

Importance sampling

Shift the distribution from which random draws are sampled to concentrate probability mass on "important" regions (e.g., near-the-money strikes). Payoffs from these regions are re-weighted to maintain unbiasedness. This is particularly effective for OTM options.

Quasi-random numbers (low-discrepancy sequences)

Replace pseudo-random numbers with Sobol or Halton sequences, which fill the unit hypercube more uniformly. This often improves convergence to $\mathcal{O}(\log(M)^d/M)$, where d is the dimension—a significant practical gain for moderate d.

Practical implementation notes

Pseudo-code for vanilla European call

Monte-Carlos : European call pricing with GBM

We consider Eq.4.2.1 with step of time $\Delta_t = \frac{T}{N}$.

```
for m = 1 to M do
    for j = 1 to N do
        Draw ϵ from N(0,1)
        Calculate the asset log-return:
```

$$\Delta X_j = r\Delta_t + \sigma\sqrt{\Delta_t}\epsilon$$

```
        Update the stock price:
```

$$S_j = S_{j-1}e^{\Delta X_j}$$

```
    end for
    payoff = exp(-r*T) * max(S - K, 0)
    payoffs.append(payoff)
end for
call˙price = mean(payoffs)
std˙error = std(payoffs) / √M
return call˙price, std˙error
```

Note the use of the log-normal form $S_{t+dt} = S_t \exp(\ldots)$ rather than additive differences; this ensures $S > 0$ always.

Data requirements for calibration

To price using Monte-Carlo in practice, one must:

- Estimate or calibrate r from the yield curve (Nelson-Siegel-Svensson or bootstrapping).
- Estimate or calibrate σ from implied volatility surface.
- For Heston/MJD: calibrate model parameters (mean-reversion rate, volatility of volatility, jump intensity) via MSE minimization on observed option prices.

These calibration steps are identical to those used in PDE approaches; Monte-Carlo does not change the input parameterization.
Hereafter, we give discretization schemes to simulate paths for *"Geometric Brownian schemes"*,*"Heston model"*, *"Merton-Jump-Diffusion"* and for two correlated underlyings according to the risk-neutral dynamics we defined previously.

Monte-Carlos : Brownian motion with drift

We consider Eq.2.1.1 with step of time $\Delta_t = \frac{T}{m}$. If $S_j := S_{j\Delta_t}$, we simulate a sample path $(S_j)_{j=0,\dots,m}$ with following loop :

1: **for** $j = 1$ to m **do**
2: Draw ϵ from $\mathcal{N}(0,1)$
3: Calculate the asset log-return:

$$\Delta X_j = r\Delta_t + \sigma\sqrt{\Delta_t}\epsilon$$

4: Update the stock price:

$$S_j = S_{j-1}e^{\Delta X_j}$$

5: **end for**

Monte-Carlos : Heston stochastic volatility

We consider Eq. 2.2.1 with step of time $\Delta_t = \frac{T}{m}$. If $S_j := S_{j\Delta_t}$ and $\nu_j := \nu_{j\Delta_t}$, we simulate a sample path $(S_j, \nu_j)_{j=0,\dots,m}$ with following loop (full truncation Euler–Maruyama scheme):

1: Initialize $\nu_0 > 0$ (e.g. $\nu_0 = \theta$)
2: **for** $j = 1$ to m **do**
3: Draw $\epsilon^S \sim \mathcal{N}(0,1)$ and $\epsilon^\nu \sim \mathcal{N}(0,1)$ independent
4: Calculate the volatility increment:

$$\Delta\nu_j = \kappa(\theta - \nu_{j-1})\Delta_t + \xi\sqrt{\nu_{j-1}^+}\sqrt{\Delta_t}\epsilon^\nu$$

5: Update the variance (full truncation):

$$\nu_j = \max(\nu_{j-1} + \Delta\nu_j,\, 0)$$

6: Calculate the asset log-return:

$$\Delta X_j = r\Delta_t + \sqrt{\nu_{j-1}}\sqrt{\Delta_t}\epsilon^S$$

7: Update the stock price:

$$S_j = S_{j-1}e^{\Delta X_j}$$

8: **end for**

Here $\nu_{j-1}^+ := \max(\nu_{j-1}, 0)$ denotes the positive part.

Monte-Carlos : Two correlated underlyings

We consider Eq. 2.3.9 with step of time $\Delta_t = \frac{T}{m}$. If $S_j^k := S_{j\Delta_t}^k$ for $k = 1, 2$, we simulate a sample path $(S_j^1, S_j^2)_{j=0,\dots,m}$ with following loop:

1: **for** $j = 1$ to m **do**
2: Draw $\epsilon^1 \sim \mathcal{N}(0,1)$ and $\epsilon^2 \sim \mathcal{N}(0,1)$
3: Calculate the correlated shocks using the diffusion matrix $\Sigma = [\sigma_{kl}]$:

$$dW_j^1 = \sqrt{\Delta_t}\epsilon^1, \qquad dW_j^2 = \sqrt{\Delta_t}(\rho\epsilon^1 + \sqrt{1-\rho^2}\epsilon^2)$$

where $\rho = \sigma_{12}/\sigma_{11}$ is the correlation (or generate $\Sigma \begin{bmatrix} \epsilon^1 \\ \epsilon^2 \end{bmatrix}$ directly)

4: Calculate the asset log-returns:

$$\Delta X_j^1 = r_1\Delta_t + \sigma_{11}dW_j^1 + \sigma_{12}dW_j^2, \qquad \Delta X_j^2 = r_2\Delta_t + \sigma_{21}dW_j^1 + \sigma_{22}dW_j^2$$

5: Update the stock prices:

$$S_j^1 = S_{j-1}^1 e^{\Delta X_j^1}, \qquad S_j^2 = S_{j-1}^2 e^{\Delta X_j^2}$$

6: **end for**

Monte-Carlos : Merton Jump diffusion

We consider Eq.2.4.1 with step of time $\Delta_t = \frac{T}{m}$. If $S_j := S_{j\Delta_t}$, we simulate a sample path $(S_j)_{j=0,\dots,m}$ with following loop :

1: **for** $j = 1$ to m **do**
2: Draw ϵ from $\mathcal{N}(0,1)$
3: Draw n from $\mathrm{Po}(\lambda\Delta_t)$
4: Draw $y_{k=1,\dots,n}$ from the Y-distribution (here, $\mathcal{N}(\mu_y, \sigma_y)$)
5: Calculate the asset log-return:

$$\Delta X_j = r\Delta_t + \sigma\sqrt{\Delta_t}\epsilon + \sum\nolimits_{k=1}^{n} y_k$$

6: Update the stock price:

$$S_j = S_{j-1}e^{\Delta X_j}$$

7: **end for**

4.2.2 Longstaff-Schwartz Algorithm

When deriving the American LCP in section 2.5, we started our reasoning with the fact that the early-exercise rule incorporated new difficulties. We will do the same here to try to give an intuition about how this algorithm works.
We will also work for american option that can be exercised only at discrete time : $t_0, t_1, \dots, t_M = T$. As M grows and becomes sufficiently large, we will come closer and closer to an american option that can be exercised at any time.
We will rely on original paper [18] and work done in [45] to introduce and explain this elegant algorithm. Illustrations in this section are computed using code made by Dialid Santiago [44].
Let us first define a "matrix of stopping times". We will call this matrix M. This matrix

will have dimensions $(N \times M)$. Every time this matrix has an entry equal to one, this mean that the holder of the american option should exercise its right. Each row will represents one simulation path of a monte-carlo simulation. Each column represents a time-step in this simulation.

Example of stopping time matrix and intuition :
We have a stopping time matrix M as :

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

In this example, it is optimal for the holder to exercise its option on 3-rd time step of the 3-rd simulation path. It is also optimal to exercise its right at first time step of fourth and fifth simulations.

If we know what were the simulated values of the asset $X(t_i)$ and the payoff function $\phi(X)$, we can create a very similar matrix that we will call "cashflow matrix" M_c. This cashflow matrix does only take into account the value of the cashflow and not its discounted value.

Example of cashflow matrix :
We have a cashflow matrix M_c as :

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \phi(X_3) \\ \phi(X_1) & 0 & 0 \\ \phi(X_1) & 0 & 0 \end{bmatrix}$$

In this example, it is optimal for the holder to exercise its option at same times as previous example and we have the values of the cashflows these exercises generated.

If we can build this cashflow matrix, the only step needed to obtain the value of our american option is to discount every entries of M_c with the corresponding risk-free rate to the inception date t_0. So let us move to how we can build this matrix.
Let us imagine that we have already computed N paths of length M using Monte-Carlo simulations as described previously. We would end up with something like :

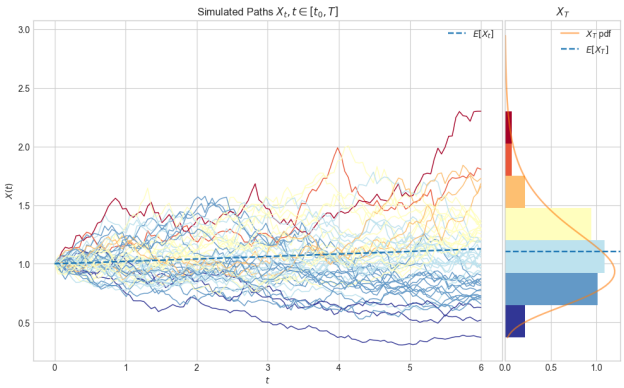


Figure 18: LSM - Initial paths

How can we determine a price for an american option based on these paths ? By using a backward and recursive algorithm that computes least-square regression at each time step. First thing to notice is that, at maturity, the option value is entirely determined by the payoff function[7]. We define $U(t_i, \omega_s)$ as the option value at time t_i for a scenario (single path) ω_s, $s = 1, \ldots, N$. At maturity, we thus have :

$$U(t_M, \omega_s) = \phi(t_M, \omega_s) = (K - X(t_M, \omega_s))_+ \qquad \text{(for a Put)}$$

At maturity, the option value $U(t_M)$ is the average across all scenarios :

$$U(t_M) = \frac{1}{N} \sum_{s=1}^{N} U(t_M, \omega_s)$$

At time t_{M-1}, let us see what choice the holder is facing. Either it exercise the option immediately, or it holds and option survives until t_M. So, the optimal choice can be resumed as either gaining the payoff at t_{M-1} or gaining the expectation of the option at t_M.

$$U(t_{M-1}) = \max\left(\phi(t_{M-1}), C(t_{M-1})\right)$$

Where $C(t_{M-1})$ is called the "continuation" and can be described as :

$$C(t_{M-1}) = e^{-r(t_M - t_{M-1})} \mathbb{E}^{\mathbb{Q}} \left[U(t_M) | \mathcal{F}_{t_{M-1}}\right] \tag{4.2.2}$$

We now need to estimate, via our set of paths, the second term :$\mathbb{E}^{\mathbb{Q}} \left[U(t_M) | \mathcal{F}_{t_{M-1}}\right]$. The idea of the Least-Square Monte-Carlo algorithm is to approximate this function $C(t_{M-1})$ with a least-square regression and to make this function a linear-combination of basis functions L (Hermite, Jacobi, ...). What is the set of points that we are trying to describe here ?
If we define the discounted cashflows as :

$$V(t_i, t_{i+1}) = e^{-r(t_{i+1} - t_i)} \phi(X(t_{i+1})) \tag{4.2.3}$$

Then, we have a set of points

$$\{X(t_i, \omega_s), V(t_i, t_{i+1}, \omega_s)\}_{s=1}^{N}$$

[7]Note : for what follows, original paper makes hypothesis about this payoff function. We will not present them here.

Using a Least-Square approach to find the coefficients α_{ij}, we describe our function at time t_i as

$$C : \mathbb{R} \mapsto \mathbb{R} : C(X_{t_i}) \approx \sum_{j=0}^{\infty} \alpha_{ij} L_j(X(t_i))$$

Note that the basis functions are the same for every simulation. Of course, we will restrict ourselve to an approximation made of only J basis functions :

$$C(t_{M-1}) \approx \sum_{j=0}^{J} \alpha_{ij} L_j(X(t_i))$$

The next figure illustrates the fit of the set of points at time t_{M-1} :

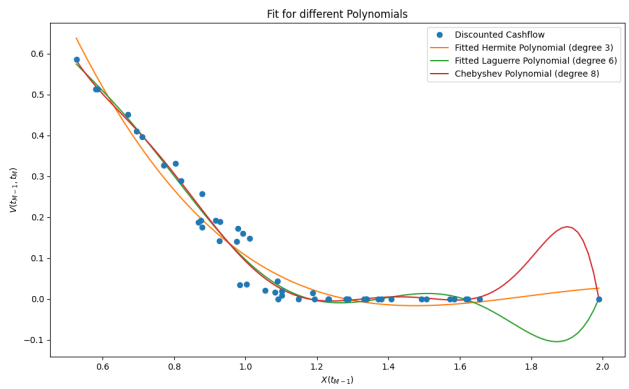


Figure 19: LSM - Fit for three different basis

After fitting the polynomial, one needs to approximate the points of the dataset by their projects on the fitted polynomial. The choice for the projection rule is not discussed here.
In reality, and according to the analysis done in [18], the parameters α_{ij} are estimated via a Least-Square regression of the set of points $\left(X(t_i, \omega), e^{-r(t_{i+1}-t_i)}\phi(t_{i+1}, \omega)\right)$ that are **in-the-money**.

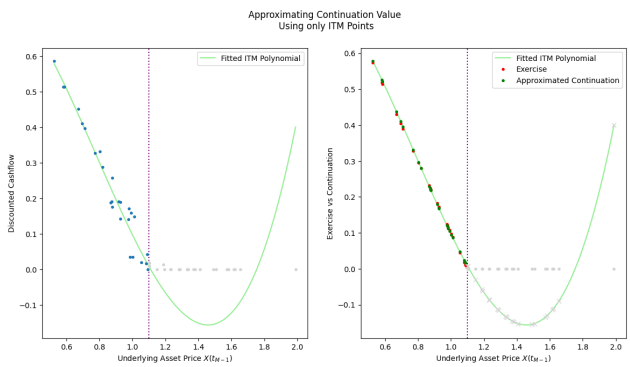


Figure 20: LSM - Fit for ITM points only

We can see on left sub-figure of figure 20 the fitted polynomial and their projections on the right sub-figure. One can also see that the fit only good for points ITM.
What do we need to do now that we have our fit ? Having our fit means we actually managed to estimate the "continuation" $C(t_i)$. So, we can update our optimal choice $U(t_i) =$

$\max(\phi(t_i), C(t_i))$. If the optimal choice is to exercise the option, we put a 1 in our cashflow matrix defined above.

Update in cashflow matrix example :
We are working with a cashflow matrix made for 5 simulated paths of length 10. Imagine that we have done three steps of our algorithm and found this cashflow matrix :

$$M_c^{3\,it} = \begin{bmatrix} / & / & / & / & / & / & / & 0 & 0 & 1 \\ / & / & / & / & / & / & / & 0 & 0 & 0 \\ / & / & / & / & / & / & / & 0 & 1 & 0 \\ / & / & / & / & / & / & / & 0 & 0 & 0 \\ / & / & / & / & / & / & / & 0 & 0 & 0 \end{bmatrix}$$

At the fourth iteration, we find that it is optimal to exercise our option on first and second path. The cashflow matrix must be updated as :

$$M_c^{4\,it} = \begin{bmatrix} / & / & / & / & / & / & 1 & 0 & 0 & 0 \\ / & / & / & / & / & / & 1 & 0 & 0 & 0 \\ / & / & / & / & / & / & 0 & 0 & 1 & 0 \\ / & / & / & / & / & / & 0 & 0 & 0 & 0 \\ / & / & / & / & / & / & 0 & 0 & 0 & 0 \end{bmatrix}$$

In first row, as the option can be exercised once and we found it was more optimal to exercise our option at time step 6 that time step 10, we replaced the entry $M_c(1, 10)$ by 0 and the entry $M_c(1, 6)$ by 1.

If we apply this algorithm on a larger set of paths of larger length, we can retrieve the optimal stopping time for our simulations. An illustration is given in next figure :

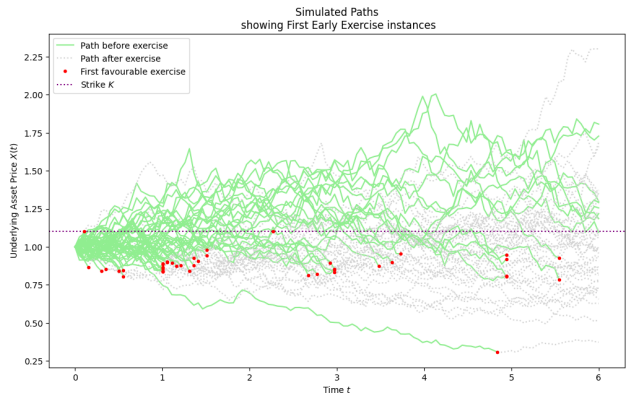


Figure 21: LSM - First Optimal Stopping Time on simulations

We found our stopping times (or optimal exercising times). If we denote the optimal stopping time for one scenario ω_s by τ_{i,ω_s}, the final value of our american option is :

$$Am(t_0) = \frac{1}{N} \sum_{s=1}^{N} e^{-r(t_i - t_0)} \phi(X_{\tau_{i,\omega_s}}, \omega_s) \tag{4.2.4}$$

Note Equation (4.2.2) holds for constant risk-free rate and can be generalised for stochastic risk-free rate and for any time $t_k, 0 < t_k < T$ as :

$$C(t_k) = \mathbb{E}^{\mathbb{Q}}\left[\sum_{j=k+1}^{M} \exp\left(-\int_{t_k}^{t_j} r(s)ds\right)\tilde{C}(t_j; t_k, T)|\mathcal{F}_{t_k}\right]$$

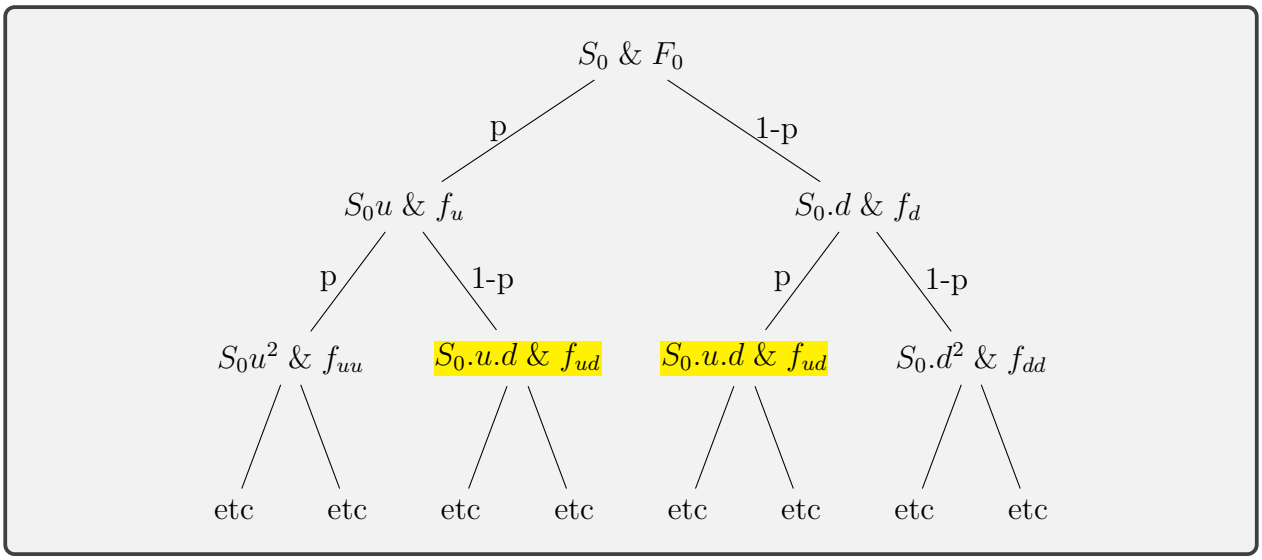
Where $\tilde{C}(t_j; t_k, T)$ is the path of cash-flows (ie, $= 0$ if not exercised, $= \phi(t_s)$ if option exercised at time t_s) for the option at time t_j, $t_k < t_j < t_M = T$. Same reasoning holds for the interest rates of equations (4.2.3) and (4.2.4)

4.3 Binomial and Trinomial trees

There exists a wide range of so-called binomial and trinomial trees methods. Here we will first present how these methods generally work, what is the idea behind the "trees". Then we will simply give the mathematical framework needed for implementation of a selection of these techniques. The models we will present and their mathematical framework are taken from the review made in [28] and a special attention for trinomial trees for american option will be presented according to the work done in [10].

Trees for option pricing

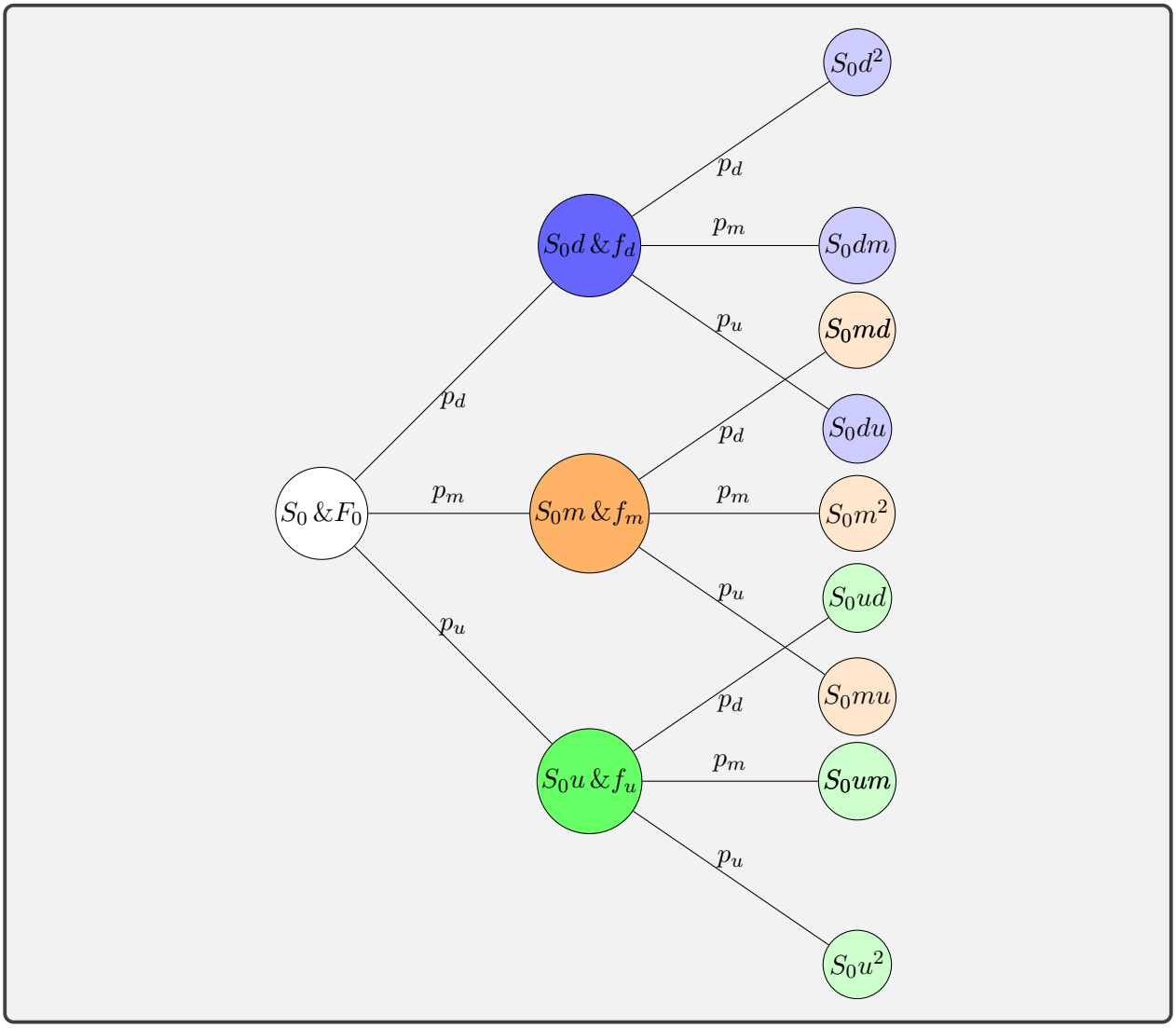
Binomial tree: at each step the price moves up by factor u with probability p, or down by factor d with probability $1-p$, where $0 < d < 1 < u$. F_0 represents the option value at inception that we are trying to estimate and f_u, f_d are the option value with corresponding underlying stock value. In order to determine the initial value of the option F_0, we use backward induction from the final nodes. The value of the option is known at each final node.



Trinomial tree: at each step the price moves up by factor u with probability p_u, has a medium movement with probability p_m, or moves down by factor d with probability p_d, where $p_u + p_m + p_d = 1$. Again, F_0 reoresents the option value at inception we are trying to estimate and f_u, f_m, f_d are the option value with corresponding underlying stock value.
A trinomial tree is defined by five parameters :

- p_u : probability of the stock moving up
- p_d : probability of the stock moving down
- d : multiplier of the stock when moving down
- u : multiplier of the stock when moving up
- m : multiplier of the stock for medium movement.

The sixth "parameter" p_m is deduced from the relationship $p_d + p_u + p_m = 1$. Intuitively, we can think that $m = 1$ means the stock stays at the same level. Even if this sounds like a reasonable assumption, we will see some models that do not make this assumption later on.



While this could look like a mistake, it is intended that at the second step of our trinomial tree, the paths do not end up on the same point. In fact, nothing guarantees that u and d will be chosen in a way that makes our tree a so-called "recombining tree". A tree is called recombining if $ud = m^2$. All the models presented in this report have this "recombining" property but the source code fournished with this report has been designed to manage non-recombining models.

Binomial models

Cox,Ross, Rubinstein (CRR) : This is probably the most known binomial model. It gives the following expressions :

$$\begin{cases} u = \exp\left(\sigma\sqrt{t}\right) \\ d = \exp\left(-\sigma\sqrt{t}\right) \\ 1 - p = \frac{1+r-d}{u-d} \end{cases}$$

Rendleman and Bartter : Based on a choice of making the up probability $p = \frac{1}{2}$, they state :

$$\begin{cases} u = \exp\left(\left(r - \frac{\sigma^2}{2}\right)\Delta + \sigma\sqrt{\Delta}\right) \\ d = \exp\left(\left(r - \frac{\sigma^2}{2}\right)\Delta - \sigma\sqrt{\Delta}\right) \\ p = 1/2 \end{cases}$$

Tian : The idea of the author was to match moments that binomial model produced with the moments in Black-Scholes framework. Reader will find the details in [4]. This results in following parameters expression :

$$\begin{cases} V = e^{\sigma^2 \Delta t} \\ p = \frac{e^{r\Delta t} - d}{u - d} \\ u = \frac{1}{2} e^{r\Delta t} V \left(V + 1 + \sqrt{V^2 + 2V - 3}\right) \\ d = \frac{1}{2} e^{r\Delta t} V \left(V + 1 - \sqrt{V^2 + 2V - 3}\right) \end{cases}$$

Trinomial models

Tian : In his work [4], Tian also proposed different version of the trinomial tree. One of these versions gives :

$$\begin{cases} M = \exp(r\Delta t) \\ V = \exp(\sigma^2 \Delta t) \\ K = \frac{M}{2}(V^4 + V^3) \\ p_u = \frac{md - M(m+d) + M^2 V}{(u-d)(u-m)} \\ p_m = \frac{M(u+d) - ud - M^2 V}{(u-m)(m-d)} \\ p_d = \frac{um - M(u+m) + M^2 V}{(u-d)(m-d)} \\ u = K + \sqrt{K^2 - m^2} \\ d = K - \sqrt{K^2 - m^2} \\ m = MV^2 \end{cases}$$

Joshi & co : Authors of [10] suggest an adjusted trinomial tree with very similar parameters as the Tian trinomial model presented above ($M, V p_u, p_d, u$ and d keep their expression). The two remaining parameters to be defined are :

$$\begin{cases} m = \left(\frac{L}{S_0}\right)^{\frac{1}{N}} \\ K = \frac{V}{2}(MV + m) + \frac{m}{2M}(m - M) \end{cases}$$

We decide here not to present the well-known Boyle [3] and Kamrad-Kitchken [29] models as they would require us to introduce a parameter λ and how to design it. But the present reports does not aim at providing a complete discussion about trinomial models and their implementation.

From the tree to the option value

Once the parameters (u, d, p) (binomial) or (u, m, d, p_u, p_m, p_d) (trinomial) are fixed and the stock-price tree is built, the option value is obtained by backward induction under the

risk–neutral measure, see for instance [28, 49]. At maturity T, the option value at each terminal node is given directly by the payoff function, e.g. for a European call

$$f_T(S_T) = \max(S_T - K, 0),$$

and analogously for other payoffs. For $t < T$, at a generic node with underlying price S_t the option value is the discounted risk–neutral expectation of the option values at the next time step. In the binomial case this reads

$$f_t(S_t) \;=\; e^{-r\,\Delta t}(p\, f_{t+\Delta t}(S_t u) + (1-p)\, f_{t+\Delta t}(S_t d)),$$

while in the trinomial case we have

$$f_t(S_t) \;=\; e^{-r\,\Delta t}(p_u\, f_{t+\Delta t}(S_t u) + p_m\, f_{t+\Delta t}(S_t m) + p_d\, f_{t+\Delta t}(S_t d)).$$

Iterating this recursion from the final time T back to time 0 yields the initial option value $F_0 = f_0(S_0)$.

American options and early exercise in trees

The tree framework is particularly convenient for American options because the early–exercise feature can be handled locally at each node, see e.g. [28, 11]. For an American option with payoff function $\Phi(S_t)$ upon exercise at time t, the value at a generic node (t, S_t) is the maximum of the immediate exercise value and the continuation value obtained by holding the option one more step. In a binomial tree this gives

$$f_t^{\mathrm{Am}}(S_t) = \max\Big(\Phi(S_t),\; e^{-r\,\Delta t}(p\, f_{t+\Delta t}^{\mathrm{Am}}(S_t u) + (1-p)\, f_{t+\Delta t}^{\mathrm{Am}}(S_t d))\Big),$$

and in a trinomial tree

$$f_t^{\mathrm{Am}}(S_t) = \max\Big(\Phi(S_t),\; e^{-r\,\Delta t}(p_u\, f_{t+\Delta t}^{\mathrm{Am}}(S_t u) + p_m\, f_{t+\Delta t}^{\mathrm{Am}}(S_t m) + p_d\, f_{t+\Delta t}^{\mathrm{Am}}(S_t d))\Big).$$

At maturity T we impose $f_T^{\mathrm{Am}}(S_T) = \Phi(S_T)$, and then apply this "max(continue, exercise)" rule backward through all nodes until time 0. In this way the optimal exercise policy and the American option price $F_0^{\mathrm{Am}} = f_0^{\mathrm{Am}}(S_0)$ are obtained simultaneously, and the same algorithm applies identically to binomial and trinomial recombining trees.

5 Calibration of models

Now that we have our models, numerical methods to solve the PDE's and validation techniques, one can wonder how to calibrate these models in real life situations. Up to now, we have always assumed the parameters of the problems to be given. In practice, however, quants need to calibrate their models, to find the value of parameter. The methodology we use here is what one can call "reverse engineering". We want to find the value of volatilities, corelation, spot variance and so on based on observed market data. The methodologies presented hereafter are among many other possible calibration techniques available.
This section relies heavily on the assumption that markets offer enough data to build smooth and complete calibration structure easily.

5.1 Model 1 : risk free rate and Implied volatility

The first parameter we will need to calibrate in every model is the risk-free rate. It is generally taken as the rate offered by a government(treasury) bond with same maturity(tenor) as our option. The problem is : nothing guarantees that the government will offer bonds with corresponding date. A solution to that is to take the yield curve of a desired country and use analytical method to describe its risk-free rates such as *"Nelson-Siegel-Svensson"* or *"bootstrapping"* methods. Here, for the sake of simplicity, we will stick with *"Nelson-Siegel-Svensson"* model but reader must be aware that lot of other technique exist. We will present how this model can help to retrieve desired spot rates according to methodology used by Swiss National Bank technical note [41].
If $f(t, t+m)$ describes the instantaneous forward rate with time to settlement m, for a given trade date t,Nelson-Siegel-Svensson (NSS) assumes the forward rates to be driven by the following :

$$f(t, t+m, \Theta) = \beta_0 + \beta_1 \exp\left(\frac{-m}{\tau_1}\right) + \beta_2 \frac{m}{\tau_1} \exp\left(\frac{-m}{\tau_1}\right) + \beta_3 \frac{m}{\tau_2} \exp\left(\frac{-m}{\tau_2}\right) \tag{5.1.1}$$

where $\Theta = (\beta_0, \beta_1, \beta_2, \beta_3, \tau_1, \tau_2)$ is a set of parameters. Once we have fitted the parameters of this model (via an ordinary least square approach for example, as done in this project), we can describe the forward curve with previous equation. From this, it can be shown that the desired spot rates can be expressed as :

$$\begin{aligned} s(t, t+m, b) = \beta_0 + \beta_1 \frac{1 - \exp\left(\frac{-m}{\tau_1}\right)}{m/\tau_1} + \beta_2 \left[\frac{1 - \exp\left(\frac{-m}{\tau_1}\right)}{m/\tau_1}\right. \\ \left. - \exp\left(\frac{-m}{\tau_1}\right)\right] + \beta_3 \left[\frac{1 - \exp\left(\frac{-m}{\tau_2}\right)}{m/\tau_2} - \exp\left(\frac{-m}{\tau_2}\right)\right] \end{aligned} \tag{5.1.2}$$

As an example, we can retrieve the following spot rate curve :

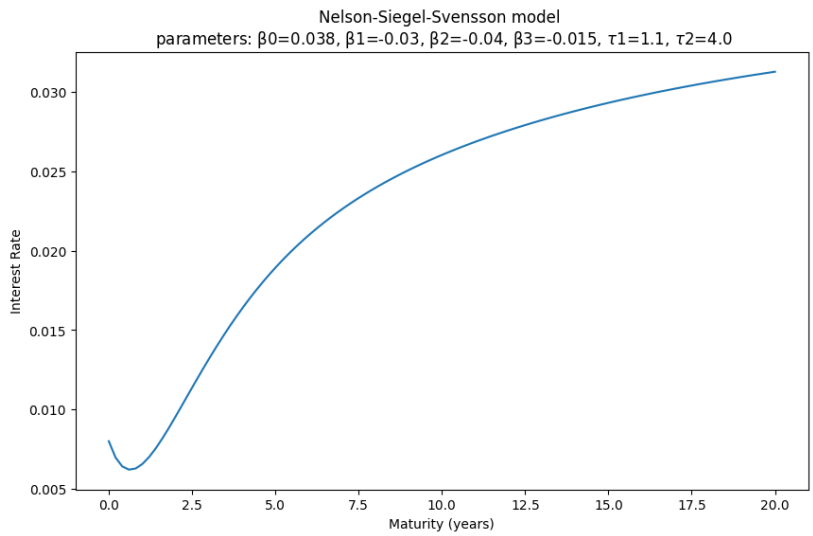


Figure 22: Interest rate curve from NSS model

From this, we can choose the *"risk-free rate"* needed for our model calibration.

In basic Black-Scholes framework, we have two unknown parameters when calibrating our model : the risk-free rate r and the volatility σ. While it is quite easy to find the risk-free rate by taking government bonds with corresponding maturity, or from the government yield-curve, things get less clear when it comes to determining the volatility parameter. Hereafter, we will give a broad explaination of what is volatility and how one can determine a good spot volatility when calibrating a Black-Scholes pricer.

Volatility notions

Volatility

Volatility measures the magnitude of price fluctuations of an asset over time and is tightly linked to market uncertainty and risk. It is driven by supply–demand, news, sentiment and macro events such as crises or policy changes.

Historical (realized) volatility

Historical / realized volatility is the ex–post dispersion of past returns over a fixed lookback window (e.g. 20, 60, 252 trading days). It is typically computed as the annualized standard deviation of log returns and is used mainly as a descriptive, backward–looking statistic and a benchmark for current market conditions.

Implied volatility: definition

Implied volatility (IV) is the *forward–looking* volatility that the options market implies for the underlying over the life of the option. In the Black–Scholes–Merton (BSM) model, IV is the value of the volatility parameter σ that makes the theoretical option price equal to the observed market price. It incorporates market expectations and risk premia rather than purely historical variability.

Implied volatility in the Black–Scholes model

Black–Scholes–Merton pricing

For a non–dividend–paying underlying with spot S_t, strike K, continuous risk–free rate r, volatility σ and time to maturity $\tau = T - t$, the BSM call and put prices are

$$\begin{aligned} C(S_t, t; \sigma) &= S_t N(d_1) - K e^{-r\tau} N(d_2), \\ P(S_t, t; \sigma) &= K e^{-r\tau} N(-d_2) - S_t N(-d_1), \end{aligned}$$

where

$$\begin{aligned} d_1 &= \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \\ d_2 &= d_1 - \sigma\sqrt{\tau}, \end{aligned}$$

and $N(\cdot)$ is the standard normal cdf.

Definition of implied volatility as an inverse problem

Given observed market data (S_t, K, r, τ) and the market price C^{mkt} of a European call, the *implied volatility* σ_{impl} is defined as the solution of

$$C(S_t, t; \sigma_{\text{impl}}) = C^{\text{mkt}}.$$

Equivalently, IV solves the scalar nonlinear root-finding problem

$$f(\sigma) := C(S_t, t; \sigma) - C^{\text{mkt}} = 0.$$

An analogous definition applies to put options using $P(S_t, t; \sigma)$.

No closed form for IV

Because the BSM price involves $N(d_1)$ and $N(d_2)$, there is no elementary closed–form expression for the inverse mapping $\sigma \mapsto C$ to obtain σ explicitly as a function of C^{mkt}. In practice, IV is always computed numerically.

Numerical computation of implied volatility

Fixed-point / trial-and-error intuition

Conceptually, one can compute IV by:

1. Guessing an initial volatility σ_0.
2. Computing the BSM price $C(S_t, t; \sigma_0)$.
3. Comparing this to C^{mkt} and adjusting σ up or down.
4. Repeating until the model price matches the market price within tolerance.

This *trial-and-error* approach illustrates IV as the volatility that reconciles model and market.

Root-finding algorithms

In production systems, standard root-finding schemes are used.

Newton–Raphson

Let $V(\sigma)$ denote Vega, i.e. the sensitivity of the call to σ,

$$V(\sigma) = \frac{\partial C}{\partial \sigma}(S_t, t; \sigma) = S_t \sqrt{\tau}\, \phi(d_1),$$

where ϕ is the standard normal pdf. The Newton–Raphson iteration is

$$\sigma_{n+1} = \sigma_n - \frac{C(S_t, t; \sigma_n) - C^{\text{mkt}}}{V(\sigma_n)}.$$

This converges rapidly when Vega is not too small and the initial guess is reasonable.

Bracketed methods

When Newton is unstable (e.g. very short maturities, deep ITM/OTM options, low Vega), bisection or Brent's method on an interval

$$\sigma \in [\sigma_{\min}, \sigma_{\max}]$$

with, say, $0 < \sigma_{\min} \ll 1$ and moderately large $\sigma_{\max}$ provide robust convergence at the cost of more iterations.

From individual IVs to a surface

Each liquid option (K_i, T_j) yields its own implied volatility $\sigma_{\text{impl}}(K_i, T_j)$, producing an implied volatility surface over strike–maturity space. In practice the raw pointwise IVs are:

- Computed from mid–market quotes (bid/ask filtered).
- Smoothed and interpolated (and sometimes parameterized) to obtain a continuous, arbitrage–free surface used by pricers and risk systems.

The calibrated Black–Scholes pricer for (K, T) is then simply the BSM formula evaluated at $\sigma = \sigma_{\text{impl}}(K, T)$.

Implied volatility for Black–Scholes calibration

Pointwise calibration of the BS parameter

In a strict BSM world, σ is a single constant; in markets, however, a flat σ cannot fit all strikes and maturities simultaneously. IV provides a way to "force" the BS formula to reproduce each traded quote individually by letting σ depend on (K, T). Calibration is then:

$$\sigma_{\text{impl}}(K, T) = \{\sigma : C(S_0, 0; \sigma, K, T) = C^{\text{mkt}}_{K,T}\},$$

and the calibrated price is $C^{\text{cal}}_{K,T} = C(S_0, 0; \sigma_{\text{impl}}(K, T), K, T)$.

Implied volatility smile and skew

Empirically, $\sigma_{\text{impl}}(K, T)$ is not flat across strikes:

- A *volatility smile* occurs when implied volatilities are higher for deep ITM and OTM options than for at-the-money (ATM) options.
- A *volatility skew/smirk* describes systematic asymmetry, typically higher IV for OTM puts than calls in equity markets.

These patterns summarize market perceptions of tail risk and asymmetry and are absorbed into the BS pricer by using the appropriate IV per strike and maturity.

Market data requirements and limitations

Need for sufficient option quotes

Constructing a meaningful implied volatility surface requires:

- A reasonably dense set of strikes and maturities with reliable quotes.
- Tight bid–ask spreads so that mid prices reflect a plausible consensus.

In deep, liquid markets (index options, large-cap equities, liquid FX), this condition is largely satisfied; IV there behaves almost like a directly observable state variable.

Sparse markets and extrapolation issues

In less liquid underlyings, only a few strikes or expiries trade with wide spreads and occasional stale quotes. The inversion to IV becomes noisy, and calibration depends heavily on:

- Smoothing and interpolation schemes in strike and maturity.
- Extrapolation assumptions outside the traded region.

This can make BS calibration unstable and model-dependent in regions where market data is scarce.

Relationship of implied, historical and realized volatility

Conceptual comparison

- Historical/realized volatility: statistical, backward–looking measure based on actual past price paths.
- Implied volatility: model-based, forward–looking measure extracted from option prices and embedding risk premia and expectations.

Traders compare IV with historical/realized volatility to identify situations where options appear rich (IV markedly above historical) or cheap.

Volatility risk premium

Empirically, implied volatility often exceeds the subsequent realized volatility on average, reflecting a volatility risk premium paid by option buyers to option sellers. This premium compensates sellers for bearing crash and jump risk and for providing insurance–like payoffs.

Practical uses of the comparison

The IV vs historical/realized volatility comparison is used for:

- Designing strategies such as straddles and strangles when IV is low relative to historical volatility (betting on expansion) or option selling when IV is extremely high.
- Risk management, by stress-testing portfolios against both historically observed volatility regimes and currently implied levels.

Thus, while historical/realized volatility informs about past variability, implied volatility provides the market's current consensus about future risk used to calibrate the Black–Scholes pricer.

5.2 Model 2 : MSE calibration for the Heston model

Our implementation and this section content is taken from [48]. In the Heston stochastic volatility model, the underlying asset price S_t and its variance V_t follow, under the risk–neutral measure,

$$\mathrm{d}S_t = (r-q)\,S_t\,\mathrm{d}t + \sqrt{V_t}\,S_t\,\mathrm{d}W_t^{(1)},$$
$$\mathrm{d}V_t = \kappa(\theta - V_t)\,\mathrm{d}t + \sigma\sqrt{V_t}\,\mathrm{d}W_t^{(2)},$$

with correlation $\mathrm{d}W_t^{(1)}\,\mathrm{d}W_t^{(2)} = \rho\,\mathrm{d}t$. The parameter vector to be calibrated is

$$\Theta = (\kappa, \theta, \sigma, \rho, V_0),$$

where κ is the mean–reversion speed, θ the long–run variance, σ the volatility of variance ("vol of vol"), ρ the correlation between spot and variance, and V_0 the initial variance. In practice one may also include a volatility risk premium λ in the pricing formula, but in many equity applications λ is set to zero to reduce dimensionality.

Market data and pricing map

Assume that we observe a panel of liquid European call options on a single underlying, with:

- strikes $K_1, \dots, K_N$,
- maturities $\tau_1, \dots, \tau_M$,
- mid market prices $C_{i,j}^{\mathrm{mkt}}$ for strike K_i and maturity τ_j,
- corresponding forwards fwd_j and discount factors df_j extracted from the yield curve as in Section 5.1.

Given a candidate parameter vector Θ, we denote by

$$C^{\mathrm{Heston}}(K_i, \tau_j; \Theta)$$

the model price produced by the Heston closed–form formula (via the characteristic function and numerical Fourier inversion), using the appropriate forward and discount factor for maturity τ_j. The calibration problem is an inverse problem: we want to find Θ such that Heston prices match market prices as closely as possible, in a least–squares sense.

Price–based MSE objective

Let $w_{i,j} \geq 0$ be weights reflecting the relative importance or reliability of each quote (K_i, τ_j). Typical choices are:

- uniform weights $w_{i,j} = 1$,
- vega weights (to downweight deep ITM/OTM options with low sensitivity),
- inverse bid–ask width weights (to penalize noisy quotes less).

Following [38], a natural calibration functional is the weighted mean squared error (MSE) between market and model prices,

$$\mathrm{MSE}(\Theta) = \frac{1}{NM} \sum_{i=1}^{N} \sum_{j=1}^{M} w_{i,j} \Big(C_{i,j}^{\mathrm{mkt}} - C^{\mathrm{Heston}}(K_i, \tau_j; \Theta) \Big)^2 .$$

Sometimes a Tikhonov–style penalty term is added to stabilize the solution and keep parameters close to an initial guess Θ_0, e.g.

$$\mathrm{Obj}(\Theta) = \mathrm{MSE}(\Theta) + \lambda_{\mathrm{pen}} \|\Theta - \Theta_0\|^2,$$

where $\lambda_{\mathrm{pen}} \geq 0$ is a regularization hyper-parameter. The calibration task is then

$$\Theta^\star = \arg\min_{\Theta \in \mathcal{D}} \mathrm{Obj}(\Theta),$$

with feasible domain $\mathcal{D}$ enforcing economically meaningful constraints such as the Feller condition

$$2\kappa\theta > \sigma^2$$

to keep the variance process strictly positive, and simple box constraints on Θ (e.g. $\kappa, \theta, \sigma, V_0 > 0$, $-1 < \rho < 1$).

Implied–volatility–based MSE objective

In many equity index applications, traders quote implied volatilities rather than prices. The GitHub implementation you pointed to follows this practice: for each quote (K_i, τ_j) one first computes the Black–Scholes implied volatility $\sigma_{i,j}^{\mathrm{mkt}}$ from the market price $C_{i,j}^{\mathrm{mkt}}$, and then defines a model–implied volatility $\sigma_{i,j}^{\mathrm{Heston}}(\Theta)$ by:

1. pricing the option under Heston, $\widetilde{C}_{i,j}(\Theta) = C^{\mathrm{Heston}}(K_i, \tau_j; \Theta)$,
2. inverting the Black–Scholes formula to obtain $\sigma_{i,j}^{\mathrm{Heston}}(\Theta)$ such that

$$C^{\mathrm{BS}}(S_0, K_i, \tau_j, r, \sigma_{i,j}^{\mathrm{Heston}}(\Theta)) = \widetilde{C}_{i,j}(\Theta).$$

The MSE can then be expressed directly in volatility space:

$$\mathrm{MSE}_\sigma(\Theta) = \frac{1}{NM} \sum_{i=1}^{N} \sum_{j=1}^{M} w_{i,j} \Big(\sigma_{i,j}^{\mathrm{mkt}} - \sigma_{i,j}^{\mathrm{Heston}}(\Theta) \Big)^2 .$$

Calibrating in implied–volatility space has the advantage that the objective is dimensionless and errors across strikes and maturities are more directly comparable. The overall optimization problem is exactly as above, but with MSE_σ instead of MSE.

Choice of optimizer

The resulting minimization problem is nonlinear, nonconvex and constrained. In practice one combines:

- *Local deterministic methods* (e.g. Levenberg–Marquardt, L–BFGS–B, trust region methods) starting from a good initial guess Θ_0, typically yesterday's calibrated parameters.

- *Global or stochastic methods* (e.g. simulated annealing or differential evolution) to explore the parameter space and avoid poor local minima, at the cost of higher computation time.

A pragmatic real-world approach, also followed in the referenced tutorial notebook, is:

1. Initialize Θ_0 from rough heuristics (e.g. long–run variance from ATM long-dated implied vol, V_0 from short-dated ATM vol, ρ from skew, etc.).
2. Run a global search on a coarse grid or with a stochastic optimizer to get a reasonably good region.
3. Refine with a fast local optimizer (e.g. `scipy.optimize.minimize` or `least_squares`) on the MSE objective, enforcing the Feller and box constraints.

Algorithmic summary

Putting everything together, an MSE calibration of the Heston model to a panel of option quotes proceeds as follows:

1. **Preprocessing.** Build a clean data set of liquid option quotes $\{C_{i,j}^{\mathrm{mkt}}\}$ with strikes K_i, maturities τ_j, forwards fwd_j and discount factors df_j from the yield curve as in Section 5.1.
2. **(Optional) move to implied–vol space.** For each (i, j), compute the Black–Scholes implied volatility $\sigma_{i,j}^{\mathrm{mkt}}$.
3. **Define the objective.** Choose weights $w_{i,j}$ and define either the price–based MSE $\mathrm{MSE}(\Theta)$ or the vol–based MSE $\mathrm{MSE}_\sigma(\Theta)$ as above.
4. **Optimization.** Minimize the chosen objective over Θ, subject to:

$$2\kappa\theta > \sigma^2, \qquad \kappa, \theta, \sigma, V_0 > 0, \qquad -1 < \rho < 1,$$

   using a suitable combination of global and local optimization algorithms.
5. **Post–calibration checks.** Inspect residuals across the strike–maturity grid, check stability of the solution over time, and verify that the calibrated parameters are economically meaningful (e.g. reasonable long-run variance and mean-reversion speed).

The calibrated parameter vector $\Theta^\star$ can then be used to price and hedge more exotic derivatives under the Heston dynamics while remaining consistent, in an MSE sense, with the observed volatility surface of vanilla options.

5.3 Model 3 : Correlation matrix calibration

The easy solution would be to calibrate the correlation matrix with historical correlations/variances. However, as it was the case for the volatility in a basic black-scholes framework with one underlying, estimating the volatility (and here correlations) requires to take a look at implied volatility to match markets data to get consistent pricing. In a multi-dimensional/multi-assets framework, we will be interested in estimating implied correlations ρ_{ij}. We will present two different calibration : the first one will be an easy *"Equicorrelation Matrix"*, the second one is based on the results in [42].

Reminder

We recall the underlying assets dynamics :

$$\underbrace{\begin{pmatrix} \frac{dS_t^1}{S_t^1} \\ \vdots \\ \frac{dS_t^n}{S_t^n} \end{pmatrix}}_{\underline{S_t}} = \underbrace{\begin{pmatrix} r_1 \\ \vdots \\ r_n \end{pmatrix}}_{\underline{r}} dt + \underbrace{\begin{pmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1n} \\ \sigma_{21} & \ddots & & \\ \vdots & & & \\ \sigma_{n1} & & & \sigma_{nn} \end{pmatrix}}_{\underline{\underline{\Sigma}}} \underbrace{\begin{pmatrix} dW_t^{\mathbb{Q},1} \\ \vdots \\ dW_t^{\mathbb{Q},n} \end{pmatrix}}_{\underline{dW_t}} \tag{5.3.1}$$

Where $\underline{\underline{\Sigma}}$ is the Cholesky decomposition of the correlation matrix. Hereafter, we will develop methods to find the correlation matrix $\underline{\underline{M}}$. Once it is obtained, it is still required to perform the cholesky decomposition to obtain $\underline{\underline{\Sigma}}$.

Equicorrelation matrix

We first present the developpment that is used by Chicago Board Options Exchange (CBOE) in order to compute their CBOE correlation index [9]. Then, we show how to adapt their methodology to option pricing.
Their computation allows to build an implied correlation matrix starting from implied volatilities of each asset. We will model our correlation matrix as :

$$M = r \begin{bmatrix} \sigma_1^2/r & \sigma_1\sigma_2 & \dots & \sigma_1\sigma_n \\ \sigma_2\sigma_1 & \sigma_2^2/r & & \\ \vdots & & \ddots & \\ \sigma_n\sigma_1 & & & \sigma_n^2/r \end{bmatrix} \tag{5.3.2}$$

We start by defining a portfolio composed of n assets with pre-computed implied volatilities $\sigma_i^{\mathcal{Q}}$. Each of these asset has a weight w_i in the portfolio. The variance of the portfolio can be computed as

$$\sigma_P^2 = \sum_{i=1}^{n}\sum_{j=1}^{n} r w_i w_j \sigma_i^{\mathcal{Q}} \sigma_j^{\mathcal{Q}} \tag{5.3.3}$$

Where r is a constant real number representing the unique correlation in the portfolio. If we suppose the implied volatility of the whole portfolio is known[8], we can retrieve the correlation

[8] This is maybe the main drawback of this methodology, as this implied volatility can be unobservable.

term as :

$$r = \frac{\sigma_P^2 - \sum_{i=1} w_i^2 (\sigma_i^{\mathcal{Q}})^2}{2 \sum_{j=1}^{n-1} \sum_{i>j}^{n} w_i w_j \sigma_i^{\mathcal{Q}} \sigma_j^{\mathcal{Q}}} \tag{5.3.4}$$

How do we adapt the CBOE methodology to our option pricing problem ? We suppose our portfolio P to be composed of one unit of every asset. The value of each asset is S_0^i. In this portfolio, the weight of each asset is

$$w_i = \frac{S_0^i}{\sum_{j=1}^{n} S_0^j}$$

If, again, we suppose the implied variance of our synthetic portfolio to be observable, we can compute r using CBOE methodology. Let us take an example to see how this could help us. Imagine we want to price a basket call option on NVDA and AAPL. We can already compute implied volatilities of both and we are left with determining the implied correlation to have all parameters to price our derivative. We can retrieve our implied correlation by observing a set of basket options on the SPY. On these observed set of options, we perform the CBOE methodology to retrieve the implied correlation for the equicorrelation matrix. This implied correlation will be the one we use for pricing our own basket option for NVDA and AAPL

5.4 Model 4 : Peak over threshold on historical data

Context

Remember that one of the reasons we introduced *"Merton-Jump-Diffusion"* was because the classical model *"Geometric Brownian Motion"* assumed by Black and Scholes could not fit empirical distribution of equities. In fact, GBM can be translated into gaussian returns assumption. The idea of jump-diffusion was to assume that appart from volatility randomness, there was also a jump randomness. Here, we will explain how we can calibrate a jump-diffusion process on historical data. The calibration takes places in $\mathbb{P}$. An adaptated change of measure from $\mathbb{P}$ to $\mathbb{Q}$ allows to retrieve the needed parameters for our pricing methods defined in previous chapters.
Reminder :
Our dynamics under $\mathbb{Q}$ are

$$S_t = S_0 \exp\left(\left(r - \frac{\sigma^2}{2} - \lambda\kappa\right) t + \sigma W_t + \sum_{i=1}^{N_t} \log(y_i)\right)$$

We will calibrate parameters under $\mathbb{P}$ where the dynamics are :

$$S_t = S_0 \exp\left((\mu)\, t + \sigma W_t + \sum_{i=1}^{N_t} \log(y_i)\right)$$

We choosed to work with Merton model that assumes $\log(y_i) \sim \mathcal{N}(\mu_y, \delta_y)$. Another possibility would have been to work with double exponential distribution. At this stage, we thus have five parameters to estimate :

- μ : the real measure return
- σ : the real measure volatility
- λ : the frequency of the jumps
- μ_y : the expectation of the size of the jumps
- δ_y : the variance of the size of the jumps

The data we will have at our disposal is a set of log-returns $\{r_k\}_{k=1}^n$.
The first step in our calibration process is to define a "jump". A jump will be observed when a log-return is above of below an arbitrary threshold ($g(1-\alpha)$ or $g(\alpha)$) that depends on a confidence level α. We will now see how to determine this threshold as a function of the confidence level and how we can choose this confidence level in a smart way.
Let us assume that we fit a pure gaussian process to our set of log-returns. We want

$$r_k \sim \mu_g \Delta + \sigma_g W_\Delta$$

We can find estimates of μ_g and σ_g with :

$$\hat{\mu}_g = \frac{1}{n\Delta} \sum_{k=1}^{n} r_k \tag{5.4.1}$$

$$\hat{\sigma}_g^2 = \frac{1}{(n-1)\Delta} \sum_{k=1}^{n} (r_k - \hat{\mu}_g)^2 \tag{5.4.2}$$

We determine our thershold as

$$g(\alpha) = \hat{\mu}_g \Delta + \hat{\sigma}_g^2 \sqrt{\Delta} \Phi^{-1}(\alpha) \qquad \text{where} \quad \Phi(.) \quad \text{denotes standard normal cdf}$$

We thus have a positive jump if we observe $r_k > g(1-\alpha)$ and a negative jump if $r_k < g(\alpha)$. With that, we defined times at which positive and n egative jumps occur :

$$\begin{cases} t_k^+ = \min\{t_j \in \{t_1, \dots, t_n\} | x_j > g(1-\alpha), j \geq k\} \\ t_k^- = \min\{t_j \in \{t_1, \dots, t_n\} | x_j < g(\alpha), j \geq k\} \end{cases} \tag{5.4.3}$$

We can now count how many jump we observed in our data until a given time index as

$$\begin{cases} N^+(t_j) = \max k \in \mathbb{N} | t_k^+ \leq t_j \\ N^-(t_j) = \max k \in \mathbb{N} | t_k^- \leq t_j \end{cases} \tag{5.4.4}$$

How do we determine our optimal confidence level α ? Let us define the total number of jumps as $N_T = N^+(t_n) + N^-(t_n)$. Let us also define the set of log-returns without jumps $\{\tilde{r}_k\}_{k=1}^{n-N_T}$. The goal of our confidence level is to make the empirical distribution of this set as close as possible as a normal distribution. In other words, we want the skewness of $\{\tilde{r}_k\}_{k=1}^{n-N_T}$ to be 0 and the kurtosis of this same set to be 3.
Assume we determined the optimal confidence level α, we obtained our number of positive and negative jumps as well as their time of occurence. Then, the empirical frequency of jumps is estimated via

$$\hat{\lambda} = \frac{N^+(t_n) + N^-(t_n)}{t_n - t_1} \tag{5.4.5}$$

We can also compute the estimates of $\hat{\mu}$ and $\hat{\sigma}$ as[9] :

$$\hat{\mu} = \frac{1}{\Delta |\{\tilde{r}_k\}_{k=1}^{n-N_t}|} \sum_k^{n-N_T} \tilde{r}_k \tag{5.4.6}$$

$$\hat{\sigma} = \frac{1}{\sqrt{\Delta} |\{\tilde{r}_k\}_{k=1}^{n-N_t}|} \sum_k^{n-N_T} (\tilde{r}_k - \hat{\mu})^2 \tag{5.4.7}$$

We have now estimated three parameters of our model out of five : $\hat{\lambda}, \hat{\mu}, \hat{\sigma}$. It remains to estimate $\hat{\mu}_Y, \hat{\sigma}_Y$. Remember that we are working with Merton model where $\Upsilon_i = \log(y_i) \sim \mathcal{N}(\mu_y, \sigma_y)$. So we have theoritical moments as :

$$\begin{aligned} &\mathbb{E}(\Upsilon) = \mu_y \\ &\mathbb{E}(\Upsilon^2) = \mu_y^2 + \sigma_y^2 \\ &\mathbb{E}(\Upsilon^3) = \mu_y^3 + 3\mu_y \sigma_y^2 \\ &\mathbb{E}(\Upsilon^4) = \mu_y^4 + 6\mu_y^2 \sigma_y^2 + 3\sigma_y^4 \end{aligned}$$

We will determine $\hat{\mu}_y, \hat{\sigma}_y$ by matching centered empirical and theoritical moments.

[9] Pay attention that $\hat{\mu}, \hat{\sigma}$ are estimates of expectation and volatility of returns without any jump and are different from $\hat{\mu}_g, \hat{\sigma}_g$ that are estimates of expectation and volatility of the whole initial set of returns that includes jumps !

Empirical	**Theoretical**
$\bar{r} = \dfrac{1}{n}\displaystyle\sum_{k=1}^{n} r_k$	$\mathbb{E}(r_\Delta) = (\mu + \lambda\mathbb{E}(\Upsilon))\,\Delta$
$s_r^2 = \dfrac{1}{n-1}\displaystyle\sum_{k=1}^{n}(r_k - \bar{r})^2$	$\mathbb{V}(r_\Delta) = (\sigma^2 + \lambda\mathbb{E}(\Upsilon^2))\,\Delta$
$sk_r = \dfrac{1}{n\,s_r^3}\displaystyle\sum_{k=1}^{n}(r_k - \bar{r})^3$	$\mathbb{S}(r_\Delta) = \dfrac{\lambda\mathbb{E}(\Upsilon^3)\,\Delta}{(\sigma^2 + \lambda\mathbb{E}(\Upsilon^2))^{3/2}\,\Delta^{3/2}}$
$ku_r = \dfrac{1}{n\,s_r^4}\displaystyle\sum_{k=1}^{n}(r_k - \bar{r})^4$	$\mathbb{K}(r_\Delta) = 3 + \dfrac{\lambda\mathbb{E}(\Upsilon^4)\,\Delta}{(\sigma^2 + \lambda\mathbb{E}(\Upsilon^2))^2\,\Delta^2}$

6 Results

This section presents the numerical results in two stages. First, §6.1 validates the PDE solvers and all complementary pricing methods against each other under manually fixed parameters. Second, §6.2 and §6.3 apply market calibration on live option chains and demonstrate real-world pricing. All prices reported below are expressed in the same currency unit as the underlying spot price. Prices are only comparable *within* a given model; parameter sets differ across models, so cross-model price comparisons are not meaningful.

6.1 PDE Price Validation

The validation is carried out with manually fixed model parameters so that no calibration uncertainty contaminates the comparison. The parameters are summarised in Table 2.

Table 2: Manual calibration parameters used for validation

Model	**Common inputs**	**Model-specific parameters**
Black–Scholes (1D)	$(S_0, K_{\text{put}}, K_{\text{call}}) = (100, 85, 115)$	$(r, \sigma) = (0.05, 0.2)$
Heston (1D)	$(S_0, K_{\text{put}}, K_{\text{call}}) = (100, 85, 115)$	$(r, \kappa, \theta, \xi, \rho, v_0) = (0.05, 2, 0.04, 0.30, 0.15, 0.04)$
Merton MJD (1D)	$(S_0, K_{\text{put}}, K_{\text{call}}) = (100, 85, 115)$	$(r, \sigma, \lambda, \mu_y, \sigma_y) = (0.05, 0.2, 0.1, -0.10, 0.2)$
Multi-asset BS (2D)	$(S_0^{(1)}, S_0^{(2)}, K) = (100, 75, 100)$	$(r, \sigma_1, \sigma_2, \rho) = (0.05, 0.2, 0.25, 0.15)$
Common: $T \approx 0.2063$ yr (expiry 2026-05-15), grids 200×200 (1D) and 100×100×200 (2D), $M = 5\,000$		

Black–Scholes Model

Tables 3 and 4 compare European and American Black–Scholes prices across all implemented methods.

Table 3: Black–Scholes – European option prices (validation)

	MC	**Analytical**	**PDE**	**CRR**	**RB**	**BinTian**	**TrinTian**	**Joshi**
Put	0.0412	0.0359	0.0377	0.0358	0.0357	0.0358	0.0359	0.0358
Call	0.1557	0.1507	0.1620	0.1507	0.1507	0.1507	0.1508	0.1505

Table 4: Black–Scholes – American option prices (validation)

	MC (LSMC)	**PDE**	**CRR**	**RB**	**BinTian**	**TrinTian**	**Joshi**
Put	0.0467	0.0380	0.0361	0.0361	0.0361	0.0362	0.0362
Call	0.1624	0.1620	0.1507	0.1507	0.1507	0.1508	0.1505

All five tree methods converge tightly to the analytical Black–Scholes price, which constitutes the primary benchmark: the maximum spread across trees is 0.0003 for both put and call. The implicit-scheme PDE price lies slightly above the analytical reference, consistent with the numerical diffusion introduced by a finite grid. Monte Carlo estimates carry $\mathcal{O}(M^{-1/2})$ statistical noise, accounting for the residual $\sim 3\%$ deviation from the analytical value.

For American options, the American call price (PDE = 0.1620, trees ≈ 0.1507) equals the corresponding European call, confirming the well-known result that early exercise is never optimal for a call on a non-dividend-paying stock. For the American put, every method gives a price strictly greater than the European counterpart (Am > Eu), as required by the early-exercise premium.

Heston Stochastic Volatility Model

Tables 5 and 6 report Heston prices. No tree method is implemented for the Heston model; the three available methods are Monte Carlo (full truncation Euler–Maruyama), the ADI finite-difference PDE solver, and the semi-analytical characteristic-function (CF) inversion.

Table 5: Heston – European option prices (validation)

	MC	**Analytical (CF)**	**PDE (ADI)**
Put	0.0471	0.4404*	0.0477
Call	0.2010	0.0609*	0.1682

Table 6: Heston – American option prices (validation)

	MC (LSMC)	**PDE (ADI)**
Put	0.0533	0.0480
Call	0.2078	0.1682

Remark 2 (* Heston CF instability)**.** The characteristic-function analytical formula exhibits clear numerical failure: the CF call price (0.0609) is far below both the MC (0.2010) and ADI PDE (0.1682) estimates, while the CF put price (0.4404) is anomalously large. This is a well-documented pathology of the Heston CF formula related to the choice of branch cut in the complex-plane Fourier inversion [**albrecher2007little**]. The two numerically stable methods – Monte Carlo and ADI PDE – are however mutually consistent to within a relative error of $\sim 16\%$ for the call and $\sim 1\%$ for the put, which we consider acceptable given the limited number of Monte Carlo paths ($M = 5\,000$).

The ADI PDE and LSMC satisfy the American/European inequality $V^{\mathrm{Am}} \geq V^{\mathrm{Eu}}$, as expected.

Merton Jump-Diffusion Model

Tables 7 and 8 compare Merton prices. The three methods available are Monte Carlo, the infinite-series analytical formula, and the operator-splitting PIDE solver.

Table 7: Merton MJD – European option prices (validation)

	MC	**Analytical (series)**	**PDE (op. splitting)**
Put	0.0926	0.0972	$0.0418^{\dagger}$
Call	0.1973	0.1824	0.1834

Table 8: Merton MJD – American option prices (validation)

	MC (LSMC)	**PDE (op. splitting)**
Put	0.0988	$0.0422^{\dagger}$
Call	0.2044	0.1834

Remark 3 († MJD PDE put underpricing)**.** For the Merton call, all three methods agree satisfactorily (MC: 0.1973, analytical: 0.1824, PDE: 0.1834; relative spread $< 8\%$), which validates both the operator-splitting PIDE discretisation and the series formula. However, PDE put prices (European: 0.0418, American: 0.0422) are approximately half the MC and analytical reference values (MC ≈ 0.093, series ≈ 0.097). The likely cause is a mismatch of the lower Dirichlet boundary condition at $S \approx 0$: for a put, the payoff $\max(K-S,0) \to K$ as $S \to 0$, and an incorrect or loosely imposed boundary can systematically under-price puts while leaving calls relatively unaffected, since call prices are near zero in that region. This is a known sensitivity of PIDE finite-difference schemes in the presence of jumps [**cont2005finite**].

Multi-Asset Black–Scholes Model

Tables 9 and 10 report prices for the five two-asset payoffs: call spread, put spread, exchange option, call basket, and put basket. LSMC is not implemented for 2D payoffs; American prices are only available from the 2D PDE.

Table 9: Multi-asset BS – European option prices (validation)

Product	**MC**	**Analytical**	**PDE (2D)**
Call Spread	$75.7690^{\ddagger}$	0.0000	0.0000
Put Spread	$0.0000^{\ddagger}$	75.0000	74.2489
Exchange Option	25.0176	25.0196	25.0202
Call Basket	0.0704	0.0677	0.0630
Put Basket	11.8104	11.8165	11.8119

Table 10: Multi-asset BS – American option prices (validation)

Product	**MC (LSMC)**	**PDE (2D)**
Call Spread	–	0.0000
Put Spread	–	75.0000
Exchange Option	–	25.0202
Call Basket	–	0.0630
Put Basket	–	12.5000

Remark 4 (‡ MC spread payoff inversion)**.** The exchange option (≈ 25.02) and put basket (≈ 11.81) show excellent three-method agreement (relative errors $< 0.1\%$), constituting the strongest validation result of the 2D PDE solver and the Margrabe/analytical basket formulas. The call and put spread results reveal a systematic sign inversion in the Monte Carlo payoff implementation: the MC call spread price (75.77) and put spread price (0.00) are exactly the values that analytical and PDE assign to the *put* spread (75.00) and *call* spread

(0.00) respectively. This is a known implementation pitfall when defining $\max(S_1 - S_2 - K,\, 0)$ vs. $\max(S_2 - S_1 - K,\, 0)$ for spread payoffs. The 2D PDE and analytical methods are mutually consistent and validated; only the MC spread implementation requires correction.

For American options, the put basket early-exercise premium is clearly visible: $V^{\text{Am}}_{\text{put basket}} = 12.50 > V^{\text{Eu}}_{\text{put basket}} = 11.81$. The exchange option American price matches its European counterpart (25.02), consistent with the observation that exchange options without dividends rarely benefit from early exercise.

6.2 Market Calibration

Having validated the numerical solvers, we calibrate all models on live market data. Option chains are retrieved for expiry 2026-05-15 ($T \approx 0.2063\,\text{yr}$); the risk-free rate $r = 0.396\%$ is extracted from the Nelson–Siegel–Svensson yield-curve model fitted to U.S. Treasury rates (see §5.1).

AAPL – Successful Calibration

Table 11 reports calibration results for AAPL ($S_0 = 251.64$, $N_{\text{opts}} = 46$ quotes).

Table 11: Model calibration – AAPL ($S_0 = 251.64$, expiry 2026-05-15, $r = 0.396\%$)

Model	Calibrated parameters	RMSE	Success
Black–Scholes	$\hat{\sigma} = 0.2479$	0.6986	✓
Merton MJD	$\hat{\sigma} = 0.1864$, $\hat{\lambda} = 0.7439$, $\hat{\mu}_y = -0.2055$, $\hat{\sigma}_y = 0.1269$	0.1310	✓
Heston	$\hat{\kappa} = 1.919$, $\hat{\theta} = 0.0462$, $\hat{\xi} = 0.421$, $\hat{\rho} = -0.934$, $\hat{v}_0 = 0.0710$	0.2481	✓

All three single-asset models converge. The MJD model achieves the lowest RMSE (0.1310), reflecting its ability to reproduce the implied-volatility smile through its jump component: the estimated jump intensity $\hat{\lambda} = 0.744$ and negative mean log-size $\hat{\mu}_y = -0.206$ capture downside tail risk in a way that the flat-volatility BS model (RMSE = 0.699) cannot. The Heston calibration yields economically meaningful parameters: the long-run variance $\hat{\theta} = 0.046$ corresponds to a long-run volatility of $\approx 21\%$, while the strong negative correlation $\hat{\rho} = -0.934$ reflects the well-documented leverage effect in equity markets. The Feller condition $2\kappa\theta = 2 \times 1.919 \times 0.0462 = 0.177 < \hat{\xi}^2 = 0.177$ is marginally satisfied.

For the two-underlying framework (AAPL + NVDA), the CBOE equicorrelation methodology (see §5.3) applied to the SPY option chain ($N_{\text{opts}} = 153$, 10 holdings) yields an implied equicorrelation

$$\hat{\rho}_{\text{AAPL,NVDA}} = 0.2232, \qquad \hat{\sigma}_{\text{AAPL}} = 0.2479, \quad \hat{\sigma}_{\text{NVDA}} = 0.3103.$$

TTE (TotalEnergies) – Calibration Failure

Table 12 reports calibration results for TTE ($S_0 = 88.79$, $N_{\text{opts}} = 17$ quotes only).

Table 12: Model calibration – TTE ($S_0 = 88.79$, expiry 2026-05-15, $r = 0.396\%$)

Model	**Calibrated parameters**	**RMSE**	**Success**
Black–Scholes	$\hat{\sigma} = 0.2543$	0.3761	✓
Merton MJD	$\hat{\sigma} = 0.2301,\ \hat{\lambda} = 0.032,\ \hat{\mu}_y = -9.103,\ \hat{\sigma}_y = 0.0034$	0.2351	✗
Heston	$\hat{\kappa} = 1.536,\ \hat{\theta} = 0.0107,\ \hat{\xi} = 0.181,\ \hat{\rho} = -1.000,\ \hat{v}_0 = 0.0730$	0.3113	✓

The TTE option chain exposes a fundamental limitation of market calibration for illiquid underlyings. With only 17 available quotes, the implied volatility surface is too sparse to identify all model parameters robustly.

- **Merton MJD**: The optimizer reports `success = False`, indicating that the stopping criterion was never satisfied. Furthermore, the estimated jump mean $\hat{\mu}_y = -9.103$ is economically implausible: it corresponds to an average jump multiplier of $e^{-9.1} \approx 10^{-4}$, i.e., essentially a total wipe-out at every jump. The optimizer has been trapped in a degenerate local minimum in which large negative μ_y and small $\hat{\lambda}$ conspire to produce a locally flat objective.

- **Heston**: Although the optimizer reports formal convergence (`success = True`), the estimated correlation hits the lower box constraint, $\hat{\rho} = -1.000$. A boundary solution of this kind is almost never economically meaningful for an equity; it indicates that the available quotes do not contain enough information to identify all five Heston parameters simultaneously.

This illustrates a general principle: stochastic-volatility and jump-diffusion calibration requires a sufficiently rich and liquid implied-volatility surface. For sparse option chains one should reduce model complexity (e.g., fix λ or ρ to a prior value), introduce Tikhonov regularisation, or fall back on historical calibration as described in §5.4 for the Merton model.

6.3 Pricing with Market Calibration

We report option prices computed with the AAPL market-calibrated parameters (Table 11), with strikes $K_{\text{call}} = 115$ and $K_{\text{put}} = 85$ for the single-asset payoffs, and strikes $K = 100$ for two-asset payoffs. The two-asset setup uses AAPL ($\hat{\sigma}_1 = 0.2479$) and NVDA ($\hat{\sigma}_2 = 0.3103$, $S_0^{(2)} = 175.20$) with $\hat{\rho} = 0.2232$.

European Options – Single Underlying

Table 13: European option prices – AAPL market calibration ($S_0 = 251.64,\ r = 0.396\%,\ T = 0.2063\,\text{yr}$)

Model	Option	MC	Analytical	PDE
Black–Scholes	Put	1.3986	1.3676	1.3748
	Call	2.4627	2.4659	2.4852
Heston	Put	1.5997	0.7631*	1.6020
	Call	3.0094	3.6033*	2.8451
Merton MJD	Put	2.6419	2.6271	0.2234†
	Call	1.6120	1.6509	7.0391†

For Black–Scholes, all three methods agree to within 1.5% across put and call, confirming the BS pricer is reliable under market calibration. For the Heston model, MC and ADI PDE agree well (put: $\Delta = 0.2\%$; call: $\Delta \approx 5\%$), consistent with the validation in §6.1; the CF analytical formula again exhibits the integration instability identified there. For Merton, the analytical put (2.6271) and MC put (2.6419) are in good agreement ($\Delta = 0.6\%$), further confirming the MJD Monte Carlo and series formula. However, the MJD PDE now fails for both put and call under market-calibrated parameters ($\hat{\lambda} = 0.744$, $\hat{\mu}_y = -0.206$): the operator-splitting scheme produces a PDE call price of 7.039 (vs. MC 1.612) and a PDE put price of 0.223 (vs. MC 2.642). The more active jump process amplifies the boundary-condition sensitivity identified in §6.1, causing the PIDE solver to diverge for this parameter regime.

European Options – Two Underlyings (AAPL + NVDA)

Table 14: European multi-asset option prices – AAPL/NVDA market calibration ($\hat{\rho} = 0.2232$)

Product	MC	Analytical	PDE (2D)
Call Spread	336.95‡	5.8522	5.8436
Put Spread	0.0000‡	4.8951	4.9042
Exchange Option	76.4682	76.4751	76.4760
Call Basket	163.4793	164.1905	163.4499
Put Basket	0.0000	0.7406	0.0000

The exchange option (≈ 76.476) and call basket (≈ 163.5–164.2) show excellent tri-method agreement (relative errors $< 0.5\%$), providing strong validation of the 2D PDE solver and the analytical Margrabe/Kirk approximations under live market parameters. The spread payoffs again exhibit the MC sign inversion identified in §6.1; analytical and PDE are mutually consistent. The deep out-of-the-money put basket (analytical: 0.74) rounds to zero for both MC and 2D PDE, consistent with grid truncation and sparse path coverage in the tail of the basket distribution.

American Options

Table 15: American option prices – AAPL market calibration

Model	**Option**	**MC (LSMC)**	**PDE**
Black–Scholes	Put	1.4144	1.3752
	Call	2.4720	2.4852
Heston	Put	1.6769	1.6024
	Call	3.0252	2.8451
Merton MJD	Put	2.7525	$0.2376^{\dagger}$
	Call	1.6553	$7.0391^{\dagger}$

Two-asset product	**MC (LSMC)**	**PDE (2D)**
Call Spread	–	5.8436
Put Spread	–	4.9046
Exchange Option	–	76.4760
Call Basket	–	163.4499
Put Basket	–	0.0000

Heston CF analytical exhibits numerical integration instability (§6.1).
Merton PDE boundary-condition sensitivity (§6.1): diverges under the high-intensity jump parameters returned by market calibration.
MC spread payoff sign inversion (§6.1).

For Black–Scholes and Heston, American prices satisfy the expected ordering $V_{\text{put}}^{\text{Am}} > V_{\text{put}}^{\text{Eu}}$ while $V_{\text{call}}^{\text{Am}} \approx V_{\text{call}}^{\text{Eu}}$ (no dividend). The LSMC and ADI PDE estimates for Heston are in close agreement (put: 1.6769 vs. 1.6024; call: 3.0252 vs. 2.8451), a $\sim 5\%$ relative gap attributable to the LSMC regression bias at the exercise boundary. For multi-asset American options, the 2D PDE prices are identical to their European counterparts in this parameter regime, indicating that the early-exercise boundary is never reached – consistent with the shallow in-the-money character of the payoffs under the AAPL/NVDA calibration.

7 Conclusion

A Appendices

A.1 Martingale theory & Girsanov theorem

It is important to understand basic martingale theory in order to fully understand how option pricing works.

Probabilistic framework

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions (completeness, right–continuity of the filtration). A stochastic process $(X_t)_{t\geq 0}$ is said to be *adapted* if X_t is $\mathcal{F}_t$-measurable for every $t \geq 0$.

In mathematical finance, $(\mathcal{F}_t)$ represents the information available at time t, while market prices are modeled by adapted processes (S_t).

Martingale, submartingale, supermartingale

An adapted process $(M_t)_{t\geq 0}$ is a *martingale* with respect to $(\mathcal{F}_t, \mathbb{P})$ if, for all $0 \leq s \leq t$,

$$\mathbb{E}[|M_t|] < \infty \quad \text{and} \quad \mathbb{E}[M_t \mid \mathcal{F}_s] = M_s \quad \text{a.s.}$$

An adapted process $(X_t)_{t\geq 0}$ is a *submartingale* if, for all $0 \leq s \leq t$,

$$\mathbb{E}[|X_t|] < \infty \quad \text{and} \quad \mathbb{E}[X_t \mid \mathcal{F}_s] \geq X_s \quad \text{a.s.}$$

and a *supermartingale* if

$$\mathbb{E}[X_t \mid \mathcal{F}_s] \leq X_s \quad \text{a.s.}$$

Semimartingale and stochastic integral

An adapted process $(X_t)_{t\geq 0}$ is a *semimartingale* if there exist a local martingale (M_t) and an adapted finite–variation process (A_t) such that

$$X_t = X_0 + M_t + A_t, \qquad t \geq 0.$$

The class of semimartingales is the largest natural class of processes for which the Itô–type stochastic integral $\int_0^t H_s \, \mathrm{d}X_s$ is well defined for every predictable integrand satisfying appropriate integrability conditions.

Square–integrable martingale

A martingale $(M_t)_{t\geq 0}$ is called *square–integrable* if

$$\mathbb{E}[M_t^2] < \infty \quad \text{for all } t \geq 0.$$

In this case there exists an increasing predictable process $\langle M \rangle_t$, called the *(predictable) quadratic variation* of M, such that $M_t^2 - \langle M \rangle_t$ is a martingale. Square–integrable martingales play a central role in martingale representation theorems.

Link with option valuation

The Martingale Representation Theorem

In an arbitrage–free financial market, there exists a probability measure $\mathbb{Q}$, equivalent to $\mathbb{P}$, such that discounted prices of traded assets are martingales. If $(S_t)_{t\geq 0}$ is the price of an asset and $(B_t)_{t\geq 0}$ a numéraire (for example, the risk–free asset), the discounted price $\tilde{S}_t := S_t/B_t$ satisfies

$$\mathbb{E}^{\mathbb{Q}}[\tilde{S}_t \mid \mathcal{F}_s] = \tilde{S}_s, \qquad 0 \leq s \leq t, \tag{A.1.1}$$

that is, $(\tilde{S}_t)$ is a martingale under $\mathbb{Q}$.

European Option Valuation via Martingales

For a European option with payoff $H = \phi(S_T)$ at maturity T, the no–arbitrage price at time $t \leq T$ is given in martingale form by

$$V_t = B_t \, \mathbb{E}^{\mathbb{Q}}\left[\frac{H}{B_T} \,\middle|\, \mathcal{F}_t\right]. \tag{A.1.2}$$

Thus, option valuation is expressed as the conditional expectation (under $\mathbb{Q}$) of the discounted payoff, while semimartingales provide a general framework for price dynamics and for rigorously defining trading gains via stochastic integration.

Girsanov's Theorem: The Bridge Between Real-World and Risk-Neutral Dynamics

Statement of Girsanov's Theorem

Girsanov's theorem provides the mathematical foundation for transforming the real-world probability measure $\mathbb{P}$ to the risk-neutral measure $\mathbb{Q}$. This transformation is essential because it removes the unobservable drift (expected return) and isolates only the parameters required for option pricing.

Theorem 5 (Girsanov, 1960, [13])**.** Let W_t be a Brownian motion on $(\Omega, \mathcal{F}, \mathbb{P})$, and let θ_t be an adapted process satisfying the Novikov condition:

$$\mathbb{E}^{\mathbb{P}}\left[\exp\left(\frac{1}{2}\int_0^T \theta_s^2 \, ds\right)\right] < \infty. \tag{A.1.3}$$

Define the Radon–Nikodym derivative

$$\left.\frac{d\mathbb{Q}}{d\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T = \exp\left(-\int_0^T \theta_s \, dW_s - \frac{1}{2}\int_0^T \theta_s^2 \, ds\right), \tag{A.1.4}$$

where Z_t is a strictly positive $\mathbb{P}$-martingale (Doléans–Dade exponential).

Then, under the new measure $\mathbb{Q}$, the process

$$\widetilde{W}_t := W_t + \int_0^t \theta_s \, ds \tag{A.1.5}$$

is a $\mathbb{Q}$-Brownian motion (i.e., a standard Brownian motion on $(\Omega, \mathcal{F}, \mathbb{Q})$).

Remark 6. The Novikov condition ensures that Z_t is indeed a martingale (not merely a local martingale), guaranteeing that $\mathbb{Q}$ is a valid probability measure equivalent to $\mathbb{P}$.

Application to Geometric Brownian Motion: Eliminating the Drift

Consider an asset with price dynamics under the real-world measure $\mathbb{P}$:

$$\frac{dS_t}{S_t} = \mu\, dt + \sigma\, dW_t, \tag{A.1.6}$$

where μ is the expected return (unobservable, market-dependent) and σ is the volatility (observable from market prices).

The risk-free asset (numéraire) follows:

$$dB_t = r\, B_t\, dt, \tag{A.1.7}$$

where r is the constant risk-free rate.

Step 1: Apply Girsanov with the Correct Market Price of Risk

To make the discounted price $\tilde{S}_t := S_t/B_t$ a martingale under the new measure $\mathbb{Q}$, we apply Girsanov's theorem with:

$$\theta_t = \frac{\mu - r}{\sigma}. \tag{A.1.8}$$

This is the *market price of risk*—the excess return per unit of volatility.

Define the Radon–Nikodym derivative:

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp\left(-\int_0^t \frac{\mu - r}{\sigma}\, dW_s - \frac{1}{2}\int_0^t \left(\frac{\mu - r}{\sigma}\right)^2 ds \right). \tag{A.1.9}$$

Under $\mathbb{Q}$, the Brownian motion becomes:

$$\widetilde{W}_t := W_t + \int_0^t \frac{\mu - r}{\sigma}\, ds = W_t + \frac{\mu - r}{\sigma} t. \tag{A.1.10}$$

Step 2: Rewrite Asset Dynamics Under the Risk-Neutral Measure

Substituting $dW_t = d\widetilde{W}_t - \frac{\mu - r}{\sigma} dt$ into equation (A.1.6):

$$\begin{aligned} \frac{dS_t}{S_t} &= \mu\, dt + \sigma \left(d\widetilde{W}_t - \frac{\mu - r}{\sigma}\, dt \right) && \text{(A.1.11)} \\ &= \mu\, dt + \sigma\, d\widetilde{W}_t - (\mu - r)\, dt && \text{(A.1.12)} \\ &= r\, dt + \sigma\, d\widetilde{W}_t. && \text{(A.1.13)} \end{aligned}$$

Remarkably, the expected return μ has been replaced by the risk-free rate r, independent of the investor's time preferences.

Step 3: Verify the Martingale Property

The discounted price $\tilde{S}_t = S_t/B_t = S_t e^{-rt}$ satisfies:

$$\begin{aligned} d\tilde{S}_t &= d\left(\frac{S_t}{B_t}\right) && \text{(A.1.14)} \\ &= \frac{1}{B_t}\,dS_t - \frac{S_t}{B_t^2}\,dB_t && \text{(A.1.15)} \\ &= \frac{1}{B_t}\left[\frac{dS_t}{S_t}\cdot S_t\right] - \frac{S_t}{B_t^2}\cdot rB_t\,dt && \text{(A.1.16)} \\ &= \tilde{S}_t\left[r\,dt + \sigma\,d\widetilde{W}_t\right] - \tilde{S}_t\cdot r\,dt && \text{(A.1.17)} \\ &= \sigma\,\tilde{S}_t\,d\widetilde{W}_t. && \text{(A.1.18)} \end{aligned}$$

Since the drift term vanishes and only the martingale part $d\widetilde{W}_t$ remains, $(\tilde{S}_t)$ is indeed a $\mathbb{Q}$-martingale. This confirms equation (A.1.1).

Connection to PDE Formulations

Itô's Lemma and the Martingale Approach

For a European option with value $V(t, S_t)$, apply Itô's lemma under the real-world measure $\mathbb{P}$:

$$dV = \left[\frac{\partial V}{\partial t} + \mu S_t\frac{\partial V}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 V}{\partial S_t^2}\right]dt + \sigma S_t\frac{\partial V}{\partial S_t}\,dW_t. \tag{A.1.19}$$

Under the risk-neutral measure $\mathbb{Q}$, using $dW_t = d\widetilde{W}_t - \theta_t\,dt$ with $\theta_t = (\mu - r)/\sigma$:

$$dV = \left[\frac{\partial V}{\partial t} + rS_t\frac{\partial V}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 V}{\partial S_t^2}\right]dt + \sigma S_t\frac{\partial V}{\partial S_t}\,d\widetilde{W}_t. \tag{A.1.20}$$

The crucial observation: the drift coefficient μ has been replaced by r.

The Martingale Condition Implies the Black–Scholes PDE

For a self-financing replicating portfolio, the discounted option value $\tilde{V}_t = V_t/B_t$ must be a $\mathbb{Q}$-martingale. This requires:

$$d\tilde{V}_t = \frac{1}{B_t}\,dV - \frac{V}{B_t^2}\,dB_t = \text{(drift-free process)} = \text{(martingale part only)}. \tag{A.1.21}$$

The drift must be zero:

$$\frac{\partial V}{\partial t} + rS_t\frac{\partial V}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2\frac{\partial^2 V}{\partial S_t^2} - rV = 0. \tag{A.1.22}$$

This is the **Black–Scholes PDE**, derived purely from the martingale property under the risk-neutral measure.

Why Girsanov's Theorem is Essential

Existence and Uniqueness of the Risk-Neutral Measure

Proposition 7. In a complete and arbitrage-free market (with one risky asset and one risk-free asset), there exists a *unique* risk-neutral measure $\mathbb{Q}$ equivalent to $\mathbb{P}$, and the market price of risk $\theta_t = (\mu - r)/\sigma$ determines this measure uniquely via Girsanov's theorem.

Remark 8. This uniqueness is crucial: it ensures that all derivatives are priced consistently (no arbitrage opportunities), regardless of investor preferences. Two investors with different expectations ($\mu^{(1)} \neq \mu^{(2)}$) will nevertheless agree on option prices because both use the same $\mathbb{Q}$.

Practical Consequence: Parameters Observable from Market Prices

Girsanov's theorem reveals why only certain parameters appear in the PDE:

- **Observable:** Spot price S_t, volatility σ (inferred from market prices via implied volatility), risk-free rate r (bond prices).
- **Unobservable:** Expected return μ (varies by investor, irrelevant for pricing).
- **Market price of risk:** $\theta = (\mu - r)/\sigma$ is independent of specific values of μ and r—it is determined by the market's collective preference and is implicit in option prices.

The transformation $W_t \to \widetilde{W}_t$ via Girsanov absorbs μ into the new measure $\mathbb{Q}$, leaving only observable quantities in the PDE.

Extension to Multidimensional Settings and Stochastic Volatility

Multi-Dimensional Girsanov

For a portfolio with n assets, let $\boldsymbol{S}_t = (S_t^1, \ldots, S_t^n)^\top$ satisfy:

$$\frac{d\boldsymbol{S}_t}{\boldsymbol{S}_t} = \boldsymbol{\mu}\, dt + \boldsymbol{\Sigma}\, d\boldsymbol{W}_t, \tag{A.1.23}$$

where $\boldsymbol{\mu} \in \mathbb{R}^n$ and $\boldsymbol{\Sigma} \in \mathbb{R}^{n\times d}$ (with d risk factors).

The market price of risk becomes a vector:

$$\boldsymbol{\theta} = \boldsymbol{\Sigma}^{-\top}(\boldsymbol{\mu} - r\mathbf{1}). \tag{A.1.24}$$

Girsanov's theorem still applies: define $\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp\left(-\int_0^t \boldsymbol{\theta}_s^\top d\boldsymbol{W}_s - \frac{1}{2}\int_0^t \|\boldsymbol{\theta}_s\|^2 ds\right)$, and the transformed Brownian motion $\widetilde{\boldsymbol{W}}_t$ becomes a standard $\mathbb{Q}$-Brownian motion in $\mathbb{R}^d$.

Heston Model: Stochastic Volatility

In the Heston model, the variance ν_t (not volatility) follows:

$$\frac{dS_t}{S_t} = \mu\, dt + \sqrt{\nu_t}\, dW_t^{(1)}, \tag{A.1.25}$$

$$d\nu_t = \kappa(\theta - \nu_t)\, dt + \xi\sqrt{\nu_t}\, dW_t^{(2)}, \tag{A.1.26}$$

where $W_t^{(1)}$ and $W_t^{(2)}$ are correlated Brownian motions (with correlation ρ).

Applying Girsanov with market price of risk $\theta_S = (\mu - r)/\sqrt{\nu_t}$ (for the stock) and a market price θ_ν for the variance risk:

$$\frac{dS_t}{S_t} = r\,dt + \sqrt{\nu_t}\,d\widetilde{W}_t^{(1)}, \tag{A.1.27}$$

$$d\nu_t = (\kappa(\theta - \nu_t) - \xi\sqrt{\nu_t}\,\lambda)\,dt + \xi\sqrt{\nu_t}\,d\widetilde{W}_t^{(2)}, \tag{A.1.28}$$

where λ is the market price of variance risk. The resulting Heston PDE is:

$$\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \kappa(\theta - \nu)\frac{\partial V}{\partial \nu} + \frac{1}{2}\nu S^2\frac{\partial^2 V}{\partial S^2} + \frac{1}{2}\xi^2\nu\frac{\partial^2 V}{\partial \nu^2} + \rho\xi\nu S\frac{\partial^2 V}{\partial S\partial \nu} = rV. \tag{A.1.29}$$

Again, μ is absent; only r (observable) and the market price of variance risk λ (implicit in derivative prices) appear.

Summary: Why Girsanov's Theorem Matters

1. **Mathematical existence:** Girsanov guarantees that a risk-neutral measure $\mathbb{Q}$ exists and is unique in a complete, arbitrage-free market.

2. **Measure transformation:** It provides an explicit construction of $\mathbb{Q}$ via the Radon–Nikodym derivative, showing how the real-world drift μ is replaced by the risk-free rate r.

3. **PDE derivation:** The martingale property of discounted prices under $\mathbb{Q}$ (guaranteed by Girsanov) directly implies the Black–Scholes PDE and its generalizations.

4. **Model independence:** The risk-neutral dynamics depend only on observable market parameters (σ, r), not on unobservable investor expectations (μ). This makes option pricing universal across different investors.

5. **Computational efficiency:** Girsanov allows us to replace a complex martingale expectation problem with a deterministic PDE, amenable to numerical methods (finite differences, Monte Carlo).

Remark 9. Girsanov's theorem is the mathematical linchpin connecting:

- Real-world price dynamics $\to$ Risk-neutral dynamics,
- Investor-specific expectations $\to$ Market-wide consensus prices,
- Expectation formulas $\to$ Deterministic PDEs,
- Probabilistic models $\to$ Numerical algorithms.

Without it, the entire edifice of modern derivatives pricing would lack rigorous foundations.

A.2 Merton Jump-Diffusion processes

The author acknowledges that the content that you will find here is heavily based on and taken from the great work done by Kazuhisa Matsuda [36], Cont, Rama and Tankov, Peter [15].
This section takes place in real-world measure $\mathbb{P}$ and we denote the instantaneous return with α to match literature habits. If one was to compute risk-neutral dynamics, the instantaneous return would be replace by risk-free rate r.

Model type

We will model our stock price as $S_t = S_0\, e^{L_t}$ where L_t will be a Lévy Process.
It is important to define precisely the "jump". Suppose that you see the price moving suddenly from S_t to $y_t S_t$, we will call y_t the *Absolute Price Jump Size* and we can observe that :

$$\frac{dS_t}{S_t} = \frac{y_t S_t - S_t}{S_t} = y_t - 1$$

Sometimes, we will use the log-size of this jump and this will be denoted by $\Upsilon_i = \ln(y_i)$.
We made the assumption that jumps are iid *(ie, independent and identically distributed)*, that the intensity λ is constant over time[10] ($\lambda \perp\!\!\!\perp t$) and that we have :

$$\begin{cases} \mathbb{P}(\text{one price jump in } dt) = \mathbb{P}(dN_t = 1) \simeq \lambda\, dt \\ \mathbb{P}(\text{more than one jump in } dt) = \mathbb{P}(dN_t \geq 2) \simeq 0 \\ \mathbb{P}(\text{no jump in } dt) = \mathbb{P}(dN_t = 0) \simeq 1 - \lambda\, dt \end{cases}$$

We define the quantity k as the expectation of this quantity : $k = \mathbb{E}[(y_t - 1)]$. As the size of the jumps (y_i) and their occurencies (N_t) are assumed to be independent, we have :

$$\mathbb{E}[(y_t - 1)dN_t] = \mathbb{E}[y_t - 1]\mathbb{E}[dN_t] = k\lambda$$

Now, we look at the quantity $\mathbb{E}\left[\frac{dS_t}{S_t}\right]$ if we were to model the returns dynamics as

$$\frac{dS_t}{S_t} = \alpha\, dt + \sigma\, dW_t + (y_t - 1)dN_t$$

We would have :

$$\begin{aligned} \mathbb{E}\left[\frac{dS_t}{S_t}\right] &= \mathbb{E}\left[\alpha\, dt + \sigma\, dW_t + (y_t - 1)dN_t\right] \\ \Leftrightarrow &= \mathbb{E}[\alpha\, dt] + \mathbb{E}[\sigma dW_t] + \mathbb{E}[(y_t - 1)dN_t] \\ \Leftrightarrow &= \alpha\, dt + 0 + \lambda k\, dt \\ \Leftrightarrow &= (\alpha + \lambda k)dt \end{aligned}$$

One can see we need to adjust the drift of our dynamics in order to cancel the "predictable part of the jumps". Our Lévy process will then be compensated by $-\lambda k$.

[10] While constant λ is convenient and appropriate for an introduction as this report aims to provide, real markets show jump clustering. Extensiens to self-exciting processes (Hawkes) or regime-switching models capture this clustering phenomenon.

- Under the real-world measure $\mathbb{P}$, the expected return is $\alpha + \lambda k$. Under the risk-neutral measure $\mathbb{Q}$ (via Girsanov's theorem), we must adjust the drift to ensure the expected return equals the risk-free rate r, which is achieved by setting the drift to $\alpha - \lambda k$ (with appropriate market price of jump risk). This ensures $\mathbb{E}^{\mathbb{Q}}[dS_t/S_t] = r\,dt$.
- The studied Lévy process will be a Brownian motion with drift (which will be a usual continuous diffusion process) on which we will add a compound Poisson process (which will be a discontinuous jump process). This Levy process will be written as :

$$L_t = \left(\alpha - \frac{\sigma^2}{2} - \lambda k\right) t + \sigma W_t + \sum_{i=1}^{N_t} \Upsilon_i \tag{A.2.1}$$

- We can observe that the discontinuous part (Compound Poisson jump process) has two different sources of randomness :
  - Poisson Process dN_t with intensity λ where λ represents the average number of jumps per unit of time as it is usually the case.
  - A random jump size ;
    - Merton : $\Upsilon \sim \mathcal{N}(\mu, \delta^2)$
    - Kou ; $\Upsilon \sim \text{dbl.expo}(p, \alpha_+, \alpha_-)$ where p represents the probability of the jump to be positive (or negative depending on the definition).

From this simple modelization of the asset, we can infer the log-returns modelization :

$$\ln\left(\frac{S_t}{S_0}\right) = L_t = \left(\alpha - \frac{\sigma^2}{2} - \lambda k\right) t + \sigma W_t + \sum_{i=1}^{N_t} \Upsilon_i$$

Model derivation

Log-Normal properties

$$\ln x \sim \mathcal{N}(a, b) \Leftrightarrow \tag{A.2.2}$$

$$x \sim LogNorm\left(\exp\left(a + \frac{b^2}{2}\right), \exp\left(2a + b^2\right).(e^{b^2} - 1)\right) \tag{A.2.3}$$

Merton theory assumes that the log-size of the jumps follow a normal distribution : $\Upsilon_i \sim \text{iid}\ \mathcal{N}(\mu, \delta^2)$. Due to log-normal properties (A.2.3) we have that :

$$\rightarrow \mathbb{E}[y_t] = \exp\left(\mu + \frac{\delta^2}{2}\right)$$

$$\rightarrow \mathbb{E}\left[(y_t - \mathbb{E}[y_t])^2\right] = \exp\left(2\mu + \delta^2\right).\left(e^{\delta^2} - 1\right)$$

We can describe the ***Merton Jump Diffusion SDE***[11] :

$$\frac{dS_t}{S_t} = (\alpha - \lambda k)dt + \sigma\, dW_t + (y_t - 1)dN_t \tag{A.2.4}$$

We give some short explaination of each term :

[11] As already introduced in 2.4

- W_t is a standard brownian motion
- N_t follows a Poisson process with intensity λ
- α is the instantaneous return on asset
- σ is the instantaneous volatility conditional on that no jump occurs
- W_t, N_t, y_t are mutually independent
- $(y_t - 1)$ is the relative price jump size of S_t. Its properties are studied below
- $k \equiv \exp\left(\mu + \frac{\delta^2}{2}\right) - 1$

We give some more description on the relative jump size $(y_t - 1)$:

- k is defined as the expectation of this quantity : $\mathbb{E}[(y_t - 1)] = \exp\left(\mu + \frac{\delta^2}{2}\right) - 1$
- $\mathbb{E}\left[(y_t - 1 - \mathbb{E}[(y_t - 1)])^2\right] = \exp\left(2\mu + \delta^2\right).\left(e^{\delta^2} - 1\right)$
- $y_t \sim \text{iid } \log\mathcal{N}\left(\exp\left(\mu + \frac{\delta^2}{2}\right), \exp\left(2\mu + \delta^2\right).\left(e^{\delta^2} - 1\right)\right)$
- $y_t - 1 \sim \text{iid } \log\mathcal{N}\left(k, \exp\left(2\mu + \delta^2\right).\left(e^{\delta^2} - 1\right)\right)$
- $\Upsilon_t = \ln(y_t) \sim \text{iid}\mathcal{N}(\mu, \delta^2)$

Solution of SDE $\frac{dS_t}{S_t} = (\alpha - \lambda k)dt + \sigma dW_t + (y_t - 1)dN_t$

Itô's Lemma for Lévy Processes : For $dX_t = K_t dt + H_t dW_t + J_t dN_t$ and $f \in C^2$:

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + K_t \frac{\partial f}{\partial X_t} + \frac{1}{2} H_t^2 \frac{\partial^2 f}{\partial X_t^2}\right) dt + \frac{\partial f}{\partial X_t} H_t dW_t + \left[f(X_{t_-} + J_t) - f(X_{t_-})\right] \tag{A.2.5}$$

Where the following must hold :

$$\int_0^T |K_s| ds < +\infty \tag{A.2.6}$$

$$\int_0^T ||H_s||^2 ds < +\infty \tag{A.2.7}$$

In order to solve this problem, we need the Ito's Lemma for Levy Processes given by equation (A.2.5). We apply this Lemma to $f(t, X_t) = \ln(X_t)$.

$$d\ln(S_t) = \frac{\partial \ln(S_t)}{\partial t} dt + \left[(\alpha - \lambda k) S_t \frac{\partial \ln(S_t)}{\partial S_t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 \ln(S_t)}{\partial S_t^2}\right] dt + \sigma S_t \frac{\partial \ln(S_t)}{\partial S_t} dW_t + (\ln(y_t S_t) - \ln(S_t))$$

We use :

$$\frac{\partial \ln(S_t)}{\partial S_t} = \frac{1}{S_t} \qquad \frac{\partial^2 \ln(S_t)}{\partial S_t^2} = \frac{-1}{S_t^2}$$

And en up with :

$$d\ln(S_t) = 0\,dt + (\alpha - \lambda k)dt - \frac{\sigma^2}{2}dt + \sigma dW_t + \ln(y_t) \tag{A.2.8}$$

By integrating the expression between 0 and t, we get :

$$\ln(S_t) = \ln(S_0) + \left(\alpha - \frac{\sigma^2}{2} - \lambda k\right)t + \sigma W_t + \sum_{i=1}^{N_t} \ln(y_i) \tag{A.2.9}$$

Which is equivalent to both following formulations that one can often find in the literature:

$$\begin{cases} S_t = S_0 \exp\left(\left(\alpha - \frac{\sigma^2}{2} - \lambda k\right)t + \sigma W_t\right) \exp\left(\sum_{i=1}^{N_t} \ln(y_i)\right) \\ S_t = S_0 \exp\left(\left(\alpha - \frac{\sigma^2}{2} - \lambda k\right)t + \sigma W_t\right) \Pi_{i=1}^{N_t} y_i \end{cases}$$

It is worth mentioning how we transition from (A.2.8) to (A.2.9) and especially how $\int_0^t \ln(y_i)dN_s = \sum_{i=1}^{N_t} \ln(y_i)$. When the Poisson process N_t jumps by 1 (it can only jump by natural numbers 0,1,2,...) at time τ_i, the jump integral picks up $\ln(y_i)$.

Comparison with GBM Black-Scholes

In MJD, we have $S_t = S_0 e^{L_t}$ where L_t is a Lévy process defined by a Brownian motion with drift and a compound Poisson Process :

$$L_t = \left(\alpha - \frac{\sigma^2}{2} - \lambda k\right)t + \sigma W_t + \sum_{i=1}^{N_t} \ln(y_i)$$

We thus have log-returns modeled as a Lévy process : $\ln\left(\frac{S_t}{S_0}\right) = L_t$.
While, in a GBM, log-returns are normally distributed :

$$S_t = S_0 \exp\left(\left(\alpha - \frac{\sigma^2}{2}\right)t + \sigma dW_t\right)$$
$$\rightarrow \ln\left(\frac{S_t}{S_0}\right) \sim \mathcal{N}\left(\left(\alpha - \frac{\sigma^2}{2}\right)t; \sigma^2 t\right)$$

Characteristic function of Merton log-returns

Here, we would like to retrieve the CF of the distribution of the log-returns.The characteristic function is essential for efficient option pricing via Fast Fourier Transform (FFT) methods. For MJD, the CF admits a closed-form expression, enabling rapid numerical computation of option values and Greeks via Fourier inversion techniques. For what follows, we will assume that the jump sizes are normally distributed : $\Upsilon \sim \mathcal{N}(\mu, \delta^2)$. Let us now analyze the probability of our log-return $x_t = \ln\left(\frac{S_t}{S_0}\right)$ to be in a gevion interval A :

$$\mathbb{P}(x_t \in A) = \sum_{i=0}^{\infty} \mathbb{P}(N_t = i).\mathbb{P}(x_t \in A | N_t = i)$$

From which we can infer our probability density:

$$p(x_t) = \sum_{i=0}^{\infty} \underbrace{\frac{e^{-\lambda t}(\lambda t)^i}{i!}}_{*_1} . \underbrace{N\left(x_t; \left(\alpha - \frac{\sigma^2}{2} - \lambda k\right)t + i\mu, \sigma^2 t + i\delta^2\right)}_{*_2}$$

where :

- $*_1$ is the probability that asset price jumps i times during the interval of length t
- $*_2$ is a black-scholes-like normal density of log-return ($N(x;\mu,\sigma^2)$ is the Gaussian density) assuming that the asset price jumps i times in the same interval.

And we can now infer the log-returns density function with a Fourier Transform :

$$\begin{aligned}\phi(\omega) &\equiv \int_{-\infty}^{\infty} \exp(i\,\omega\,x_t)p(x_t)dx_t \\ &= \exp\left(\lambda\,t\,\exp\left(\frac{1}{2}\omega\left(2\,i\,\mu-\delta^2\omega\right)\right)-\lambda t(1+i\omega k)-\frac{1}{2}t\omega\left(-2i\alpha+\sigma^2(i+\omega)\right)\right)\end{aligned}$$

If we define the characteristic exponent as $\varphi(\omega)$ in $\phi(\omega)=e^{t\,\varphi(\omega)}$, we can find that :

$$\varphi(\omega)=\lambda\left(\exp\left(i\omega\mu-\frac{\delta^2\omega^2}{2}\right)-1\right)+i\omega\left(\alpha-\frac{\sigma^2}{2}-\lambda k\right)-\frac{\sigma^2\omega^2}{2}$$

where we have $k=\exp\left(\mu+\frac{\delta^2}{2}\right)-1$.

Log stock Price process for Merton Jump-Diffusion Model

We start by reminding equation (A.2.9) :

$$\ln(S_t)=\ln(S_0)+\left(\alpha-\frac{\sigma^2}{2}-\lambda k\right)t+\sigma W_t+\sum_{i=1}^{N_t}\Upsilon_i \tag{A.2.10}$$

From which we obtained :

$$p(\ln(S_t))=\sum_{i=0}^{\infty}\underbrace{\frac{e^{-\lambda t}(\lambda t)^i}{i!}}_{*_1}.\underbrace{N\left(\ln(S_t);\ln(S_0)+\left(\alpha-\frac{\sigma^2}{2}-\lambda k\right)t+i\mu,\sigma^2t+i\delta^2\right)}_{*_2}$$

Where

$$\begin{aligned}&N\left(\ln(S_t);\ln(S_0)+\left(\alpha-\frac{\sigma^2}{2}-\lambda k\right)t+i\mu,\sigma^2t+i\delta^2\right)\\ &=\frac{1}{\sqrt{2\pi(\sigma^2t+i\delta^2)}}\exp\left[-\frac{\left\{\ln(S_t)-\left(\ln(S_0)+\left(\alpha-\frac{\sigma^2}{2}-\lambda k\right)t+i\mu\right)\right\}^2}{2(\sigma^2t+i\delta^2)}\right]\end{aligned}$$

A.3 MJD PIDE proof

This appendix will give a revisit of the proof made in [15]. This is intended to be made more affordable for people new to Lévy and jump processes. The author does not claim the following proof to be his own.

We start this appendix by stating the proposition giving the final result.

Proposition - Backward PIDE for European option

For a European option with a payoff $H(S_T)$, the payoff function is assumed to be Lipschitz continuous. We thus have :

$$|H(x_1) - H(x_2)| \leq c|x_1 - x_2| \tag{A.3.1}$$

Our underlying asset is assumed to have following dynamics :

$$S_t = S_0 \exp(rt + L_t) \tag{A.3.2}$$

We define L_t as a Lévy-Process satisfying following properties :

1. L_t is a Lévy-Process with triplet (ν, γ, σ^2) such that, under risk-free measure $\mathbb{Q}$,

   $$\hat{S}_t = e^{-rt} S_t = e^{X_t}$$

   is a Martingale.

2. We assume the following to hold :

   $$\int_{|y|\geq 1} e^{2y} \nu(dy) < \infty \tag{A.3.3}$$

If one of the following holds :

$$\sigma > 0 \tag{A.3.4}$$

$$\exists \beta \in]0, 2[, \qquad \liminf_{\epsilon \downarrow 0} \epsilon^{-\beta} \int_{-\epsilon}^{\epsilon} |x|^2 d\nu(x) > 0 \tag{A.3.5}$$

Then, the value of a european style option with maturity payoff $H(S_T)$ is given by $V(t, S_t)$ where :

- $V : [0, T] \times]0, \infty[\to \mathbb{R}$
- $(t, S_t) \mapsto V(t, S_t) = \mathbb{E}[H(S_T)|S_t = S]$

is continuous on its domain, is once differentiable in time and twice w.r.t S_t (ie, is $C^{1,2}$) on $]0, T[\times]0, \infty[$ and verifies the PIDE :

$$\frac{\partial V}{\partial t} + rS_t \frac{\partial V}{\partial S_t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2} - rV + \int \left[V(t, S_t e^y) - V(t, S_t) - S_t(e^y - 1) \frac{\partial V}{\partial S_t} \right] \nu(dy) = 0 \tag{A.3.6}$$

on $[0, T[\times]0, \infty[$ with the condition at maturity :

$$\forall S > 0, \quad V(T, S) = H(S_T)$$

Proof

We start this proof by giving the dynamics assumed for the underlying asset (and how it is related to the dynamics we described in 2.4) as well as the Itô's lemma for Jump-diffusion processes.

Underlying asset dynamics :

$$S_t = S_0 + \int_0^t rS_u\,du + \int_0^t S_u\sigma dW_u + \int_0^t \int_{-\infty}^{\infty} (e^x - 1)S_u\tilde{J}(du, dx) \tag{A.3.7}$$

Where

- $\tilde{J}(du, dx) = J(du, dx) - \nu(dx)dt$
- $J(du, dx)$ is the Poisson random measure
- $\nu(dx)$ is the Lévy-measure

Itô lemma for Jump-diffusion processes Let X be a diffusion process with jumps, defined as :

$$X_t = X_0 + \int_0^t b_s ds + \int_0^t \sigma_s dW_s + \sum_{i=1}^{N_t} \Delta X_i \tag{A.3.8}$$

where

$$\int_0^T |b_s| ds < +\infty$$
$$\int_0^T |\sigma_s|^2 ds < +\infty$$

Then, for any function twice differentiable w.r.t underlying asset and once w.r.t time (ie, $C^{1,2}$) $f : [0, T] \times \mathbb{R} \to \mathbb{R}$, the process $V_t = f(t, X_t)$ can be represented as :

$$dV_t = \left(\frac{\partial f}{\partial t} + b_t \frac{\partial f}{\partial X_t} + \frac{\sigma_t^2}{2}\frac{\partial^2 f}{\partial X_t^2}\right) dt + \sigma_t \frac{\partial f}{\partial X_t} dW_t + [f(X_{t^-} + \Delta X_t) - f(X_{t^-})] \tag{A.3.9}$$

Itô's lemma for semi-martingales Let $(X_t)_{t\geq 0}$ be a semi-martingale. For any $C^{1,2}$ function $f : [0, T] \times \mathbb{R} \to \mathbb{R}$,

$$\begin{aligned} f(t, X_t) - f(0, X_0) = &\int_0^t \frac{\partial f}{\partial s}(s, X_s)ds + \int_0^t \frac{\partial f}{\partial x}(s, X_{s^-})dX_s + \frac{1}{2}\int_0^t \frac{\partial^2 f}{\partial x^2}(s, X_{s^-})d\langle S, S\rangle\Big|_s^c \\ &+ \sum_{0\leq s\leq t}\left[f(s, X_s) - f(s, X_{s^-}) - \Delta X_s \frac{\partial f}{\partial x}(s, X_{s^-})\right] \end{aligned} \tag{A.3.10}$$

We assume $V(t, S_t)$ to be differentiable function of t and smooth function of S_t. A proof to assess both assumptions is available at *Proposition 12.1* of [15].

We apply the Itô's lemma for semi-martingales as in (A.3.10) to $V(t, S_t)$.

$$dV(t,S_t) = \frac{\partial V}{\partial t}(t,S_{t^-}) + \frac{\partial V}{\partial S_t}(t,S_{t^-})dS_t + \frac{1}{2}\frac{\partial^2 V}{\partial S_t^2}(t,S_{t^-})d\langle S,S\rangle\Big|_t^c + \underbrace{\sum_{jumps}\left[V(t,S_t) - V(t,S_{t^-}) - \frac{\partial V}{\partial S_t}(t,S_{t^-})\Delta S_t\right]}_{*_1}$$

where

$$\begin{aligned}
&dS_t = rS_t + \sigma S_t dW_t + \int_{\mathbb{R}} S_t(e^x - 1)\tilde{J}(dx,dt)\\
&d\langle S,S\rangle\Big|_t^c = \sigma^2 S_t^2 dt\\
&\Delta S_t = S_t(e^x - 1) \quad \text{for a jump of size } x\\
&*_1 = \int_{\mathbb{R}} V(t,S_t e^x) - V(t,S_{t^-}) - \frac{\partial V}{\partial S_t}(t,S_{t^-})S_{t^-}(e^x-1)J(dx,dt)
\end{aligned}$$

Using $\tilde{J}(dx,dt) = J(dx,dt) - \nu(dx)dt$ and combinin term accordingly, we get :

$$\begin{aligned}
dV = &\left[\frac{\partial V}{\partial t} + rS_{t^-}\frac{\partial V}{\partial S_t} + \frac{\sigma^2 S_{t^-}^2}{\partial S_t^2} + \int_{\mathbb{R}}\left(V(t,S_{t^-}e^x) - V(t,S_{t^-}) - \frac{\partial V}{\partial S_t}S_{t^-}(e^x-1)\nu(dx)\right)\right]dt\\
&+ \frac{\partial V}{\partial S_t}\sigma S_{t^-}dW_t + \int_{\mathbb{R}}\left[V(t,S_{t^-}e^x) - V(t,S_{t^-})\right]\tilde{J}(dx,dt)
\end{aligned} \tag{A.3.11}$$

We now define $\hat{V}(t,S_t) = e^{-rt}V(t,S_t)$ which resumes to :

$$\tilde{V}(t,S_t) = \mathbb{E}[e^{-rt}H|\mathcal{F}_t]$$

We would like to have that $\tilde{V}(t,S_t)$ is a martingale by construction. Using product rule :

$$d\tilde{V} = e^{-rt}dV_t - re^{-rt}V_t dt$$

Replacing dV_t by (A.3.11), we get :

$$d\hat{V}_t = a(t)dt + dM_t$$

where

$$\begin{aligned}
a(t) = e^{-rt}&\left[-rV_t + \frac{\partial V}{\partial t} + \frac{\sigma^2 S_{t^-}^2}{2}\frac{\partial^2 V_t}{\partial S_t^2} + rS_{t^-}\frac{\partial V_t}{\partial S_t}\right](t,S_{t^-}\\
&+ \int_{\mathbb{R}} e^{-rt}\left[V(t,S_{t^-}e^x) - V(t,S_{t^-}) - S_{t^-}(e^x-1)\frac{\partial V_t}{\partial S_t}(t,S_{t^-})\right]
\end{aligned}$$

and

$$dM_t = e^{-rt}\left[\frac{\partial V}{\partial S_t}(t,S_{t^-})\sigma S_{t^-}dW_t + \int_{\mathbb{R}}(V(t,S_{t^-}e^x) - V(t,S_{t^-}))\tilde{J}(dx,dt)\right]$$

As we would like $\tilde{V}(t,S_t)$ to be a martingale, we will first show dM_t is a martingale and then set $a(t)$ to zero. The desired PIDE will come from setting $a(t)$ to zero.

Why is dM_t a martingale ?

We will need the following propositions in our demonstration.

Compensated Poisson integrals
For any *Predictable random function* $\phi : \Omega \times [0,T] \times \mathbb{R}^d \to \mathbb{R}$ verifying

$$\mathbb{E}\left[\int_0^T \int_{\mathbb{R}^d} |\phi(t,y)|^2 \mu(dy,dt)\right] < \infty \tag{A.3.12}$$

the following properties hold :

$$t \mapsto \int_0^t \int_{\mathbb{R}^d} \phi(s,y)\tilde{M}(dy,ds) \quad \text{is a square integrable martingale}$$
$$\mathbb{E}\left[\left|\int_0^t \int_{\mathbb{R}^d} \phi(s,y)\tilde{M}(dy,ds)\right|^2\right] = \mathbb{E}\left[\int_0^t \int_{\mathbb{R}^d} |\phi(s,y)|^2 \mu(dy,ds)\right]$$

Isometry formula for Brownian integrals
Let ϕ be a *predictable process* verifying

$$\mathbb{E}\left[\int_O^T |\phi_t|^2 dt\right] < \infty \tag{A.3.13}$$

Then $\int_0^t \phi\, dW$ is a *square integrable martingale* and

$$\mathbb{E}\left[\int_0^T \phi_t dW_t\right] = 0 \tag{A.3.14}$$
$$\mathbb{E}\left[\left|\int_0^T \phi_t\, dW_t\right|^2\right] = \mathbb{E}\left[\int_0^T |\phi_t|^2 dt\right] \tag{A.3.15}$$

First, we note that our Lévy process X_t is such that e^{X_t} a martingale and we denote the time to maturity as $\tau = T - t$

Then, we infer that our option value function is Lipschitz continuous because the payoff function was assumed to be Lipschitz continuous.

$$\begin{aligned} C(t,x) - C(t,y) &= e^{-r\tau}\left(\mathbb{E}\left[H\left(xe^{r\tau+X_\tau}\right) - H\left(ye^{r\tau+X_\tau}\right)\right]\right) \\ &\leq c|x-y|\mathbb{E}\left[e^{X_\tau}\right] = c|x-y| \end{aligned}$$

We used the fact that e^{X_t} was a martingale on last equality.
Now, we analyze the "predictable random function" $\psi(t,x) \triangleq V(t, S_{t^-}e^x) - V(t, S_{t^-})$. We show that this predictable random function checks sufficient condition (A.3.12).

$$\begin{aligned} \mathbb{E}\left[\int_0^T \int_{\mathbb{R}} |\psi(t,x)|^2 \nu(dx)\, dt\right] &= \mathbb{E}\left[\int_0^T \int_{\mathbb{R}} |V(t,S_{t^-}e^x) - V(t,S_{t^-})|^2 \nu(dx)\, dt\right] \\ &\leq \mathbb{E}\left[\int_0^T \int_{\mathbb{R}} x^2(e^{2x}+1)S_{t^-}^2 \nu(dx)\, dt\right] \\ &\leq c^2 \int_{\mathbb{R}} (e^{2x}+1)\nu(dx)\, \mathbb{E}\left[\int_0^T S_{t^-}^2 dt\right] < \infty \end{aligned}$$

We can apply Proposition of ***Compensated Poisson integrals*** to asses that the equation (A.3.16) is square-integrable martingale :

$$\int_0^t \int_{\mathbb{R}} e^{-rt} \left[V(t, S_{t^-} e^x) - V(t, S_{t^-}) \right] \tilde{J}(dx, dt) \tag{A.3.16}$$

Due to the Lipschitz continuousity of V, we have

$$\left\| \frac{\partial V}{\partial S_t}(t, .) \right\| \leq c$$

That allows to say that

$$\mathbb{E}\left[\int_0^T S_{t^-}^2 \left| \frac{\partial V}{\partial S_t}(t, S_{t^-}) \right| dt \right] \leq c^2 \mathbb{E}\left[\int_0^T S_{t^-}^2 dt \right] < \infty$$

So, by proposition of ***Isometry formula for Brownian integrals***, $\int_0^t \sigma S_t \frac{\partial V}{\partial S_t}(t, S_{t^-}) dW_t$ is also a (square-integrable) martingale.
Since $V_t - \int_0^t a(s) ds$ is a martingale (by the martingale properties shown above), the drift term must vanish: $a(t) = 0$ for all $t \in [0, T)$.

Setting a(t) = 0 from the expression above yields the desired PIDE:

$$\frac{\partial V}{\partial t} + r S_t \frac{\partial V}{\partial S_t} + \frac{\sigma^2 S_t^2}{2} \frac{\partial^2 V}{\partial S_t^2} - rV + \int_{\mathbb{R}} \left[V(t, S_t e^y) - V(t, S_t) - S_t (e^y - 1) \frac{\partial V}{\partial S_t} \right] \nu(dy)$$

which matches equation (A.3.6).

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